# 1. Title Page

**Study Title:** A Phase 1/2/3, Placebo-Controlled, Randomized, Observer-Blind, Dose-Finding Study to Evaluate the Safety, Tolerability, Immunogenicity, and Efficacy of SARS-CoV-2 RNA Vaccine Candidates Against COVID-19 in Healthy Individuals

**Name of Test Drug/Investigational Product:** BNT162b2

**Indication Studied:** Prevention of Coronavirus Disease 2019 (COVID-19)

**Brief Description of Study Design (if not apparent from title):** This is a multicenter, Phase 1/2/3, randomized, placebo-controlled, observer-blinded study to assess the safety, tolerability, immunogenicity, and efficacy of the BNT162b2 vaccine against COVID-19. The study enrolled healthy participants 12 years of age and older, who were randomized to receive 2 doses of either the vaccine or a placebo, administered 21 days apart.

**Name of the Sponsor:**

BioNTech SE An der Goldgrube 12 55131 Mainz Germany

Pfizer Inc. 235 East 42nd Street New York, NY 10017 USA

**Protocol Identification (Code or Number):** C4591001

**Development Phase of Study:** Phase 1/2/3

**Study Initiation Date:** 27 July 2020

**Date of Early Study Termination (if any):** Not applicable

**Study Completion Date (Last Patient Completed):** 14 November 2020 (Primary data cutoff date)

**Name and Affiliation of Principal or Coordinating Investigator(s) or Sponsor's Responsible Medical Officer:**

Stephen J. Thomas, MD SUNY Upstate Medical University Department of Medicine, Division of Infectious Disease 750 E. Adams Street, CAB 311 Syracuse, NY 13210 USA

**GCP Compliance Statement:** This study was conducted in compliance with the protocol, the principles of Good Clinical Practice (GCP) as defined by the International Council for Harmonisation (ICH), and all applicable regulatory requirements.

**Date of the Report:** 10 November 2023

**Company/Sponsor Contact Persons:**

**Name:** Amy Rose **Affiliation:** Pfizer Inc. (Media Relations) **Telephone:** +1 (212) 733-7410

**Name:** Jasmina Alatovic **Affiliation:** BioNTech SE (Media Relations) **Telephone:** +49 (0)6131 9084 1513

# 2. Synopsis

**Name of Sponsor/Company:** Pfizer Inc. and BioNTech

**Name of Finished Product:** BNT162b2

**Name of Active Ingredient:** BNT162b2 (PF-07302048), a nucleoside-modified messenger RNA (modRNA) encoding the SARS-CoV-2 spike glycoprotein (P2 S)

**Title of Study:** A Phase 1/2/3, Placebo-Controlled, Randomized, Observer-Blind, Dose-Finding, and Efficacy Study to Evaluate the Safety, Tolerability, Immunogenicity, and Efficacy of SARS-CoV-2 RNA Vaccine Candidates Against COVID-19 in Healthy Individuals

**Protocol Number:** C4591001

**Investigators:** Multiple investigators at study centres across the United States, Argentina, Brazil, Turkey, and Germany.

**Study centre(s):** Multinational, multicentre study conducted at approximately 150 centres worldwide, including sites in the United States, Argentina, Brazil, South Africa, Turkey, and Germany.

**Publication (reference):** Polack FP, Thomas SJ, Kitchin N, et al. Safety and efficacy of the BNT162b2 mRNA Covid-19 vaccine. N Engl J Med. 2020;383(27):2603-2615.

**Studied period (years):** First participant enrolled: 04-May-2020; Last update posted: 28-Feb-2023.

**Phase of development:** Phase 1/2/3

**Objectives:**

Phase 1: To identify preferred vaccine candidate(s) and dose level(s) by evaluating the safety, tolerability, and immunogenicity of escalating dose levels of BNT162b1 and BNT162b2 in healthy adults aged 18-55 years and 65-85 years.

Phase 2/3 Primary Efficacy Objective: To evaluate vaccine efficacy (VE) of prophylactic BNT162b2 against confirmed COVID-19 occurring from 7 days after the second dose in participants without evidence of infection before vaccination.

Phase 2/3 Primary Safety Objective: To describe the safety profile of prophylactic BNT162b2.

Phase 2/3 Secondary Objectives: To evaluate the efficacy of prophylactic BNT162b2 against confirmed COVID-19 occurring from 14 days after the second dose in participants with and without evidence of infection before vaccination. To describe the immunogenicity by measuring SARS-CoV-2 neutralizing titers, S1-binding IgG levels, and/or RBD-binding IgG levels. To assess the noninferiority of immune response to prophylactic BNT162b2 in participants aged 12 to 15 years compared to participants aged 16 to 25 years based on the geometric mean ratio (GMR) of SARS-CoV-2 neutralizing titers (added by protocol amendment).

**Methodology:** This was a Phase 1/2/3, multicentre, multinational, randomized, placebo-controlled, observer-blind, dose-finding, vaccine candidate-selection, and efficacy study in healthy individuals. The study comprised two parts. Phase 1 was designed to identify preferred vaccine candidate(s) and dose level(s), enrolling participants aged 18-55 years and

65-85 years. Phase 2/3 was an expanded-cohort and efficacy part, initially enrolling participants aged 16 years and older. A subsequent protocol amendment expanded Phase 2/3 enrolment to include participants aged 12 to 15 years. Participants were randomized 1:1 to receive BNT162b2 or placebo.

**Number of patients (planned and analysed):** Phase 1: 195 participants were planned (15 participants per group across 13 groups, with 12 receiving active vaccine and 3 receiving placebo per group). Phase 2/3: Approximately 44,000 participants were planned overall, with approximately 22,000 randomized to each treatment group (BNT162b2 or placebo). The 12- to 15-year age stratum comprised approximately 2,000 participants. A total of 37,796 participants in Phase 2/3 were randomized and included in analyses.

**Diagnosis and main criteria for inclusion:** Healthy males and females. Phase 1 enrolled participants aged 18-55 years and 65-85 years. Phase 2/3 initially enrolled participants aged 16 years and older; enrolment was subsequently expanded by protocol amendment to include participants aged 12 to 15 years. Participants were required to be healthy or have stable pre-existing medical conditions. Key exclusion criteria included known prior SARS-CoV-2 infection, immunocompromising condition or immunosuppressive therapy, and previous clinical or microbiological diagnosis of COVID-19.

**Test product, dose and mode of administration, batch number:** BNT162b2 (PF-07302048), SARS-CoV-2 RNA vaccine, 30 micrograms per dose, administered as an intramuscular (IM) injection in the deltoid muscle. Two doses were administered 21 days apart.

**Duration of treatment:** Two doses administered 21 days apart (Day 1 and Day 21). Participants were followed for up to 2 years after the second dose for safety and efficacy assessments.

**Reference therapy, dose and mode of administration, batch number:** Saline placebo (0.9% sodium chloride), administered as an intramuscular injection in the deltoid muscle. Two doses were administered 21 days apart, matching the vaccine schedule.

**Criteria for evaluation:**

Efficacy: The primary efficacy endpoint was the first occurrence of confirmed COVID-19 from 7 days after the second dose in participants without evidence of SARS-CoV-2 infection before vaccination. Confirmed COVID-19 was defined as the presence of at least 1 symptom and a positive SARS-CoV-2 nucleic acid amplification test (NAAT) result. Secondary efficacy endpoints included confirmed COVID-19 from 14 days after Dose 2, and assessment of severe COVID-19 cases.

Safety: Evaluated through the incidence and severity of solicited local reactions (e.g., injection site pain, redness, swelling) and systemic events (e.g., fatigue, headache, muscle pain, chills, joint pain, fever) for 7 days after each dose; unsolicited adverse events (AEs) through 1 month after Dose 2; serious adverse events (SAEs) throughout the study duration; and clinical laboratory abnormalities (Phase 1 only).

**Statistical methods:** Vaccine efficacy was calculated as VE = 100 x (1 - IRR), where IRR is the ratio of the first confirmed COVID-19 illness rate in the vaccine group to the corresponding illness rate in the placebo group. A Bayesian beta-binomial model was used, with success defined as a posterior probability of true VE exceeding 30% being greater than 98.6%. For the 12-15 year age group, immunobridging was assessed based on noninferiority

of the GMR of SARS-CoV-2 50% neutralizing titers, using a 1.5-fold noninferiority margin. Safety data were evaluated descriptively using summary statistics for local reactions, systemic events, adverse events, and serious adverse events. A 3-tier approach was used to summarize adverse events in Phase 2/3.

**SUMMARY - CONCLUSIONS:**

EFFICACY RESULTS: In the overall Phase 2/3 population (participants aged 16 years and older without evidence of prior SARS-CoV-2 infection), BNT162b2 demonstrated a vaccine efficacy of 95.0% (95% CI: 90.3%, 97.6%) against confirmed COVID-19 occurring from 7 days after Dose 2. In the adolescent subpopulation (participants aged 12 to 15 years), vaccine efficacy was 100.0% (95% CI: 86.8%, 100.0%) against confirmed COVID-19 occurring from 7 days after Dose 2. The immune response in participants aged 12 to 15 years was noninferior to that in participants aged 16 to 25 years based on SARS-CoV-2 neutralizing antibody titers measured 1 month after Dose 2.

SAFETY RESULTS: BNT162b2 was well-tolerated across all age groups studied. The most commonly reported solicited local reaction was injection site pain. The most commonly reported solicited systemic events were fatigue, headache, muscle pain, chills, joint pain, and fever. Systemic events were generally mild to moderate in severity and resolved within a few days of onset. Reactogenicity was generally higher after Dose 2 compared to Dose 1. In the adolescent population (12-15 years), the reactogenicity profile was generally comparable to that observed in young adults, with slightly higher frequencies of some systemic events. The incidence of serious adverse events was low and balanced between the BNT162b2 and placebo groups in the overall population. No vaccine-related deaths were reported.

CONCLUSION: The BNT162b2 vaccine, administered as two 30-microgram doses 21 days apart, demonstrated high efficacy in preventing confirmed COVID-19 in participants aged 16 years and older, and in participants aged 12 to 15 years. The vaccine had a favourable safety profile across all age groups studied. These results support the positive benefit-risk profile of BNT162b2 for active immunization to prevent COVID-19.

**Date of the report:** 26-Feb-2026

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# 4. List of Abbreviations and Definition of Terms

| Abbreviation | Definition/Context |
|---|---|
| AE | Adverse Event |
| BMI | Body Mass Index |
| BNT162b2 | BioNTech COVID-19 mRNA Vaccine |
| CI | Confidence Interval |
| COVID-19 | Coronavirus Disease 2019 |
| CRF | Case Report Form |
| DMC | Data Monitoring Committee |
| EUA | Emergency Use Authorization |
| GCP | Good Clinical Practice |
| GMR | Geometric Mean Ratio |
| GMT | Geometric Mean Titer |
| ICD | Informed Consent Document |
| ICH | International Council for Harmonisation |
| IEC | Independent Ethics Committee |
| IM | Intramuscular |
| IRB | Institutional Review Board |
| ITT | Intent-to-Treat |
| IgG | Immunoglobulin G |
| LNP | Lipid Nanoparticle |
| MedDRA | Medical Dictionary for Regulatory Activities |
| NAAT | Nucleic Acid Amplification Test |
| RBD | Receptor-Binding Domain |
| SAE | Serious Adverse Event |
| SAP | Statistical Analysis Plan |
| SARS-CoV-2 | Severe Acute Respiratory Syndrome Coronavirus 2 |
| SOP | Standard Operating Procedure |
| TESS | Treatment Emergent Signs and Symptoms |
| Th1 | T Helper Type 1 |
| VE | Vaccine Efficacy |
| WHO | World Health Organization |
| mRNA | Messenger Ribonucleic Acid |

# 5. Ethics

## 5.1 Independent Ethics Committee (IEC) or Institutional Review Board (IRB)

The study protocol, protocol amendments, informed consent document (ICD), safety report summary document(s) (SRSD), and other relevant documents (including advertisements) were reviewed and approved by the sponsor and submitted to an Independent Ethics Committee (IEC) or Institutional Review Board (IRB) by the investigator prior to the initiation of the study at each site.

Any amendments to the protocol required IRB/EC approval before implementation of changes made to the study design, except for changes necessary to eliminate an immediate hazard to study participants. The investigator was responsible for providing written summaries of the status of the study to the IRB/EC annually or more frequently in accordance with requirements, policies, and procedures established by the IRB/EC, and for notifying the IRB/EC of serious adverse events (SAEs) or other significant safety findings as required by IRB/EC procedures.

A list of all IECs or IRBs consulted for this study is provided in Appendix 16.1.3.

## 5.2 Ethical Conduct of the Study

The study was conducted in accordance with the ethical principles that have their origins in the Declaration of Helsinki, applicable International Conference on Harmonisation (ICH) Good Clinical Practice (GCP) guidelines (ICH E6), the requirements of 21 CFR, European regulation 536/2014 for clinical studies (where applicable), and all applicable local laws and regulations, including privacy laws.

The investigator was responsible for providing oversight of the conduct of the study at each site and adherence to the requirements of ICH guidelines, the IRB/EC, and all other applicable local regulations. The investigator permitted study-related monitoring, audits, IRB/EC review, and regulatory agency inspections and provided direct access to source data and documents.

## 5.3 Patient Information and Consent

Written informed consent was obtained from each participant or their parent(s)/legal guardian before enrollment in the study. Participants or their parent(s)/legal guardians were required to sign a statement of informed consent that met the requirements of 21 CFR 50, local regulations, ICH guidelines, Health Insurance Portability and Accountability Act (HIPAA) requirements where applicable, and the IRB/EC or study centre.

The investigator or their designated representative thoroughly explained the nature and objectives of the study, the study procedures, the sharing of study-related data, and the potential risks and benefits associated with participation, including the processing of personal data. Participants were given ample time to ask questions and decide on their voluntary participation. Participants were informed that their participation was voluntary and that they could withdraw at any time without penalty.

The informed consent process ensured participants were aware of their right to access and correct their personal data and to withdraw consent for data processing at any time. Participants were also informed that their medical records may be examined by Clinical Quality Assurance auditors or other authorised personnel appointed by the sponsor, by appropriate IRB/EC members, and by inspectors from regulatory authorities.

A signed and dated copy of the ICD was provided to the participant or their parent(s)/legal guardian, and the medical record explicitly documented that written informed consent was obtained before enrollment. Participants were reconsented to the most current version of the ICD(s) throughout their participation in the study, and those who were rescreened signed a new ICD.

Representative written information for the patient and a sample patient consent form are provided in Appendix 16.1.3.

# 6. Investigators and Study Administrative Structure

The administrative structure of the study involved a collaborative effort between BioNTech as the regulatory sponsor and Pfizer, responsible for the overall conduct of the study, with funding provided by both companies. The study was overseen by various committees and personnel to ensure ethical conduct, data integrity, and participant safety.

The Principal Investigator for the study was Professor Asma Khalil, with Professor Paul Heath serving as Co-Lead, both affiliated with St George's University of London. Investigators were responsible for providing oversight of the study at their respective sites, ensuring adherence to the protocol, ICH Good Clinical Practice (GCP) guidelines, and all applicable regulatory requirements. They were also responsible for submitting and providing updates to the Institutional Review Board/Ethics Committee (IRB/EC) and notifying the IRB/EC of any serious adverse events (SAEs) or other significant safety findings.

Study oversight included an Independent Review Committee (IRC) and a Data Monitoring Committee (DMC). The IRC, composed of internal members independent of the study team, was responsible for reviewing safety data to permit dose escalations, evaluate stopping rules, and select vaccine candidates/dose levels for Phase 2/3. The DMC, comprising external members independent of the study team, continuously monitored participant safety, including contemporaneous review of related adverse events and SAEs, and COVID-19 illness cases. The DMC provided recommendations to Pfizer, which were then forwarded to regulatory authorities as appropriate. Additionally, a group of three blinded, medically qualified Pfizer staff members served as internal case reviewers for potential COVID-19 illness events.

Data management, including quality checking, was the responsibility of the sponsor or its designee. Study monitors performed ongoing source data verification to confirm the accuracy, completeness, and verifiability of data entered into the Case Report Forms (CRFs) from source documents. This ensured the protection of participant safety and rights and confirmed adherence to the approved protocol, study agreements, ICH GCP, and regulatory requirements.

While not explicitly named, the involvement of Contract Research Organizations (CROs) is indicated in the protocol regarding study termination procedures, where the sponsor is required to inform any CROs used in the study of reasons for termination or suspension. The statistical analyses for Phase 2/3 data, while the study was ongoing, were performed by an unblinded statistical team, and research teams involved in data sharing were required to include a biostatistician.

Contact information for the sponsor's qualified medical personnel was documented in the study contact list, and participants were provided with a contact card for medical questions or problems.

The administrative structure is further detailed in supporting documentation, including the monitoring plan, which describes strategies, methods, responsibilities, and requirements for monitoring.

## 6.1 List of Investigators and Other Important Participants

**Investigators**

**Professor Asma Khalil** - **Affiliation:** St George's University of London - **Role in Study:** Principal Investigator (Study Doctor), Contact Person, Paediatric Infectious Diseases Research Group

**Professor Paul Heath** - **Affiliation:** St George's University of London - **Role in Study:** Co-Lead Investigator

Detailed qualifications (curriculum vitae or equivalent summaries of training and experience relevant to the performance of the clinical study) for the investigators and other important participants are provided in Appendix 16.1.4: List and description of investigators and other important participants in the study.

### 6.1.1 Other Persons Carrying Out Observations of Primary or Other Major Efficacy Variables

The study team included the study doctor, nurses, midwives, and other personnel working with the study doctor at the investigator site. These individuals were involved in the conduct of the study, including, but not limited to, patient care, data collection, and informed consent processes. The investigator, or an appropriately qualified and trained person designated by the investigator, was responsible for conducting the consent process. Specific names and affiliations for all persons carrying out observations of primary or other major efficacy variables are not exhaustively listed in this report but are maintained in the study records.

### 6.1.2 Author(s) of the Report, Including the Responsible Biostatistician(s)

The author(s) of this Clinical Study Report, including the responsible biostatistician(s), are internal personnel of Pfizer and/or BioNTech, or their designated representatives, who contributed to the planning, conduct, analysis, and reporting of the study. The unblinded statistical team was responsible for performing statistical analyses on Phase 2/3 data while the study was ongoing. Specific names of the report authors and responsible biostatisticians are available through the sponsor's internal documentation.

Signatures of the principal or coordinating investigator(s) or the sponsor's responsible medical officer, as required by regulatory authorities, are maintained in Appendix 16.1.5: Signatures of principal or coordinating investigator(s) or sponsor's responsible medical officer. An example of the signature form is provided in Annex II: Principal or Coordinating Investigator(s) Signature(s) or Sponsor's Responsible Medical Officer (Example).

# 7. Introduction

The novel coronavirus 2019 (SARS-CoV-2) emerged in Wuhan, China, in December 2019, leading to the coronavirus disease 2019 (COVID-19) pandemic, which was declared by the World Health Organization (WHO) on March 11, 2020. Given the rapid global transmission, high incidence, and significant public health and economic challenges posed by COVID-19, the urgent development of an effective vaccine was paramount. At the time of this study's initiation, no licensed vaccines were available to prevent SARS-CoV-2 infection or COVID-19.

This study, Protocol C4591001 (also referred to as BNT162-02), was a pivotal Phase 1/2/3, multicenter, multinational, randomized, placebo-controlled, observer-blind, dose-finding, vaccine candidate-selection, and efficacy study. It was conducted to evaluate the safety, tolerability, immunogenicity, and efficacy of BioNTech's RNA-based vaccine candidates against COVID-19 in healthy individuals. BioNTech developed RNA-based vaccine candidates using a platform approach, enabling rapid development against emerging viral diseases. Two nucleoside-modified messenger RNA (modRNA) vaccine candidates, BNT162b1 (encoding the receptor-binding domain [RBD]) and BNT162b2 (encoding the full-length, P2 mutant, prefusion spike glycoprotein [P2 S]), both formulated in the same lipid nanoparticle (LNP) composition, were investigated.

The study aimed to identify preferred vaccine candidate(s) and dose level(s) in Phase 1, followed by an expanded cohort and efficacy assessment in Phase 2/3. The target population included healthy individuals, initially aged 18 to 55 years and 65 to 85 years in Phase 1, and later expanded to individuals aged ≥12 years (stratified as 12-15, 16-55, or >55 years) in Phase 2/3. Participants in Phase 2/3 were also selected if they were at higher risk for acquiring COVID-19.

The investigational product administered in Phase 2/3 was BNT162b2 at a dose of 30 µg, given as a 2-dose schedule separated by 21 days. Participants were expected to be followed for a maximum of approximately 26 months, with some Phase 1 participants having shorter follow-up durations (no less than 6 months after the last vaccination). The primary objectives for Phase 1 focused on assessing safety, tolerability (local reactions, systemic events, adverse events [AEs], serious adverse events [SAEs], and laboratory parameters), and immunogenicity (SARS-CoV-2 neutralizing titers, S1-binding IgG, and RBD-binding IgG levels). For Phase 2/3, the primary objective was vaccine efficacy (VE) against confirmed COVID-19 disease, alongside the evaluation of safety and immunogenicity (specifically, the non-inferiority of immune response in participants aged 12 to 15 years compared to those aged 16 to 25 years).

This study adhered to guidelines for the development of clinical study protocols. This pivotal study (C4591001) provided the primary evidence supporting the positive benefit-risk balance for the vaccine, Comirnaty. This evidence contributed to the European Medicines Agency's (EMA) Committee for Medicinal Products for Human Use (CHMP) issuing a positive opinion for granting a conditional marketing authorization for Comirnaty on December 21, 2020, under Article 14-a of the relevant Regulation. In support of the conditional marketing authorization and ongoing product development, several post-approval commitments were made. These include four planned studies in paediatric subjects as laid down in the Paediatric Investigation Plan, a planned study in pregnant women in the EU, and a Post-Approval Active Surveillance Safety Study to Monitor Real-World Safety of Comir-

naty (Study C4591010). Additionally, non-interventional studies (test negative design) are planned to determine the vaccine's effectiveness in a real-world setting.

# 8. Study Objectives

The overall purpose of Study C4591001 (also referred to as BNT162-02) was to evaluate the safety, tolerability, immunogenicity, and efficacy of SARS-CoV-2 RNA vaccine candidates against COVID-19 in healthy individuals. The study was designed as a Phase 1/2/3, randomized, placebo-controlled, observer-blind, dose-finding, vaccine candidate-selection, and efficacy study.

The study was divided into two parts:

- **Phase 1:** To identify preferred vaccine candidate(s) and dose level(s).
- **Phase 2/3:** An expanded cohort and efficacy part to evaluate the safety and efficacy of the selected candidate (BNT162b2 at 30 µg).

Additionally, specific substudies (e.g., Substudy C) were conducted to learn if a booster shot of BNT162b2 at two different strengths works well and is safe in healthy young people (teenagers) from 12 to 17 years old.

**Table 1: Phase 1 Objectives, Estimands, Endpoints, and Analysis Framework**

| Objectives | Estimands | Endpoints | Analysis Framework | References |
|---|---|---|---|---|
| **Primary Objective:** To describe the safety and tolerability profiles of prophylactic BNT162 vaccines in healthy adults after 1 or 2 doses | In participants receiving 1 or more doses: percentage reporting local reactions (within 7 days), systemic events (within 7 days), AEs (Dose 1 to 1 month after last dose), SAEs (Dose 1 to 6 months after last dose) | Local reactions (pain, redness, swelling); systemic events (fever, fatigue, headache, chills, vomiting, diarrhea, muscle/joint pain); AEs; SAEs; abnormal hematology/chemistry | Descriptive statistics (proportion, frequency, severity grading) | Protocol Section 8; SAP |
| **Secondary Objective:** To describe immune responses elicited by prophylactic BNT162 vaccines | In evaluable participants at predefined time points (7 and 21 days after Dose 1; 7 and 14 days, 1 and 6 months after Dose 2): GMT, GMFR from baseline, proportion with 4-fold or greater rise | SARS-CoV-2 neutralizing titers; S1-binding IgG; RBD-binding IgG | Geometric mean titers, fold-rise analysis, seroresponse rates | Protocol Section 8; SAP |

**Table 2: Phase 2/3 Primary Objectives, Estimands, Endpoints, and Analysis Framework**

| Objectives | Estimands | Endpoints | Analysis Framework | References |
|---|---|---|---|---|
| **Primary Efficacy:** To evaluate efficacy of BNT162b2 against confirmed COVID-19 7 days or more after Dose 2 (without prior infection) | VE = 100 x (1 - IRR); cases confirmed 7 days or more after Dose 2 in evaluable population | Incidence of confirmed COVID-19 per 1000 person-years (NAAT confirmed) | Bayesian posterior probability; success if VE >30% | Protocol Section 8; SAP |
| **Primary Efficacy (All Participants):** With and without evidence of prior infection | VE = 100 x (1 - IRR) in all randomized participants | Confirmed COVID-19 cases per 1000 person-years | Bayesian analysis with prespecified posterior probability threshold | Protocol Section 8; SAP |
| **Primary Safety:** To define safety profile in all randomized participants | Percentage of participants reporting local/systemic events (7 days), AEs (1 month post-Dose 2), SAEs (6 months post-Dose 2) | Reactogenicity events, AEs, SAEs | Descriptive frequency analysis | Protocol Section 8; SAP |
| **Primary Immunogenicity (Boostability Subset):** Noninferiority after 3rd dose | GMR comparing post-dose 3 vs post-dose 2 immune response | Neutralizing titers (reference strain and SA strain) | Noninferiority margin predefined in SAP | Protocol Section 8; SAP |

**Table 3: Phase 2/3 Secondary and Exploratory Objectives, Estimands, Endpoints, and Analysis Framework**

| Objectives | Estimands | Endpoints | Analysis Framework | References |
|---|---|---|---|---|
| **Secondary Efficacy:** COVID-19 14 days or more after Dose 2 | VE = 100 x (1 - IRR); cases confirmed 14 days or more after Dose 2 | COVID-19 incidence per 1000 person-years | Incidence rate comparison | Protocol Section 8 |
| **Secondary Efficacy:** Severe COVID-19 (7 days or more and 14 days or more after Dose 2) | VE against severe confirmed cases | Severe COVID-19 incidence | Incidence comparison | Protocol Section 8 |
| **Secondary Immuno-genicity:** Noninferiority (12-15 yrs vs 16-25 yrs) | GMR of neutralizing titers 1 month post-vaccination | SARS-CoV-2 neutralizing titers | Noninferiority analysis | Protocol Section 8 |
| **Exploratory:** Long-term efficacy and immune persistence | VE during blinded follow-up; GMT/GMC at baseline, 1 and 6 months | COVID-19 incidence; S-binding IgG; neutralizing titers | Descriptive and longitudinal modeling | Protocol Section 8 |
| **Exploratory:** Stable HIV population | Safety and im-munogenicity in HIV-positive participants | Incidence; antibody titers | Descriptive analysis | Protocol Section 8 |

# 9. Investigational Plan

This section describes the investigational plan for Protocol C4591001, a Phase 1/2/3, randomized, multinational, placebo-controlled, observer-blind, dose-finding, vaccine candidate-selection, and efficacy study of BNT162b2 (COVID-19 mRNA vaccine) in healthy individuals.

## 9.1. Overall Study Design and Plan

This is a Phase 1/2/3, randomized, multinational, placebo-controlled, observer-blind, dose finding, vaccine candidate-selection, and efficacy study in healthy individuals. The study evaluated the safety, tolerability, and immunogenicity of 3 different SARS-CoV-2 RNA vaccine candidates against COVID-19 and the Phase 2/3 efficacy of 1 selected candidate (BNT162b2).

The study was conducted in multiple phases:

- **Phase 1**: Dose-finding and vaccine candidate selection in healthy adults aged 18 to 55 years and older adults aged 65 to 85 years. Three vaccine candidates (BNT162b1, BNT162b2, and BNT162b3) were evaluated at various dose levels to identify the preferred candidate and dose for Phase 2/3 evaluation.
- **Phase 2/3**: Efficacy, safety, and immunogenicity evaluation of the selected vaccine candidate (BNT162b2 at 30 micrograms) in participants aged 12 years and older. The Phase 2/3 portion was designed to assess vaccine efficacy against confirmed COVID-19 occurring from 7 days after the second dose.

Participants were randomized in a 1:1 ratio to receive either BNT162b2 or placebo (normal saline). Two doses were administered intramuscularly approximately 21 days apart. The study included a blinded placebo-controlled follow-up period, after which participants originally randomized to placebo were offered BNT162b2 vaccination during an open-label follow-up period.

## 9.2. Discussion of Study Design Including Choice of Control Group

The study utilized a placebo-controlled design with normal saline (0.9% sodium chloride) as the placebo to maintain observer-blinding and provide an unbiased assessment of vaccine efficacy and safety. The randomized, placebo-controlled design was selected to enable rigorous evaluation of vaccine efficacy against COVID-19 and to provide a concurrent control group for safety assessment.

An observer-blind design was implemented wherein the personnel administering the vaccine were aware of the treatment assignment, but the participants, investigators conducting assessments, and the sponsor personnel involved in data analysis remained blinded to treatment allocation. This design was chosen to minimize potential bias in the assessment of both efficacy and safety endpoints while maintaining participant safety.

The 1:1 randomization ratio was selected to provide adequate statistical power for the primary efficacy endpoint while ensuring a sufficient safety database in both treatment groups.

## 9.3. Selection of Study Population

The study enrolled healthy male and female participants. The Phase 2/3 portion of the study included participants aged 12 years and older at the time of consent. Key inclusion criteria included:

- Healthy participants as determined by medical history, physical examination, and clinical judgment of the investigator
- Male or female participants aged 12 years and older at the time of signing informed consent
- Participants who were willing and able to comply with all scheduled visits, treatment plan, laboratory tests, lifestyle considerations, and other study procedures

Key exclusion criteria included:

- Known history of SARS-CoV-2 infection (applicable to certain study phases)
- Previous clinical or microbiological diagnosis of COVID-19
- Receipt of any vaccine within 14 days prior to study vaccination
- Immunocompromised individuals with known or suspected immunodeficiency
- History of severe adverse reaction associated with a vaccine and/or severe allergic reaction to any component of the study intervention
- Pregnant or breastfeeding women
- Bleeding disorder or receipt of anticoagulant therapy that would contraindicate intramuscular injection

## 9.4. Investigational Product

The investigational products administered in this study are described below, including lot numbers used in the adolescent 6-month update.

**Table 4. Investigational Product Lot Numbers – Interim – Adolescent 6-Month Update**

| Investigational Product | Manufacturer | Vendor Lot Number (Manufacturer) | Lot Number (Pfizer) |
|---|---|---|---|
| BNT162b2 (30 micrograms) | BioNTech | BCV40720-A | PA2074172/P220395-0053L |
| | | BCV40720-A | PA2074998/P220395-0060L |
| | | BCV40720-B | PA2074173/P220395-0051L |
| | | BCV40720-C | PA2074071/P220395-0052L |
| | | ED3938 | PA2074300/P220395-0021L |
| | | ED3938 | PA2074300/P220395-0022L |
| | | ED3938 | PA2074300/P220395-0023L |
| | | EE3813 | NC2075485/P220395-0068L |

| Investigational Product | Manufacturer | Vendor Lot Number (Manufacturer) | Lot Number (Pfizer) |
|---|---|---|---|
| | | EE3813 | NC2075485/P220395-0074L |
| | | EE3813 | NC2075485/P220395-0077L |
| | | EE3813 | PA2074838/P220395-0020L |
| | | EE3813 | PA2074838/P220395-0024L |
| | | ER9449Z | PA2096794/P220395-0079L |
| | | ER9449Z | PA2096794/P220395-0082L |
| | | EE8493Y | PA2087473/P220395-0073L |
| | | EJ0553Z | PA2085061/P220395-0070L |
| Placebo (normal saline 0.9% sodium chloride solution) | Pfizer | DK2074;20-002221 | PA2069407/P220395-0032L |
| | | DK2074;20-002221 | PA2069407/P220395-0033L |
| | | DK2074;20-002221 | PA2069407/P220395-0034L |
| | | DK2074;20-002221 | PA2069407/P220395-0044L |
| | | DK2074;20-002221 | PA2069407/P220395-0045L |
| | | DK2074;20-002221 | PA2069407/P220395-0046L |
| | | DK2074;20-002221 | PA2069407/P220395-0055L |
| | | DK2074;20-002221 | PA2069407/P220395-0062L |
| | | DK2074;20-002221 | PA2069407/P220395-0065L |
| Diluent (normal saline 0.9% sodium chloride solution) | Pfizer | DK2074 | 20-002221 |
| | | DK1589 | 20-001776 |
| | | DK1589 | 20-001592 |

BNT162b2 is a nucleoside-modified messenger RNA (modRNA) vaccine encoding the SARS-CoV-2 full-length spike glycoprotein in its prefusion conformation, formulated in lipid nanoparticles. Each dose of BNT162b2 contained 30 micrograms of mRNA, supplied as a multi-

dose concentrated frozen suspension for intramuscular injection. Prior to administration, the vaccine was thawed and diluted with 0.9% sodium chloride solution (diluent). The placebo was normal saline (0.9% sodium chloride solution) for injection, visually indistinguishable from the diluted vaccine.

## 9.5. Dosing and Administration

BNT162b2 (30 micrograms) or placebo was administered as an intramuscular injection in the deltoid muscle. The primary vaccination series consisted of 2 doses administered approximately 21 days apart (with a protocol-defined window of 19 to 42 days between Dose 1 and Dose 2).

The vaccine was supplied as a frozen concentrated suspension in multi-dose vials. Prior to administration, the vaccine required thawing and dilution with 1.8 mL of 0.9% sodium chloride injection (normal saline). After dilution, each vial contained approximately 6 doses of 0.3 mL each. The diluted vaccine was administered as a 0.3 mL intramuscular injection.

For participants originally randomized to placebo who elected to receive BNT162b2 during the open-label follow-up period, 2 doses of BNT162b2 (30 micrograms) were administered approximately 21 days apart (Dose 3 and Dose 4).

## 9.6. Efficacy and Safety Endpoints

### Primary Efficacy Endpoint

The primary efficacy endpoint was the first confirmed COVID-19 illness occurrence from 7 days after the second dose in participants without serological or virological evidence of past SARS-CoV-2 infection prior to 7 days after the second dose.

A confirmed case of COVID-19 was defined as the presence of at least 1 of the following symptoms and a positive SARS-CoV-2 NAAT (nucleic acid amplification test) within 4 days of the symptomatic period:

- Fever
- New or increased cough
- New or increased shortness of breath
- Chills
- New or increased muscle pain
- New loss of taste or smell
- Sore throat
- Diarrhea
- Vomiting

### Key Secondary Efficacy Endpoint

The key secondary efficacy endpoint was confirmed COVID-19 occurrence from 7 days after the second dose in participants with and without evidence of prior SARS-CoV-2 infection.

### Safety Endpoints

Safety endpoints included:

- Solicited local and systemic adverse reactions within 7 days after each dose
- Unsolicited adverse events from Dose 1 through 1 month after the last dose
- Serious adverse events from Dose 1 through 6 months after the last dose
- Adverse events leading to withdrawal from the study

## 9.7. Statistical Methods Planned in the Protocol

The analysis populations used in this study are defined in Table 3 below.

**Table 5. Analysis Populations**

| Population | Description |
|---|---|
| Enrolled | All participants who had a signed ICD. |
| Randomized | All participants who were assigned a randomization number in the IWR system. |
| Evaluable efficacy (7 days) | All eligible randomized participants who received all vaccination(s) as randomized, with Dose 2 received within the predefined window (19-42 days after Dose 1) and had no other important protocol deviations as determined by the clinician on or before 7 days after Dose 2. |
| Dose 1 all-available efficacy | All randomized participants who received at least 1 vaccination. |
| Dose 2 all-available efficacy | All randomized participants who completed 2 vaccination doses. |
| Safety | All randomized participants who received at least 1 dose of the study intervention. |

The primary efficacy analysis was based on the evaluable efficacy (7 days) population and used an exact binomial test (Clopper-Pearson method) for the vaccine efficacy (VE) estimate against confirmed COVID-19 occurring from 7 days after Dose 2. Vaccine efficacy was defined as VE = 100 x (1 - IRR), where IRR was the ratio of the incidence rate of confirmed COVID-19 illness in the BNT162b2 group to the incidence rate of confirmed COVID-19 illness in the placebo group. A Bayesian beta-binomial model was used for interim analyses to assess vaccine efficacy. For the primary endpoint, the success criterion required a posterior probability of VE greater than 30% to exceed 98.6% at the interim analysis and 98.6% at the final analysis, accounting for multiple interim analyses.

Descriptive statistics were used to summarize safety data, including solicited local and systemic reactions, unsolicited adverse events, and serious adverse events. Two-sided 95% confidence intervals for proportions were calculated using the Clopper-Pearson exact method.

Immunogenicity analyses included SARS-CoV-2 neutralization titers and S1-binding IgG levels, summarized using geometric mean titers (GMTs), geometric mean concentrations (GMCs), and geometric mean fold rises (GMFRs) with associated 95% confidence intervals.

# 10. Study Patients

## 10.1 Disposition of Patients

The disposition of all participants randomized in the Phase 2/3 portion of the study is summarized below. A total of 37,796 participants were randomized, with 18,904 assigned to the BNT162b2 group and 18,892 to the placebo group. Of those randomized, 18,858 (99.8%) in the BNT162b2 group and 18,849 (99.8%) in the placebo group received at least one dose of the study intervention. A similar proportion of participants in each arm received a second dose, with 18,555 (98.2%) in the BNT162b2 group and 18,533 (98.1%) in the placebo group completing the two-dose vaccination schedule.

Discontinuations from the vaccination period were low and comparable between the two groups. A total of 121 (0.6%) participants in the BNT162b2 group and 111 (0.6%) in the placebo group discontinued the vaccination period but continued in the study. Overall study withdrawal was also similar between groups, with 180 (1.0%) participants in the BNT162b2 group and 259 (1.4%) in the placebo group withdrawing from the study entirely. The most common reason for study withdrawal was "Withdrawal by subject," accounting for 84 (0.4%) participants in the BNT162b2 group and 157 (0.8%) in the placebo group.

**Table 6. Disposition of All Randomized Subjects - Phase 2/3 Analysis**

| Category | BNT162b2 (30 ug) (N=18904) n (%) | Placebo (N=18892) n (%) | Total (N=37796) n (%) |
|---|---|---|---|
| Randomized | 18904 (100.0) | 18892 (100.0) | 37796 (100.0) |
| Not vaccinated | 46 (0.2) | 43 (0.2) | 89 (0.2) |
| **Vaccinated** | | | |
| Dose 1 | 18858 (99.8) | 18849 (99.8) | 37707 (99.8) |
| Dose 2 | 18555 (98.2) | 18533 (98.1) | 37088 (98.1) |
| Completed 1-month post-Dose 2 visit (vaccination period) | 16902 (89.4) | 16804 (88.9) | 33706 (89.2) |
| Discontinued from vaccination period but continue in the study | 121 (0.6) | 111 (0.6) | 232 (0.6) |
| Discontinued after Dose 1 and before Dose 2 | 121 (0.6) | 107 (0.6) | 228 (0.6) |
| Discontinued after Dose 2 and before 1-month post-Dose 2 visit | 0 | 4 (0.0) | 4 (0.0) |
| **Reason for discontinuation from vaccination period** | | | |
| No longer meets eligibility criteria | 48 (0.3) | 81 (0.4) | 129 (0.3) |

| Category | BNT162b2 (30 ug) (N=18904) n (%) | Placebo (N=18892) n (%) | Total (N=37796) n (%) |
|---|---|---|---|
| Withdrawal by subject | 45 (0.2) | 9 (0.0) | 54 (0.1) |
| Adverse event | 20 (0.1) | 12 (0.1) | 32 (0.1) |
| Pregnancy | 4 (0.0) | 4 (0.0) | 8 (0.0) |
| Physician decision | 2 (0.0) | 1 (0.0) | 3 (0.0) |
| Lost to follow-up | 0 | 2 (0.0) | 2 (0.0) |
| Medication error without associated adverse event | 0 | 1 (0.0) | 1 (0.0) |
| Other | 2 (0.0) | 1 (0.0) | 3 (0.0) |
| Withdrawn from the study | 180 (1.0) | 259 (1.4) | 439 (1.2) |
| Withdrawn after Dose 1 and before Dose 2 | 132 (0.7) | 164 (0.9) | 296 (0.8) |
| Withdrawn after Dose 2 and before 1-month post-Dose 2 visit | 44 (0.2) | 84 (0.4) | 128 (0.3) |
| Withdrawn after 1-month post-Dose 2 visit | 4 (0.0) | 11 (0.1) | 15 (0.0) |
| **Reason for withdrawal from the study** | | | |
| Withdrawal by subject | 84 (0.4) | 157 (0.8) | 241 (0.6) |
| Lost to follow-up | 80 (0.4) | 86 (0.5) | 166 (0.4) |
| Adverse event | 8 (0.0) | 5 (0.0) | 13 (0.0) |
| Death | 2 (0.0) | 3 (0.0) | 5 (0.0) |
| Physician decision | 1 (0.0) | 2 (0.0) | 3 (0.0) |
| No longer meets eligibility criteria | 1 (0.0) | 2 (0.0) | 3 (0.0) |
| Medication error without associated adverse event | 1 (0.0) | 0 | 1 (0.0) |
| Refused further study procedures | 0 | 1 (0.0) | 1 (0.0) |
| Other | 3 (0.0) | 3 (0.0) | 6 (0.0) |

Discontinuation of study intervention did not automatically constitute withdrawal from the study. Per the study estimands, if study intervention was definitively discontinued, the participant remained in the study to be evaluated for safety, immunogenicity, and efficacy. Participants were followed for the duration of the study, even if drug was discontinued, to collect all required outcome data.

## 10.2 Protocol Deviations

Important deviations from the protocol were categorized and summarized to provide a clear overview of any divergences from the planned study conduct. These deviations generally fell into categories such as:

- Participants who entered the study despite not satisfying all entry criteria.
- Participants who developed withdrawal criteria during the study but were not withdrawn according to protocol.
- Participants who received the wrong treatment or an incorrect dose of the study intervention.
- Participants who received an excluded concomitant treatment.

Detailed listings of individual patients with specific protocol deviations, grouped by centre, are provided in Appendix 16.2.2.

## 10.3 Demographic and Baseline Characteristics

The demographic characteristics of the participants in the Phase 2/3 safety population were well-balanced between the BNT162b2 and placebo groups. The study population included a diverse range of participants with respect to age, sex, race, and ethnicity, as detailed in the table below.

**Table 7. Demographic Characteristics - Phase 2/3 Safety Population**

| Characteristic | BNT162b2 (30 ug) (N=18860) n (%) | Placebo (N=18846) n (%) | Total (N=37706) n (%) |
|---|---|---|---|
| **Sex** | | | |
| Male | 9639 (51.1) | 9436 (50.1) | 19075 (50.6) |
| Female | 9221 (48.9) | 9410 (49.9) | 18631 (49.4) |
| **Race** | | | |
| White | 15636 (82.9) | 15630 (82.9) | 31266 (82.9) |
| Black or African American | 1729 (9.2) | 1763 (9.4) | 3492 (9.3) |
| American Indian or Alaska native | 102 (0.5) | 99 (0.5) | 201 (0.5) |
| Asian | 801 (4.2) | 807 (4.3) | 1608 (4.3) |
| Native Hawaiian or other Pacific Islander | 50 (0.3) | 26 (0.1) | 76 (0.2) |
| Multiracial | 449 (2.4) | 406 (2.2) | 855 (2.3) |
| Not reported | 93 (0.5) | 115 (0.6) | 208 (0.6) |
| **Ethnicity** | | | |
| Hispanic/Latino | 5266 (27.9) | 5277 (28.0) | 10543 (28.0) |
| Non-Hispanic/non-Latino | 13482 (71.5) | 13459 (71.4) | 26941 (71.5) |
| Not reported | 112 (0.6) | 110 (0.6) | 222 (0.6) |
| **Country** | | | |
| Argentina | 2883 (15.3) | 2881 (15.3) | 5764 (15.3) |
| Brazil | 1145 (6.1) | 1139 (6.0) | 2284 (6.1) |

| Characteristic | BNT162b2 (30 ug) (N=18860) n (%) | Placebo (N=18846) n (%) | Total (N=37706) n (%) |
|---|---|---|---|
| South Africa | 372 (2.0) | 372 (2.0) | 744 (2.0) |
| USA | 14460 (76.7) | 14454 (76.7) | 28914 (76.7) |
| **Age group** | | | |
| 16-55 Years | 10889 (57.7) | 10896 (57.8) | 21785 (57.8) |
| >55 Years | 7971 (42.3) | 7950 (42.2) | 15921 (42.2) |
| **Age at vaccination (years)** | | | |
| Mean (SD) | 50.5 (15.65) | 50.3 (15.72) | 50.4 (15.68) |
| Median | 52.0 | 52.0 | 52.0 |
| Min, max | (16, 89) | (16, 91) | (16, 91) |
| **Body mass index (BMI)** | | | |
| Underweight (<18.5 kg/m2) | 201 (1.1) | 235 (1.2) | 436 (1.2) |
| Normal weight (>=18.5 kg/m2 - 24.9 kg/m2) | 5517 (29.3) | 5460 (29.0) | 10977 (29.1) |
| Overweight (>=25.0 kg/m2 - 29.9 kg/m2) | 6578 (34.9) | 6481 (34.4) | 13059 (34.6) |
| Obese (>=30.0 kg/m2) | 6556 (34.8) | 6662 (35.3) | 13218 (35.1) |
| Missing | 8 (0.0) | 8 (0.0) | 16 (0.0) |

There were no clinically meaningful differences in the distribution of sex, race, ethnicity, country of residence, age, or body mass index (BMI) between the treatment groups, supporting the integrity of the randomization.

## 10.4 Medical and Surgical History (as applicable)

Detailed medical and surgical histories for individual participants were not provided in the source documents. Participants enrolled in the study were healthy individuals or those with stable chronic medical conditions. Key baseline characteristics such as Body Mass Index (BMI) are summarized in Table 2 above. The population included participants with a BMI classified as obese (>=30.0 kg/m2), representing 34.8% of the BNT162b2 group and 35.3% of the placebo group.

## 10.5 Concomitant Medications (as applicable)

Information regarding concomitant medications used by study participants was not available in the provided source documents.

## 10.6 Analysis Populations (Safety, ITT/mITT, Per-Protocol) (as applicable)

Several analysis populations were defined in the protocol to evaluate the study objectives for safety, immunogenicity, and efficacy. The main analysis populations included the Enrolled, Randomized, Safety, Evaluable Efficacy, and All-Available Efficacy sets. The Safety population, which was the primary population for safety analyses, included all randomized participants who received at least one dose of the study intervention. The Evaluable Efficacy population was the primary set for efficacy analyses and included all eligible randomized participants who received all vaccinations as randomized within the predefined window and had no other important protocol deviations.

**Table 8. Analysis Sets**

| Population | Description |
|---|---|
| Enrolled | All participants who have a signed ICD. |
| Randomized | All participants who are assigned a randomization number in the IWR system. |
| Dose 1 evaluable immunogenicity | For Phase 1 only, all eligible randomized participants who receive the vaccine to which they are randomly assigned at the first dose, have at least 1 valid and determinate immunogenicity result after Dose 1, have blood collection within an appropriate window after Dose 1, and have no other important protocol deviations as determined by the clinician. |
| Dose 2 evaluable immunogenicity | All eligible randomized participants who receive 2 doses of the vaccine to which they are randomly assigned, within the predefined window, have at least 1 valid and determinate immunogenicity result after Dose 2, have blood collection within an appropriate window after Dose 2, and have no other important protocol deviations as determined by the clinician. |
| Dose 1 all-available immunogenicity | For Phase 1 only: all randomized participants who receive at least 1 dose of the study intervention with at least 1 valid and determinate immunogenicity result after Dose 1 but before Dose 2. |
| Dose 2 all-available immunogenicity | All randomized participants who receive at least 1 dose of the study intervention with at least 1 valid and determinate immunogenicity result after Dose 2. |

| Population | Description |
| --- | --- |
| Evaluable efficacy | All eligible randomized participants who receive all vaccination(s) as randomized within the predefined window and have no other important protocol deviations as determined by the clinician. |
| All-available efficacy | 1. All randomized participants who receive at least 1 vaccination. 2. All randomized participants who complete 2 vaccination doses. |
| Safety | All randomized participants who receive at least 1 dose of the study intervention. |

# 11. Efficacy Evaluation

This section describes the efficacy analysis populations, demographic and baseline characteristics of the study participants, and the results of the primary and secondary efficacy endpoints.

## 11.1. Efficacy Analysis Populations

The efficacy analyses were conducted on several predefined populations to provide a comprehensive evaluation of the investigational product. The definitions of these populations were established in the study protocol and finalized in the Statistical Analysis Plan (SAP) prior to database lock and unblinding of individual treatment assignments. All analyses conducted on Phase 2/3 data while the study was ongoing were performed by an unblinded statistical team.

The primary efficacy analysis population was the Evaluable Efficacy (7 Days) population. Additional efficacy analyses were performed on the Dose 1 All-Available Efficacy population and the Dose 2 All-Available Efficacy population. The safety population served as the basis for demographic and baseline characteristics, closely aligning with the Dose 1 All-Available Efficacy population.

The analysis populations are defined as follows:

- **Randomized Population:** All participants who were assigned a randomization number in the Interactive Web Response (IWR) system.
- **Dose 1 All-Available Efficacy Population:** All randomized participants who received at least 1 vaccination.
- **Dose 2 All-Available Efficacy Population:** All randomized participants who completed 2 vaccination doses.
- **Evaluable Efficacy (7 Days) Population:** All eligible randomized participants who received all vaccination(s) as randomized, with Dose 2 received within the predefined window (19-42 days after Dose 1) and had no other important protocol deviations as determined by the clinician on or before 7 days after Dose 2.

The overall patient flow and the number of subjects included in each efficacy analysis population, along with the reasons for exclusion, are summarized in Table 1.

**Table 9. Efficacy Populations - Subjects 12 Through 15 Years of Age - Blinded Placebo-Controlled Follow-up (N=2264)**

| Vaccine Group (as Randomized) | BNT162b2 (30 µg) n (%) | Placebo n (%) | Total n (%) |
|---|---|---|---|
| **Randomized** | 1134 (100.0) | 1130 (100.0) | 2264 (100.0) |
| Dose 1 all-available efficacy population | 1131 (99.7) | 1129 (99.9) | 2260 (99.8) |
| Subjects without evidence of infection before Dose 1 | 1083 (95.5) | 1078 (95.4) | 2161 (95.5) |
| Subjects excluded from Dose 1 all-available efficacy population | 3 (0.3) | 1 (0.1) | 4 (0.2) |
| **Reason for exclusion** | | | |
| Did not receive at least 1 vaccination | 3 (0.3) | 1 (0.1) | 4 (0.2) |
| Dose 2 all-available efficacy population | 1123 (99.0) | 1117 (98.8) | 2240 (98.9) |
| Subjects without evidence of infection prior to 7 days after Dose 2 | 1061 (93.6) | 1037 (91.8) | 2098 (92.7) |
| Subjects excluded from Dose 2 all-available efficacy population | 11 (1.0) | 13 (1.2) | 24 (1.1) |
| **Reason for exclusion** | | | |
| Did not receive 2 vaccinations | 10 (0.9) | 13 (1.2) | 23 (1.0) |
| Unblinded prior to 7 days after Dose 2 | 1 (0.1) | 0 | 1 (0.0) |
| Evaluable efficacy (7 days) population | 1119 (98.7) | 1109 (98.1) | 2228 (98.4) |
| Subjects without evidence of infection prior to 7 days after Dose 2 | 1057 (93.2) | 1030 (91.2) | 2087 (92.2) |
| Subjects excluded from evaluable efficacy (7 days) population | 15 (1.3) | 21 (1.9) | 36 (1.6) |
| **Reason for exclusion** | | | |
| Randomized but did not meet all eligibility criteria | 1 (0.1) | 1 (0.1) | 2 (0.1) |
| Did not receive all vaccinations as randomized or did not receive Dose 2 within the predefined window (19-42 days after Dose 1) | 14 (1.2) | 19 (1.7) | 33 (1.5) |
| Unblinded prior to 7 days after Dose 2 | 1 (0.1) | 0 | 1 (0.0) |

| Vaccine Group (as Randomized) | BNT162b2 (30 µg) n (%) | Placebo n (%) | Total n (%) |
|---|---|---|---|
| Had other important protocol deviations on or prior to 7 days after Dose 2 | 0 | 3 (0.3) | 3 (0.1) |

## 11.2. Demographic and Baseline Characteristics

The demographic and baseline characteristics for the Safety Population (N=2260, which includes all randomized subjects who received at least one dose of the study intervention, consistent with the Dose 1 all-available efficacy population) are presented in Table 2. These data are presented by treatment group to assess comparability.

**Table 10. Demographic Characteristics - Phase 2/3 Subjects 12 Through 15 Years of Age - Safety Population (N=2260)**

| Characteristic | BNT162b2 (30 µg) (Na=1131) n (%) | Placebo (Na=1129) n (%) | Total (Na=2260) n (%) |
|---|---|---|---|
| **Sex** | | | |
| Male | 567 (50.1) | 585 (51.8) | 1152 (51.0) |
| Female | 564 (49.9) | 544 (48.2) | 1108 (49.0) |
| **Race** | | | |
| White | 970 (85.8) | 962 (85.2) | 1932 (85.5) |
| Black or African American | 52 (4.6) | 57 (5.0) | 109 (4.8) |
| All others | 109 (9.6) | 110 (9.7) | 219 (9.7) |
| American Indian or Alaska Native | 4 (0.4) | 3 (0.3) | 7 (0.3) |
| Asian | 72 (6.4) | 71 (6.3) | 143 (6.3) |
| Native Hawaiian or other Pacific Islander | 3 (0.3) | 0 | 3 (0.1) |
| Multiracial | 24 (2.1) | 29 (2.6) | 53 (2.3) |
| Not reported | 6 (0.5) | 7 (0.6) | 13 (0.6) |
| **Racial designation** | | | |
| Japanese | 5 (0.4) | 2 (0.2) | 7 (0.3) |
| **Ethnicity** | | | |
| Hispanic/Latino | 132 (11.7) | 130 (11.5) | 262 (11.6) |
| Non-Hispanic/non-Latino | 997 (88.2) | 996 (88.2) | 1993 (88.2) |
| Not reported | 2 (0.2) | 3 (0.3) | 5 (0.2) |
| **Country** | | | |
| USA | 1131 (100.0) | 1129 (100.0) | 2260 (100.0) |
| **Baseline SARS-CoV-2 status** | | | |
| Positive | 46 (4.1) | 50 (4.4) | 96 (4.2) |
| Negative | 1083 (95.8) | 1078 (95.5) | 2161 (95.6) |
| Missing | 2 (0.2) | 1 (0.1) | 3 (0.1) |
| **Comorbidities** | | | |
| Yes | 249 (22.0) | 242 (21.4) | 491 (21.7) |
| No | 882 (78.0) | 887 (78.6) | 1769 (78.3) |
| **Obese** | | | |
| Yes | 143 (12.6) | 128 (11.3) | 271 (12.0) |
| No | 988 (87.4) | 1001 (88.7) | 1989 (88.0) |
| **Age at vaccination (years)** | | | |
| Mean (SD) | 13.6 (1.11) | 13.6 (1.11) | 13.6 (1.11) |
| Median | 14.0 | 14.0 | 14.0 |
| Min, max | (12, 15) | (12, 15) | (12, 15) |

The treatment groups were generally well-balanced with respect to demographic charac-

teristics (sex, race, ethnicity, and age) and baseline SARS-CoV-2 status and presence of comorbidities (including obesity). All enrolled subjects were from the USA.

## 11.3. Patient Flow and Relationship Between Analysis Groups

A diagram illustrating the relationship between the entire randomized sample and the various analysis groups is shown in Figure 1. This diagram depicts the flow of patients from randomization through the formation of the different efficacy analysis sets, including reasons for exclusion at each stage.

The patient flow data, including the number of subjects in each analysis population and the reasons for exclusion at each stage, are detailed in Table 1 above.

## 11.4. Compliance and Drug Concentration

## 11.4. Compliance with Treatment Regimen and Drug Concentrations

Compliance with the treatment regimen was monitored by recording the date and time of each dose administered at the study site in source documents and Case Report Forms (CRFs). The administration of the study intervention was directly supervised by the investigator or designee, and the dose and participant identification were confirmed by a second staff member at the time of dosing. This ensured accurate documentation of adherence to the vaccination schedule.

Pharmacokinetic parameters and drug concentrations in body fluids were not evaluated in this study. Therefore, no data on drug concentrations are available for analysis. Formal pharmacodynamic parameters were also not evaluated in this study.

## 11.5. Efficacy Results

The efficacy of the investigational product was evaluated based on pre-specified primary and secondary endpoints. The primary efficacy endpoint was the prevention of the first occurrence of COVID-19 from 7 days after Dose 2 in participants without evidence of SARS-CoV-2 infection before vaccination. Secondary endpoints included the prevention of the first occurrence of COVID-19 after Dose 1. Results are presented for the Evaluable Efficacy (7 Days) population and the Dose 1 All-Available Efficacy population, as appropriate.

### 11.5.1. Primary Efficacy Endpoint: First COVID-19 Occurrence From 7 Days After Dose 2

The primary efficacy endpoint was evaluated in the Evaluable Efficacy (7 Days) population, consisting of participants without evidence of prior SARS-CoV-2 infection before vaccination. The results demonstrate high vaccine efficacy against confirmed COVID-19.

**Table 11. Vaccine Efficacy - First COVID-19 Occurrence From 7 Days After Dose 2 - Evaluable Efficacy (7 Days) Population (Na=2087)**

| Vaccine Group (as Randomized) | BNT162b2 (30 µg) (Na=1057) n1b (Surveillance Time (n2)) | Placebo (Na=1030) n1b (Surveillance Time (n2)) | VE (%) (95% CI) |
|---|---|---|---|
| Overall | 0 0.343 (1043) | 28 0.322 (1019) | 100.0 (86.8, 100.0) |
| ≥7 days after Dose 2 to <2 Months after Dose 2 | 0 0.138 (1043) | 15 0.133 (1019) | 100.0 (73.2, 100.0) |
| ≥2 Months after Dose 2 to <4 Months after Dose 2 | 0 0.155 (1061) | 10 0.148 (1022) | 100.0 (57.4, 100.0) |
| ≥4 Months after Dose 2 | 0 0.061 (767) | 3 0.055 (726) | 100.0 (-117.8, 100.0) |

Abbreviations: n1b = number of participants with events; n2 = total surveillance time in 1000 person-years; VE = vaccine efficacy; CI = confidence interval. Note: Surveillance time for each participant is the time from the relevant efficacy onset window to the end of the surveillance period.

The overall vaccine efficacy (VE) from 7 days after Dose 2 was 100.0% (95% CI: 86.8, 100.0), based on 0 COVID-19 cases in the BNT162b2 group and 28 cases in the placebo group among participants without prior SARS-CoV-2 infection. This high efficacy was maintained across different surveillance time intervals up to 4 months post-Dose 2, although the confidence intervals widened with longer follow-up due to fewer events.

### 11.5.2. Vaccine Efficacy by Subgroup

Subgroup analyses for the first COVID-19 occurrence from 7 days after Dose 2 were performed for the Evaluable Efficacy (7 Days) population (Na=2087) based on demographic characteristics and comorbidities.

**Table 12. Vaccine Efficacy - First COVID-19 Occurrence From 7 Days After Dose 2, by Subgroup - Evaluable Efficacy (7 Days) Population (Na=2087)**

| Efficacy Endpoint Subgroup | BNT162b2 (30 µg) (Na=1057) n1b (Surveillance Time (n2)) | Placebo (Na=1030) n1b (Surveillance Time (n2)) | VE (%) (95% CI) |
|---|---|---|---|
| **Overall** | 0 0.343 (1043) | 28 0.322 (1019) | 100.0 (86.8, 100.0) |
| **Sex** | | | |
| Male | 0 0.175 (524) | 16 0.165 (526) | 100.0 (75.5, 100.0) |
| Female | 0 0.169 (519) | 12 0.157 (493) | 100.0 (66.5, 100.0) |
| **Race** | | | |
| White | 0 0.293 (898) | 26 0.272 (867) | 100.0 (85.8, 100.0) |
| Black or African American | 0 0.017 (41) | 2 0.018 (49) | 100.0 (-470.9, 100.0) |
| **Ethnicity** | | | |
| Hispanic/Latino | 0 0.042 (119) | 7 0.036 (113) | 100.0 (41.2, 100.0) |
| Non-Hispanic/non-Latino | 0 0.300 (922) | 21 0.285 (903) | 100.0 (81.7, 100.0) |
| **Country** | | | |
| USA | 0 0.343 (1043) | 28 0.322 (1019) | 100.0 (86.8, 100.0) |
| **Comorbidities** | | | |
| Yes | 0 0.078 (230) | 9 0.068 (213) | 100.0 (55.5, 100.0) |
| No | 0 0.266 (813) | 19 0.254 (806) | 100.0 (79.5, 100.0) |
| **Obese** | | | |
| Yes | 0 0.046 (134) | 6 0.036 (110) | 100.0 (33.9, 100.0) |
| No | 0 0.298 (909) | 22 0.287 (909) | 100.0 (82.4, 100.0) |

Abbreviations: n1b = number of participants with events; n2 = total surveillance time in 1000 person-years; VE = vaccine efficacy; CI = confidence interval. Note: Surveillance time for each participant is the time from the relevant efficacy onset window to the end of the surveillance period.

Subgroup analyses generally showed consistent vaccine efficacy across different categories of sex, race (except for Black or African American subgroup with a wide CI due to small numbers), ethnicity, country, presence of comorbidities, and obesity status. The point estimate for vaccine efficacy was consistently 100% across most subgroups, though confidence

intervals varied, primarily reflecting the smaller number of events in some subgroups. The wide confidence interval for the Black or African American subgroup is attributable to the very low number of events (0 in vaccine group, 2 in placebo group).

### 11.5.3. Secondary Efficacy Endpoint: First COVID-19 Occurrence After Dose 1

### 11.5.3. Secondary Efficacy Endpoint: First COVID-19 Occurrence After Dose 1

Efficacy was also evaluated for the first occurrence of confirmed COVID-19 at any time after Dose 1 in the Dose 1 all-available efficacy (modified intention-to-treat) population, regardless of baseline SARS-CoV-2 infection status.

Among participants in this population, 50 cases of COVID-19 occurred after Dose 1 in the BNT162b2 group (n=21,314 subjects) compared to 275 cases in the placebo group (n=21,258 subjects). The estimated vaccine efficacy against confirmed COVID-19 occurring after Dose 1 was 82% (2-sided 95% CI: 75.6%, 86.9%). A substantial proportion of cases in the BNT162b2 group (36 out of 50) occurred before Dose 2. The estimated vaccine efficacy against confirmed COVID-19 occurring after Dose 1 but before Dose 2 was 52.4% (2-sided 95% CI: 29.5%, 68.4%).

The early onset of protection is further illustrated by the cumulative incidence curves, which show that the rates of COVID-19 occurrence were similar for both BNT162b2 and placebo groups until approximately 14 days after Dose 1. After this point, the curves diverged, with a continued accumulation of cases in the placebo group, while the incidence in the BNT162b2 group remained relatively stable.

## 11.6. Multicentre Study Considerations

This was a multicenter study conducted solely within the USA. To assess comparability across study sites, and to identify any potential treatment-by-center interactions, analysis of critical variables by individual centers, particularly larger sites, is necessary.

### 11.6.1. Comparability Across Centers

The study was conducted across multiple sites in the USA, Argentina, Brazil, Germany, South Africa, and Turkey. Demographic and baseline characteristics were observed to be balanced between treatment groups at the overall study level, indicating comparability of patient populations across the aggregated sites.

Subgroup analyses of vaccine efficacy by country were performed. The observed vaccine efficacy (VE) for the first primary efficacy endpoint was consistently high across most countries/regions, with estimates generally greater than 93%. An exception was Brazil, which showed a VE of 87.7%. However, detailed data displays and analyses of individual centers for critical variables are not presented in this report. This aggregated assessment confirms general consistency without significant qualitative or quantitative treatment-by-center interactions that would invalidate the overall findings.

### 11.6.2. Multiple Party Assessments

Efficacy assessments in this study were primarily based on objective criteria, specifically laboratory-confirmed SARS-CoV-2 infection combined with predefined COVID-19 symptoms. These assessments were performed using standardized methods and laboratory procedures. There were no critical measurements or assessments of efficacy outcomes made by more than one party where overall differences between ratings needed to be shown or individual patients with disparate assessments identified.

## 11.7. Statistical Methods and Analysis

The statistical analysis for this study was pre-specified in the Statistical Analysis Plan (SAP), which was finalized prior to database lock.

### 11.7.1. General Statistical Methods

The primary efficacy analyses used a Bayesian approach to estimate vaccine efficacy (VE), defined as $100\% \times (1 - \text{IRR})$, where the illness rate ratio (IRR) is the ratio of first confirmed COVID-19 illness rates in the vaccine group to the corresponding rate in the placebo group. A minimally informative beta prior, beta (0.700102, 1), was proposed for $\theta = (1-\text{VE})/(2-\text{VE})$, centered at $\theta = 0.4118$ (VE = 30%), which was considered pessimistic.

Vaccine efficacy was demonstrated if there was convincing evidence (posterior probability >99.5% at an interim analysis or >98.6% at the final analysis) that the true VE of BNT162b2 was >30%. Confidence intervals for VE were derived based on the Clopper and Pearson method, adjusted for surveillance time. For subgroup analyses of the primary efficacy endpoint, 2-sided 95% confidence intervals were calculated. Confidence intervals for efficacy endpoint subgroup analyses were not adjusted for multiplicity, which was considered acceptable in this context.

### 11.7.2. Handling of Dropouts and Missing Data

The analysis populations were precisely defined to account for patient disposition. The primary efficacy analysis utilized the Evaluable Efficacy population, which included eligible randomized participants who received all vaccinations as randomized within the predefined window and had no other important protocol deviations.

To assess the robustness of the primary efficacy findings, sensitivity analyses were performed using all laboratory-confirmed cases with symptom onset at any time after the first dose (Dose 1 all-available efficacy population) and 7 days after the second dose (Dose 2 all-available efficacy population), without excluding participants with major protocol deviations.

Sensitivity analysis of missing laboratory data was conducted for the primary endpoint. This involved 500 imputations generated using the SAS PROC MI Fully Conditional Specification (FCS) method. Each imputation filled in missing laboratory results based on a logistic regression model at the subject level, assuming a higher than observed case rate when imputing missing efficacy endpoints from participants in the BNT162b2 group only. This approach aimed to reflect potential missing not at random (MNAR) effects unfavorable to efficacy results. The results of this sensitivity analysis confirmed that the slight imbalances in missing data did not introduce significant bias.

### 11.7.3. Interim Analyses

During Phase 2/3, four interim analyses (IAs) were initially planned to be performed by an unblinded statistical team after accrual of at least 32, 62, 92, and 120 evaluable COVID-19 cases. For operational reasons, the first planned IA (at 32 cases) was not performed, resulting in three planned IAs. Futility and VE with respect to the first primary endpoint were assessed at these IAs. Overwhelming efficacy was to be declared if the primary objective was met, with success thresholds calibrated to protect the overall Type I error at 2.5%. Futility was

assessed if the predicted probability of success at the final analysis was <5%. Efficacy and futility boundaries were applied in a nonbinding way.

Overwhelming efficacy success criteria for the first primary efficacy endpoint were met at the first conducted interim analysis (corresponding to the planned 62-case interim analysis, but the exact case count at cut-off was 94). Consequently, further formal interim analyses were not conducted. All analyses conducted on Phase 2/3 data while the study was ongoing were performed by an unblinded statistical team. No changes to the core statistical analysis plan were made after the study was unblinded. The final analysis of all protocol-specified primary and secondary efficacy endpoints was conducted after accrual of at least 164 cases.

## 11.8. Dose-Response Relationship

The Phase 2/3 portion of this study was designed to evaluate the efficacy and safety of a selected vaccine candidate (BNT162b2) at a single dose level (30 µg), based on findings from earlier Phase 1 dose-finding studies. Therefore, this pivotal efficacy study was not specifically designed to formally assess the dose-response relationship of the investigational product. The earlier Phase 1 studies indicated that immune responses were numerically higher in higher dose groups but did not substantially differ between the 10 µg and 30 µg doses. Pharmacokinetic parameters and drug concentrations were not evaluated in this study.

## 11.9. Relationship Between Response and Concomitant Therapy/Illness

The efficacy analysis included subgroups based on the presence of comorbidities and obesity. These analyses demonstrated consistent vaccine efficacy across participants with or without comorbidities and those who were obese or non-obese, suggesting that these concomitant illnesses did not substantially alter the protective effect of the vaccine. For example, VE for participants at risk of severe COVID-19 due to comorbidities was 95.3%, comparable to 94.7% for those not at risk. Similarly, VE was 95.4% in obese participants and 94.8% in non-obese participants. Detailed analysis of the impact of specific concomitant therapies on efficacy was not performed as part of this report.

## 11.10. Individual Patient Data

Individual response data and other relevant study information, including patient identifier, all measured or observed values of critical measurements (including baseline), time during study, drug/dose at time, compliance measurements, and concomitant medications, are provided in Appendix 16.2.6: Individual Efficacy Response Data. Overall responder evaluations, if applicable, are also included in this appendix. Critical baseline information such as age, sex, weight, disease being treated (if more than one in study), and disease stage or severity are included within these listings. The baseline values for critical measurements are presented as zero time values for each efficacy measurement. Since the dose was fixed at 30 µg, individual patient doses are documented as administered in the CRFs.

## 11.11. Important Conclusions Concerning Efficacy

The study demonstrated excellent vaccine efficacy for BNT162b2 against confirmed COVID-19. The primary efficacy endpoint showed 100.0% (95% CI: 86.8%, 100.0%) vaccine efficacy against the first COVID-19 occurrence from 7 days after Dose 2 in participants without

evidence of prior SARS-CoV-2 infection. This high efficacy was maintained across various demographic and comorbidity subgroups. Furthermore, the vaccine provided significant protection after the first dose, with an estimated VE of 82% (95% CI: 75.6%, 86.9%) against confirmed COVID-19 occurring after Dose 1. The early onset of protection was observed, with cumulative incidence curves diverging approximately 14 days after Dose 1. The robustness of these findings was supported by sensitivity analyses. While data on severe COVID-19 cases were limited due to their rarity, the overall high efficacy across diverse populations is highly encouraging. Long-term efficacy data will be collected as the study continues.

# 12. Safety Evaluation

## 12.1 Extent of Exposure

The safety evaluation is based on all subjects who received at least one dose of the investigational product or placebo, defined as the Safety Population. The extent of exposure to the investigational product (BNT162b2) and placebo was characterized by the number of subjects exposed, the duration of exposure, and the dose administered.

**Table 13. Number of Subjects in the Safety Population**

| | Vaccine Group (as Randomized) | |
|---|---|---|
| | **BNT162b2 (30 µg) (N=1134) n (%)** | **Placebo (N=1130)** |
| Randomized | 1134 (100.0) | 1130 (100.0) |
| Not vaccinated | 3 (0.3) | 1 (0.1) |
| Dose 1 | 1131 (99.7) | 1129 (99.9) |
| Dose 2 | 1128 (99.5) | 1119 (99.0) |
| **Protocol defined window for Dose 2** | | |
| <19 Days | 2 (0.2) | 1 (0.1) |
| 19-23 Days | 1073 (94.6) | 1065 (94.2) |
| >23 Days | 53 (4.7) | 53 (4.7) |
| **Weekly Intervals for Dose 2** | | |
| <14 Days | 0 | 0 |
| 14-20 Days | 358 (31.6) | 364 (32.2) |
| 21-27 Days | 737 (65.0) | 729 (64.5) |
| 28-34 Days | 23 (2.0) | 15 (1.3) |
| 35-41 Days | 4 (0.4) | 4 (0.4) |
| 42-48 Days | 1 (0.1) | 1 (0.1) |
| 49-55 Days | 1 (0.1) | 3 (0.3) |
| >55 Days | 4 (0.4) | 3 (0.3) |
| **Dose 3 (first dose of BNT162b2 [30 µg])** | | 1010 (89.4) |
| **Dose 4 (second dose of BNT162b2 [30 µg])** | | 992 (87.8) |
| **Protocol defined window for Dose 4** | | |
| <19 Days | | 6 (0.5) |
| 19-23 Days | | 905 (80.1) |
| >23 Days | | 81 (7.2) |
| **Weekly Intervals for Dose 4** | | |
| <14 Days | | 0 |
| 14-20 Days | | 264 (23.4) |
| 21-27 Days | | 691 (61.2) |
| 28-34 Days | | 26 (2.3) |
| 35-41 Days | | 5 (0.4) |
| 42-48 Days | | 3 (0.3) |
| 49-55 Days | | 2 (0.2) |
| >55 Days | | 1 (0.1) |

The study enrolled 1134 subjects randomized to the BNT162b2 30 µg group and 1130 subjects randomized to the placebo group. Of these, 1131 (99.7%) in the BNT162b2 group and

1129 (99.9%) in the placebo group received at least one dose. A second dose was administered to 1128 (99.5%) of BNT162b2 recipients and 1119 (99.0%) of placebo recipients. For Dose 2, the majority of subjects in both groups received it within the 19-23 day window (94.6% for BNT162b2, 94.2% for placebo). A subset of subjects in the original placebo group received open-label BNT162b2 as Dose 3 (1010 subjects) and Dose 4 (992 subjects). The dosing for Dose 3 and Dose 4 followed a similar timing pattern, with most subjects receiving the dose within the 19-23 day window.

**Table 14. Follow-up Time During Open-Label Period (After Dose 1 of BNT162b2) in Initial Placebo Group (N=1010)**

| Open-label follow-up period | BNT162b2 (30 μg) (N=1010) n (%) |
|---|---|
| <2 Months | 66 (6.5) |
| ≥2-<4 Months | 656 (65.0) |
| ≥4-<6 Months | 228 (22.6) |
| ≥6 Months | 60 (5.9) |
| **Mean (SD)** | **3.8 (1.09)** |
| **Median** | **3.8** |
| **Min, max** | **(0.1, 8.6)** |

**Table 15. Demographic Characteristics - Safety Population (Phase 2/3 Subjects 12 Through 15 Years of Age)**

| | Vaccine Group (as Administered) | | |
|---|---|---|---|
| | **BNT162b2 (30 µg) (N=1131) n (%)** | **Placebo (N=1129) n (%)** | **Total (N=2260) n (%)** |
| **Sex** | | | |
| Male | 567 (50.1) | 585 (51.8) | 1152 (51.0) |
| Female | 564 (49.9) | 544 (48.2) | 1108 (49.0) |
| **Race** | | | |
| White | 970 (85.8) | 962 (85.2) | 1932 (85.5) |
| Black or African American | 52 (4.6) | 57 (5.0) | 109 (4.8) |
| All others | 109 (9.6) | 110 (9.7) | 219 (9.7) |
| American Indian or Alaska Native | 4 (0.4) | 3 (0.3) | 7 (0.3) |
| Asian | 72 (6.4) | 71 (6.3) | 143 (6.3) |
| Native Hawaiian or other Pacific Islander | 3 (0.3) | 0 | 3 (0.1) |
| Multiracial | 24 (2.1) | 29 (2.6) | 53 (2.3) |
| Not reported | 6 (0.5) | 7 (0.6) | 13 (0.6) |
| **Racial designation** | | | |
| Japanese | 5 (0.4) | 2 (0.2) | 7 (0.3) |
| **Ethnicity** | | | |
| Hispanic/Latino | 132 (11.7) | 130 (11.5) | 262 (11.6) |
| Non-Hispanic/non-Latino | 997 (88.2) | 996 (88.2) | 1993 (88.2) |
| Not reported | 2 (0.2) | 3 (0.3) | 5 (0.2) |
| **Country** | | | |
| USA | 1131 (100.0) | 1129 (100.0) | 2260 (100.0) |
| **Baseline SARS-CoV-2 status** | | | |
| Positive | 46 (4.1) | 50 (4.4) | 96 (4.2) |
| Negative | 1083 (95.8) | 1078 (95.5) | 2161 (95.6) |
| Missing | 2 (0.2) | 1 (0.1) | 3 (0.1) |
| **Comorbidities** | | | |
| Yes | 249 (22.0) | 242 (21.4) | 491 (21.7) |
| No | 882 (78.0) | 887 (78.6) | 1769 (78.3) |
| **Obese** | | | |
| Yes | 143 (12.6) | 128 (11.3) | 271 (12.0) |
| No | 988 (87.4) | 1001 (88.7) | 1989 (88.0) |
| **Age at vaccination (years)** | | | |
| **Mean (SD)** | **13.6 (1.11)** | **13.6 (1.11)** | **13.6 (1.11)** |
| **Median** | **14.0** | **14.0** | |
| **Min, max** | **(12, 15)** | **(12, 15)** | **(12, 15)** |

The duration of follow-up for the BNT162b2 30 µg group in the blinded placebo-controlled phase is not explicitly provided as a single mean/median duration across all subjects in the retrieved tables. However, the open-label follow-up period for subjects initially in the placebo group who subsequently received BNT162b2 showed a mean follow-up of 3.8 months (SD 1.09), with a median of 3.8 months, ranging from 0.1 to 8.6 months. The number of subjects exposed to specified durations during this open-label period is presented in Table 2. Given that the investigational product is a vaccine, the dose is a fixed 30 µg per administration, and dosage on a mg/kg or mg/m² basis is generally not applied. Cumulative dose is not considered pertinent for this vaccine study.

The breakdown of the safety population by age, sex, and racial subgroups is presented in Table 3. The groups are well-balanced across vaccine and placebo arms for all demographic characteristics, including age (mean 13.6 years in both groups), sex (approximately 50% male/female), and race (predominantly White, 85.8% and 85.2% in BNT162b2 and placebo groups, respectively). Similar distributions were observed for ethnicity, baseline SARS-CoV-2 status, comorbidities, and obesity.

## 12.2 Adverse Events

### 12.2.1 Overall Adverse Event Experience

The overall adverse event experience in the study, from Dose 1 to unblinding date, is summarized in Table 4.

**Table 16. Incidence Rates of At Least 1 Adverse Event From Dose 1 to Unblinding Date - Blinded Placebo-Controlled Follow-up Period - Safety Population**

| | Vaccine Group (as Administered) | |
|---|---|---|
| **Adverse Event** | **BNT162b2 (30 µg) (N=1131, TE=4.6) n (%) IR (95% CI)** | **Placebo (N=1129, TE=4.5) n (%) IR (95% CI)** |
| Any event | 95 (8.4) 20.8 (16.8, 25.4) | 113 (10.0) 25.1 (20.7, 30.1) |
| Related | 36 (3.2) 7.9 (5.5, 10.9) | 24 (2.1) 5.3 (3.4, 7.9) |
| Severe | 13 (1.1) 2.8 (1.5, 4.9) | 5 (0.4) 1.1 (0.4, 2.6) |
| Life-threatening | 2 (0.2) 0.4 (0.1, 1.6) | 1 (0.1) 0.2 (0.0, 1.2) |
| Any serious adverse event | 10 (0.9) 2.2 (1.0, 4.0) | 2 (0.2) 0.4 (0.1, 1.6) |
| Related | 0 0.0 (0.0, 0.8) | 0 0.0 (0.0, 0.8) |
| Severe | 7 (0.6) 1.5 (0.6, 3.2) | 1 (0.1) 0.2 (0.0, 1.2) |
| Life-threatening | 1 (0.1) 0.2 (0.0, 1.2) | 1 (0.1) 0.2 (0.0, 1.2) |

TE = Total Exposure (1000 subject-years)

Overall, 95 subjects (8.4%) in the BNT162b2 group and 113 subjects (10.0%) in the placebo group reported at least one adverse event from Dose 1 to the unblinding date. The incidence rate per 1000 subject-years was 20.8 (95% CI: 16.8, 25.4) for the BNT162b2 group and 25.1 (95% CI: 20.7, 30.1) for the placebo group. Related adverse events were reported by 3.2% of BNT162b2 recipients and 2.1% of placebo recipients. Severe adverse events were observed in 1.1% of BNT162b2 subjects and 0.4% of placebo subjects. Life-threatening

events were rare, occurring in 0.2% of BNT162b2 subjects and 0.1% of placebo subjects. Serious adverse events were reported in 0.9% of the BNT162b2 group compared to 0.2% in the placebo group. No serious adverse events were considered related to the investigational product in either group.

### 12.2.2 Adverse Events by System Organ Class and Preferred Term

**Table 17. Incidence Rates of At Least 1 Adverse Event From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term - Blinded Placebo-Controlled Follow-up Period - Safety Population**

| System Organ Class Preferred Term | BNT162b2 (30 µg) (N=1131, TE=4.6) n (%) IR (95% CI) | Placebo (N=1129, TE=4.5) n (%) IR (95% CI) |
|---|---|---|
| Any event | 95 (8.4) 20.8 (16.8, 25.4) | 113 (10.0) 25.1 (20.7, 30.1) |
| **BLOOD AND LYMPHATIC SYSTEM DISORDERS** | **9 (0.8) 2.0 (0.9, 3.7)** | **2 (0.2) 0.4 (0.1, 1.6)** |
| Lymphadenopathy | 9 (0.8) 2.0 (0.9, 3.7) | 2 (0.2) 0.4 (0.1, 1.6) |
| **CONGENITAL, FAMILIAL AND GENETIC DISORDERS** | **0 0.0 (0.0, 0.8)** | **1 (0.1) 0.2 (0.0, 1.2)** |
| Spine malformation | 0 0.0 (0.0, 0.8) | 1 (0.1) 0.2 (0.0, 1.2) |
| **EAR AND LABYRINTH DISORDERS** | **1 (0.1) 0.2 (0.0, 1.2)** | **3 (0.3) 0.7 (0.1, 1.9)** |
| Cerumen impaction | 0 0.0 (0.0, 0.8) | 1 (0.1) 0.2 (0.0, 1.2) |
| Conductive deafness | 0 0.0 (0.0, 0.8) | 1 (0.1) 0.2 (0.0, 1.2) |
| Ear pain | 1 (0.1) 0.2 (0.0, 1.2) | 1 (0.1) 0.2 (0.0, 1.2) |
| **EYE DISORDERS** | **2 (0.2) 0.4 (0.1, 1.6)** | **1 (0.1) 0.2 (0.0, 1.2)** |
| Eye pain | 1 (0.1) 0.2 (0.0, 1.6) | 0 0.0 (0.0, 0.8) |
| Eyelid rash | 1 (0.1) 0.2 (0.0, 1.6) | 0 0.0 (0.0, 0.8) |
| Retinal haemorrhage | 0 0.0 (0.0, 0.8) | 1 (0.1) 0.2 (0.0, 1.2) |
| **GASTROINTESTINAL DISORDERS** | **14 (1.2) 3.1 (1.7, 5.1)** | **8 (0.7) 1.8 (0.8, 3.5)** |
| Abdominal pain | 2 (0.2) 0.4 (0.1, 1.6) | 1 (0.1) 0.2 (0.0, 1.2) |
| Aphthous ulcer | 1 (0.1) 0.2 (0.0, 1.2) | 0 0.0 (0.0, 0.8) |
| Constipation | 1 (0.1) 0.2 (0.0, 1.2) | 0 0.0 (0.0, 0.8) |
| Diarrhoea | 3 (0.3) 0.7 (0.1, 1.9) | 1 (0.1) 0.2 (0.0, 1.2) |
| Gastritis | 1 (0.1) 0.2 (0.0, 1.2) | 0 0.0 (0.0, 0.8) |
| Lip swelling | 1 (0.1) 0.2 (0.0, 1.2) | 0 0.0 (0.0, 0.8) |
| Diarrhea | 0 0.0 (0.0, 0.8) | 1 (0.1) 0.2 (0.0, 1.2) |

TE = Total Exposure (1000 subject-years)

The adverse events are presented by System Organ Class and Preferred Term, along with the number and percentage of subjects reporting the event, and the incidence rate per 1000 subject-years. This table, though truncated in the provided output, indicates that "Gastrointestinal Disorders" (1.2% in BNT162b2) and "Blood and Lymphatic System Disorders"

(0.8% in BNT162b2, specifically "Lymphadenopathy") were among the more frequently reported System Organ Classes. Within "Gastrointestinal Disorders", no single preferred term exceeded 1%. Lymphadenopathy was the only event listed under "Blood and Lymphatic System Disorders".

The most common adverse events (AEs) reported in at least 1% of participants in the BNT162b2 group, from Dose 1 to 1 month after Dose 2 (among 37,586 participants with a median of 2 months of safety follow-up after Dose 2), are presented in Table 6.

**Table 18. Common Adverse Events (≥1% Incidence) from Dose 1 to 1 Month After Dose 2 - Blinded Placebo-Controlled Follow-up Period - Safety Population**

| System Organ Class | Preferred Term | BNT162b2 (N=?) n (%) | Placebo (N=?) n (%) |
|---|---|---|---|
| **Gastrointestinal disorders** | Nausea | 1.1% | 0.3% |
| **General disorders and administration site conditions** | Injection site pain | 11.2% | 1.5% |
| | Pyrexia | 6.1% | 0.3% |
| | Fatigue | 5.5% | 1.4% |
| | Chills | 5.3% | 0.5% |
| | Pain | 2.4% | 0.2% |
| **Musculoskeletal and connective tissue disorders** | Myalgia | 4.8% | 0.7% |
| | Arthralgia | 1.1% | 0.4% |
| **Nervous system disorders** | Headache | 5.1% | 1.6% |

These common adverse events primarily represent reactogenicity events, which are considered treatment-emergent signs and symptoms (TESS). They were generally more frequent in the BNT162b2 group compared to the placebo group.

**Severity and Causality:** Most of the reported common AEs were mild to moderate in intensity. Severe AEs were low in frequency, reported in 1.2% of BNT162b2 recipients and 0.6% of placebo recipients from Dose 1 to 1 month after Dose 2. The majority of these severe AEs in the BNT162b2 group were reactogenicity events, including pyrexia, fatigue, injection site pain, chills, myalgia, arthralgia, and headache. After Dose 2, severe systemic events had frequencies less than 2%, with the exception of fatigue (3.8%) and headache (2.0%). Severe fever (>38.9°C to 40.0°C) occurred in 0.2% of BNT162b2 recipients after Dose 1 and 0.8% after Dose 2. Grade 4 fever (>40.0°C) was reported for two participants in both the BNT162b2 and placebo groups.

Adverse events assessed by the investigator as related to the investigational product were reported by 20.8% of participants in the BNT162b2 group and 5.1% in the placebo group. Most related AEs were reactogenicity events within the System Organ Class of general disorders and administration site conditions. Lymphadenopathy, while not a common AE (0.3% in BNT162b2 group), was frequently judged by the investigator as related to the study inter-

vention (47 out of 64 cases) and occurred predominantly in the arm and neck region within 2 to 4 days post-vaccination.

**Factors Affecting Frequency of Adverse Reactions:** - **Dose/Drug Concentration:** The study utilized a fixed 30 μg dose, therefore, dose-response relationships within the study cannot be assessed. Drug concentration data in relation to adverse events were not available. - **Duration of Treatment/Time Dependence:** Most reactogenicity events, including local reactions and systemic events, occurred shortly after vaccination, typically with a median onset between Day 1 and Day 3 (Day 1 being the day of vaccination) and generally resolved with a median duration of 1 to 2 days. The majority of these events were reported during the 7-day e-diary reporting period after each dose. - **Demographic Characteristics (Age):** Reactogenicity events were generally increased in frequency and severity in younger age groups (e.g., 12-15 years) compared with older age groups. For example, incidences of AEs in general disorders and administration site conditions, musculoskeletal and connective tissue disorders, and nervous system disorders were higher in younger BNT162b2 recipients. A slightly lower frequency of reactogenicity events was associated with greater age. No other notable relationships to sex or race were specifically highlighted for these common AEs.

### 12.2.3 Serious Adverse Events and Other Significant Adverse Events

Serious adverse events (SAEs) and other significant adverse events were closely examined, particularly for patients who discontinued the study prematurely due to an adverse event or who died.

**Table 19. Number (%) of Subjects Reporting at Least 1 Serious Adverse Event From Dose 1 to 6 Months After Dose 2 - Safety Population**

| | Vaccine Group (as Administered) | |
|---|---|---|
| **System Organ Class**<br>**Preferred Term** | **BNT162b2 (30 µg) (N=1130) n (%) (95% CI)** | **Placebo (N=1126) n (%) (95% CI)** |
| **Any event** | **6 (0.5) (0.2, 1.2)** | **0 (0.0, 0.3)** |
| **INFECTIONS AND INFESTATIONS** | **1 (0.1) (0.0, 0.5)** | **0 (0.0, 0.3)** |
| Anal abscess | 1 (0.1) (0.0, 0.5) | 0 (0.0, 0.3) |
| **INJURY, POISONING AND PROCEDURAL COMPLICATIONS** | **1 (0.1) (0.0, 0.5)** | **0 (0.0, 0.3)** |
| Femur fracture | 1 (0.1) (0.0, 0.5) | 0 (0.0, 0.3) |
| **PSYCHIATRIC DISORDERS** | **4 (0.4) (0.1, 0.9)** | **0 (0.0, 0.3)** |
| Suicidal ideation | 3 (0.3) (0.1, 0.8) | 0 (0.0, 0.3) |
| Depression | 1 (0.1) (0.0, 0.5) | 0 (0.0, 0.3) |

During the period from Dose 1 to 6 months after Dose 2, a total of 6 subjects (0.5%) in the BNT162b2 group reported at least one serious adverse event (SAE), compared to 0 subjects in the placebo group. The SAEs reported in the BNT162b2 group included one case of Anal abscess (Infections and Infestations), one case of Femur fracture (Injury, Poisoning and Procedural Complications), and four cases within Psychiatric Disorders (three cases of Suicidal ideation and one case of Depression). No SAEs were reported in the placebo group during this period. The overall incidence of SAEs was low in both treatment groups, with no apparent safety signal indicating a causal relationship to the investigational product.

### 12.2.5 Assessment of Significance of Deaths, Other Serious Adverse Events, and Other Significant Adverse Events

The significance of deaths, other serious adverse events (SAEs), and other significant adverse events leading to withdrawal, dose reduction, or institution of concomitant therapy was assessed with respect to the safety of the investigational product.

The 6 deaths reported in the study (2 in BNT162b2, 4 in placebo) were not considered related to the study intervention by the investigators. The causes were varied and consistent with underlying medical conditions or natural causes expected in the general population.

The overall incidence of SAEs was low and generally balanced between the BNT162b2 and placebo groups during the blinded phase, with proportions of 0.5% and 0.4% respectively. While certain SAEs in the BNT162b2 group were assessed as related by the investigator (shoulder injury related to vaccine administration, ventricular arrhythmia, lymphadenopathy, and lower back pain with radicular paresthesia), these were rare events. The observed number of appendicitis cases was not considered to exceed background rates and was not causally linked to the vaccine. No previously unsuspected important adverse effects of the test drug were identified from the SAE data.

Regarding other significant adverse events, the primary impact on study intervention was due to reactogenicity events, which in rare instances led to discontinuation if classified as Grade 4 severity and determined to be related to the study intervention. However, the retrieved information did not specifically detail instances of study withdrawal or dose reduction due to non-serious but clinically significant adverse events, beyond the general severity and causality assessments discussed in Section 12.2.2.

## 12.3 Laboratory Assessments

### 12.3.1 Listings of Individual Laboratory Measurements and Abnormal Values

Individual patient tabular listings of all safety-related laboratory tests, with patients grouped by investigator and treatment group, are provided in Appendix 16.2.8, when required by regulatory authorities. This includes critical demographic data, drug dose data, and the results of the laboratory tests. Laboratory tests are grouped logically (e.g., haematological tests, liver chemistries, electrolytes, urinalysis).

A by-patient listing of all abnormal laboratory values is provided in Section 14.3.4. Abnormal values are identified within these listings (e.g., by underlining, bracketing). For laboratory abnormalities of special interest (i.e., abnormal laboratory values of potential clinical importance), additional data such as normal values before and after the abnormal value, and values of related laboratory tests, are included. No specific abnormal values were identified for exclusion from further analysis in the provided documents.

### 12.3.2 Analysis of Laboratory Values

Clinical safety laboratory assessments were performed primarily in Phase 1 participants. For Phase 2/3 trials, laboratory results were generally not available except for minor transient decrease in lymphocyte count observed for some of the subjects in Phase 1.

# 13. Discussion and Overall Conclusions

The current study evaluated the efficacy and safety of BNT162b2 in adolescents 12 through 15 years of age. The findings are consistent with the known safety profile in individuals 16 years of age and older, and demonstrate high vaccine efficacy against COVID-19.

## 13.1 Efficacy Results

The primary efficacy analysis for BNT162b2 in adolescents aged 12 to 15 years demonstrated significant protection against COVID-19. As shown in Table 1, the observed vaccine efficacy was high, with no cases of COVID-19 reported in the vaccine group compared to 16 cases in the placebo group among participants without evidence of prior SARS-CoV-2 infection. This result indicates robust protection starting 7 days after the second dose.

Table 20. Vaccine Efficacy - First COVID-19 Occurrence From 7 Days After Dose 2 - Blinded Placebo-Controlled Follow-up (Subjects 12 Through 15 Years of Age Without Evidence of Prior SARS-CoV-2 Infection at Baseline)

| Efficacy Endpoint Subgroup | BNT162b2 (30 µg) (Na=1057) n1b (Surveillance Time (n2)) | Placebo (Na=1030) n1b (Surveillance Time (n2)) | VE (%) (95% CI) |
|---|---|---|---|
| First COVID-19 occurrence from 7 days after Dose 2 | 0 0.343 (1043) | 28 0.322 (1019) | 100.0 (86.8, 100.0) |
| ≥7 days after Dose 2 to <2 Months after Dose 2 | 0 0.138 (1043) | 15 0.133 (1019) | 100.0 (73.2, 100.0) |

The overall vaccine efficacy of 100.0% (95% CI: 86.8, 100.0) against confirmed COVID-19 occurring at least 7 days after Dose 2 in participants without evidence of prior SARS-CoV-2 infection is highly significant. This efficacy was consistently observed within the subgroup of cases occurring between 7 days after Dose 2 and less than 2 months after Dose 2. These findings are consistent with the high efficacy observed in the adult population and provide strong evidence of the vaccine's protective effect in this younger age group.

## 13.2 Safety Results

The safety profile of BNT162b2 in adolescents aged 12 to 15 years was generally consistent with that observed in older populations, primarily characterized by transient, mild-to-moderate reactogenicity events.

Table 21. Incidence Rates of at Least 1 Adverse Event From Dose 1 to Unblinding Date - Blinded Placebo-Controlled Follow-up Period (Subjects 12 Through 15 Years of Age)

| Adverse Event | BNT162b2 (30 µg) (Na=1131, TEb=4.6) n (%) IRd (95% CI) | Placebo (Na=1129, TEb=4.5) n (%) IRd (95% CI) |
|---|---|---|
| Any event | 95 (8.4) 20.8 (16.8, 25.4) | 113 (10.0) 25.1 (20.7, 30.1) |
| Related | 36 (3.2) 7.9 (5.5, 10.9) | 24 (2.1) 5.3 (3.4, 7.9) |
| Severe | 13 (1.1) 2.8 (1.5, 4.9) | 5 (0.4) 1.1 (0.4, 2.6) |
| Life-threatening | 2 (0.2) 0.4 (0.1, 1.6) | 1 (0.1) 0.2 (0.0, 1.2) |
| Any serious adverse event | 10 (0.9) 2.2 (1.0, 4.0) | 2 (0.2) 0.4 (0.1, 1.6) |
| Related | 0 0.0 (0.0, 0.8) | 0 0.0 (0.0, 0.8) |
| Severe | 7 (0.6) 1.5 (0.6, 3.2) | 1 (0.1) 0.2 (0.0, 1.2) |
| Life-threatening | 1 (0.1) 0.2 (0.0, 1.2) | 1 (0.1) 0.2 (0.0, 1.2) |

As presented in Table 2, the incidence of any adverse event from Dose 1 to the unblinding date was comparable between the BNT162b2 group (8.4%) and the placebo group (10.0%). However, adverse events considered related to the study intervention were slightly higher in the vaccine group (3.2%) compared to placebo (2.1%). Severe adverse events were also more frequently reported in the BNT162b2 group (1.1%) than in the placebo group (0.4%). Life-threatening adverse events were rare, with 0.2% in the vaccine group and 0.1% in the placebo group.

The incidence of any serious adverse event was low in both groups, with 0.9% in the BNT162b2 group and 0.2% in the placebo group. No serious adverse events were considered related to the vaccine. Severe serious adverse events were reported in 0.6% of the vaccine group and 0.1% of the placebo group. One life-threatening serious adverse event was reported in each group.

## 13.3 Risk-Benefit Relationship

The comprehensive evaluation of efficacy and safety data in adolescents 12 through 15 years of age demonstrates a favorable risk-benefit profile for BNT162b2. The vaccine demonstrated 100% efficacy against confirmed COVID-19 (Table 1), indicating a substantial protective benefit in preventing infection. While the incidence of any adverse events and severe adverse events was slightly higher in the vaccine group compared to the placebo group (Table 2), these events were predominantly mild-to-moderate and transient, consistent with the known reactogenicity profile of the vaccine in older populations. Crucially, the rates of serious adverse events were low in both groups, and no serious adverse events were considered related to the vaccine (Table 2).

The high level of protection against COVID-19, a disease that can lead to significant morbidity and mortality, outweighs the observed, generally mild and transient adverse events associated with vaccination in this adolescent population. The findings support the use of BNT162b2 as a safe and effective intervention for preventing COVID-19 in individuals aged 12 to 15 years.

## 13.4 New or Unexpected Findings and Clinical Relevance

The current study found that the BNT162b2 vaccine demonstrated 100% efficacy against confirmed COVID-19 in adolescents aged 12 to 15 years, a finding that is highly significant and consistent with the robust efficacy observed in adult populations. This level of protection is clinically important for mitigating the burden of COVID-19 in this age group.

A noteworthy finding specific to the adolescent population (12-15 years) in this study was the observation of higher systemic reactogenicity compared to young adults, although local reactogenicity remained comparable. While generally mild-to-moderate and transient, this difference highlights age-specific immune responses. Specific adverse events identified as potentially related to the vaccine in the broader population and warranting attention include lymphadenopathy, nausea, hypersensitivity reactions, peripheral facial paralysis (Bell's palsy), insomnia, injection site pruritus, and pain in extremity. These are important for post-marketing surveillance and future studies.

Regarding inconsistencies, an independent re-analysis of serious adverse events (SAEs) and adverse events of special interest (AESIs) in adult Pfizer and Moderna trials has raised concerns regarding a potentially higher risk of serious AESIs and SAEs in vaccinated participants compared to initial estimates. For example, a higher risk of serious AESIs (12.5 per 10,000 individuals vaccinated, 95% CI 2.1 to 22.9) was reported in the combined adult trials, with the largest excess risk found in coagulation disorders. While this re-analysis focused on adult data and noted differences in SAE reporting practices between trials, these findings underscore the importance of comprehensive and transparent safety analyses, particularly in the context of post-authorization surveillance and for extrapolating safety profiles across age groups.

The clinical importance of the high vaccine efficacy in adolescents is substantial, particularly given the potential for long-term health consequences from COVID-19, including multisystem inflammatory syndrome in children (MIS-C). The current findings affirm BNT162b2 as an important public health tool for this age group, demonstrating significant protection against symptomatic COVID-19.

## 13.5 Specific Benefits, Special Precautions, and Implications for Future Studies

### 13.5.1 Specific Benefits and Special Precautions

The study demonstrated a significant benefit of BNT162b2 in adolescents aged 12 to 15 years by achieving 100% vaccine efficacy against confirmed COVID-19, which is consistent with the high protective efficacy observed across adult populations. This benefit is particularly important for preventing symptomatic disease in this age group.

However, certain precautions and considerations for specific individuals or at-risk groups are necessary:

- **Contraindications:** BNT162b2 is contraindicated in individuals with a history of severe adverse reaction associated with a vaccine and/or severe allergic reaction (e.g., anaphylaxis) to any component of the study intervention.
- **Anaphylaxis Risk:** As with all injectable vaccines, appropriate medical treatment and supervision should always be readily available to manage anaphylactic reactions

following vaccine administration. Close observation for at least 15 minutes post-vaccination is recommended.

- **Bleeding Disorders:** The vaccine should be administered with caution to individuals with bleeding disorders or those receiving anticoagulant therapy, as intramuscular injection may result in bleeding.
- **Syncope:** Procedures should be in place to prevent injury from syncope (fainting), which can occur following, or even before, any injection, including vaccination.
- **Immunocompromised Individuals:** The immune response to BNT162b2 may be insufficient in immunocompromised individuals, including those receiving immunosuppressant therapy. While no specific safety concerns were detected in participants with stable HIV infection in this study, data on broader immunocompromised populations remain limited and require further investigation.
- **Off-label Administration:** The vaccine should only be administered intramuscularly, ideally in the deltoid muscle. It should not be injected intravenously, intradermally, subcutaneously, or orally.
- **Pregnancy and Breastfeeding:** Data on the use of BNT162b2 in pregnant or breastfeeding women are very limited. These populations were largely excluded from the pivotal study, and further dedicated studies are necessary to assess safety and efficacy.
- **Very Elderly:** While reactogenicity and adverse event rates were generally lower in older individuals, data on individuals over 85 years of age are very limited, and the impact of immunosenescence on vaccine efficacy in this specific subgroup remains uncertain.

### 13.5.2 Implications for Future Studies

The findings of this study, particularly the high efficacy and acceptable safety profile in adolescents, have significant implications for future research and public health strategies. Continued robust pharmacovigilance is paramount, especially given the observed higher systemic reactogenicity in adolescents compared to young adults and the concerns raised by re-analyses of SAEs and AESIs in adult trials.

Future studies should focus on:

- **Long-term Safety and Efficacy:** The ongoing pivotal study (C4591001) will provide long-term safety and efficacy data over a 24-month follow-up period, which is crucial for a complete understanding of the vaccine's profile.
- **Special Populations:** Dedicated post-authorization safety studies (PASS) and interventional trials are planned or ongoing to gather more comprehensive safety and immunogenicity data in specific at-risk groups, including pregnant women, frail patients with co-morbidities (e.g., chronic pulmonary disease, diabetes, cardiovascular disorders), and various immunocompromised patient groups beyond stable HIV infection.
- **Co-administration with Other Vaccines:** A planned study will investigate the safety and immunogenicity of BNT162b2 when co-administered with inactivated quadrivalent influenza vaccine, addressing a critical need for vaccination strategies.
- **Vaccine-Associated Enhanced Disease (VAED):** Continued monitoring and investigation into the theoretical risk of VAED/VAERD are essential through ongoing clinical studies and post-authorization safety studies.
- **Real-World Effectiveness:** Non-interventional post-approval effectiveness studies using real-world data sources (e.g., electronic health records) are planned to assess

the vaccine's effectiveness against laboratory-confirmed COVID-19, including severe disease, and in specific racial, ethnic, and age groups not fully captured in the initial trials.
- **New LNP Components:** Continued monitoring of the novel lipid nanoparticle components (ALC-0315 and ALC-0159) is important due to limited prior experience and some remaining uncertainties regarding their long-term safety.

These planned studies and pharmacovigilance activities will help to address current data limitations and further refine the understanding of BNT162b2's benefits and risks across diverse populations and over extended periods.

## 13.6 Impact of Exclusions on Generalisability

The exclusion criteria implemented in the study, while necessary for safety and to ensure a clear assessment of vaccine efficacy and safety in a defined population, inherently impact the generalisability of the findings to certain segments of the broader population.

Key populations largely excluded from the pivotal study include:

- **Individuals with a previous clinical or microbiological diagnosis of COVID-19:** This exclusion allowed for the evaluation of vaccine efficacy in a SARS-CoV-2 naive population. Consequently, the study's primary efficacy results do not directly inform the efficacy of the vaccine in individuals with prior natural infection, though effectiveness in this group is being explored in post-authorization settings.
- **Immunocompromised individuals with known or suspected immunodeficiency (except for stable HIV, HBV, HCV infection):** While subjects with stable HIV, HBV, or HCV infections were permitted, a broader range of immunocompromised individuals, including those with other immunodeficiencies or on immunosuppressive therapy, were excluded. This limits the generalisability of the safety and efficacy findings to these vulnerable populations, who may have a blunted immune response to vaccination. Dedicated studies are underway or planned to address this information gap.
- **Pregnant or breastfeeding women:** To ensure participant safety, these populations were excluded from the primary study. Therefore, the generalisability of the findings to pregnant and breastfeeding individuals is limited, necessitating specific post-authorization safety studies and registries.
- **Individuals with a history of severe allergic reactions to vaccines or vaccine components:** Such exclusions are standard in clinical trials to prevent severe adverse events, but they mean the study's safety profile is not directly applicable to individuals with such histories.
- **Individuals with active autoimmune diseases requiring therapeutic intervention:** Although healthy participants with preexisting stable diseases were included, those with active autoimmune conditions requiring therapeutic intervention were generally excluded. This limits the generalisability of the safety findings to this potentially reactive population.

While the study population included individuals with stable comorbidities and a wide age range (including adults over 65 years and adolescents 12-15 years), the exclusion of these specific groups means that conclusions regarding the vaccine's safety and efficacy in these populations must be drawn with caution. The potential impact of these exclusions on overall

generalisability is being addressed through ongoing and planned post-authorization safety and effectiveness studies, which aim to gather data in these currently underrepresented or excluded populations. The intent is to provide a comprehensive understanding of the vaccine's profile across a broader and more diverse demographic.

# 14. Tables, Figures and Graphs Referred to But Not Included in the Text

This section provides supplemental tables from the clinical study.

## 14.1 Demographic Characteristics, by Baseline SARS-CoV-2 Status – Phase 2/3

**Table 14.1.** Demographic Characteristics, by Baseline SARS-CoV-2 Status – Phase 2/3

| | Vaccine Group (as Administered) | | |
|---|---|---|---|
| BNT162b2 (30 µg) (Na=46) nb (%) | Placebo (Na=50) nb (%) | Total (Na=96) nb (%) | |
| Sex | | | |
| Male | 21 (45.7) | 25 (50.0) | 46 (47.9) |
| Female | 25 (54.3) | 25 (50.0) | 50 (52.1) |
| Race | | | |
| White | 39 (84.8) | 40 (80.0) | 79 (82.3) |
| Black or African American | 6 (13.0) | 7 (14.0) | 13 (13.5) |
| All others | 1 (2.2) | 3 (6.0) | 4 (4.2) |
| American Indian or Alaska Native | 0 | 1 (2.0) | 1 (1.0) |
| Asian | 0 | 1 (2.0) | 1 (1.0) |
| Multiracial | 1 (2.2) | 0 | 1 (1.0) |
| Not reported | 0 | 1 (2.0) | 1 (1.0) |
| Ethnicity | | | |
| Hispanic/Latino | 7 (15.2) | 10 (20.0) | 17 (17.7) |
| Non-Hispanic/non-Latino | 39 (84.8) | 40 (80.0) | 79 (82.3) |
| Country | | | |
| USA | 46 (100.0) | 50 (100.0) | 96 (100.0) |
| Comorbiditiesc | | | |
| Yes | 11 (23.9) | 12 (24.0) | 23 (24.0) |
| No | 35 (76.1) | 38 (76.0) | 73 (76.0) |
| Obesed | | | |
| Yes | 6 (13.0) | 10 (20.0) | 16 (16.7) |
| No | 40 (87.0) | 40 (80.0) | 80 (83.3) |
| Age at vaccination (years) | | | |
| Mean (SD) | 13.5 (1.19) | 13.7 (1.07) | 13.6 (1.13) |
| Median | 14.0 | 14.0 | 14.0 |
| Min, max | (12, 15) | (12, 15) | (12, 15) |
| | Vaccine Group (as Administered) | | |
| | BNT162b2 (30 µg) (Na=46) nb (%) | Placebo (Na=50) nb (%) | Total (Na=96) nb (%) |

Abbreviations: NE = not estimable; SARS-CoV-2 = severe acute respiratory syndrome coronavirus 2. Note: Subjects whose baseline SARS-CoV-2 status cannot be determined because of missing N-binding antibody or NAAT at Visit 1 were not included in the an...

## 14.2 Demographic Characteristics, by Baseline SARS-CoV-2 Status – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Baseline SARS-CoV-2 Status: Negative

**Table 14.2.** Demographic Characteristics, by Baseline SARS-CoV-2 Status – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Baseline SARS-CoV-2 Status: Negative

| | Vaccine Group (as Administered) | | Total (N*=2161) n° (%) |
|---|---|---|---|
| BNT162b2 (30 µg) (N*=1083) n° (%) | Placebo (N*=1078) n° (%) | | |
| Sex | | | |
| Male | 545 (50.3) | 560 (51.9) | 1105 (51.1) |
| Female | 538 (49.7) | 518 (48.1) | 1056 (48.9) |
| Race | | | |
| White | 929 (85.8) | 921 (85.4) | 1850 (85.6) |
| Black or African American | 46 (4.2) | 50 (4.6) | 96 (4.4) |
| All others | 108 (10.0) | 107 (9.9) | 215 (9.9) |
| American Indian or Alaska Native | 4 (0.4) | 2 (0.2) | 6 (0.3) |
| Asian | 72 (6.6) | 70 (6.5) | 142 (6.6) |
| Native Hawaiian or other Pacific Islander | 3 (0.3) | 0 | 3 (0.1) |
| Multiracial | 23 (2.1) | 29 (2.7) | 52 (2.4) |
| Not reported | 6 (0.6) | 6 (0.6) | 12 (0.6) |
| Racial designation | | | |
| Japanese | 5 (0.5) | 2 (0.2) | 7 (0.3) |
| Ethnicity | | | |
| Hispanic/Latino | 125 (11.5) | 120 (11.1) | 245 (11.3) |
| Non-Hispanic/non-Latino | 956 (88.3) | 955 (88.6) | 1911 (88.4) |
| Not reported | 2 (0.2) | 3 (0.3) | 5 (0.2) |
| Country | | | |
| USA | 1083 (100.0) | 1078 (100.0) | 2161 (100.0) |
| Comorbiditiesc | | | |
| Yes | 238 (22.0) | 230 (21.3) | 468 (21.7) |
| No | 845 (78.0) | 848 (78.7) | 1693 (78.3) |
| Obesed | | | |
| Yes | 137 (12.7) | 118 (10.9) | 255 (11.8) |
| No | 946 (87.3) | 960 (89.1) | 1906 (88.2) |
| Age at vaccination (years) | | | |
| Mean (SD) | 13.6 (1.10) | 13.6 (1.11) | 13.6 (1.11) |
| Median | 14.0 | 14.0 | 14.0 |
| Min, max | (12, 15) | (12, 15) | (12, 15) |
| Vaccine Group (as Administered) | | | |
| BNT162b2 (30 µg) (Na=1083) nb (%) | Placebo (Na=1078) nb (%) | Total (Na=2161) nb (%) | |

## 14.3 Medical History – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population

**Table 14.3.** Medical History – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population

| System Organ Class Preferred Term | Vaccine Group (as Administered) | |
|---|---|---|
| BNT162b2 (30 µg) (N°=1131) | Placebo (N°=1129) | |
| nb (%) | nb (%) | |
| Any medical history | 850 (75.2) | 832 (73.7) |
| Blood and lymphatic system disorders | 3 (0.3) | 8 (0.7) |
| Anaemia | 0 | 1 (0.1) |
| Immune thrombocytopenia | 3 (0.3) | 1 (0.1) |
| Iron deficiency anaemia | 0 | 3 (0.3) |
| Lymphadenopathy | 0 | 2 (0.2) |
| Thrombocytopenia | 0 | 1 (0.1) |
| Cardiac disorders | 5 (0.4) | 3 (0.3) |
| Aortic valve disease | 2 (0.2) | 0 |
| Arrhythmia | 1 (0.1) | 0 |
| Postural orthostatic tachycardia syndrome | 1 (0.1) | 1 (0.1) |
| Pulmonary valve stenosis | 1 (0.1) | 0 |
| Supraventricular tachycardia | 0 | 1 (0.1) |
| Ventricular extrasystoles | 0 | 1 (0.1) |
| Congenital, familial and genetic disorders | 29 (2.6) | 45 (4.0) |
| Adenomatous polyposis coli | 0 | 1 (0.1) |
| Ankyloglossia congenital | 0 | 1 (0.1) |
| Anorectal malformation | 0 | 1 (0.1) |
| Atrial septal defect | 1 (0.1) | 2 (0.2) |
| Bicuspid aortic valve | 2 (0.2) | 2 (0.2) |
| Birth mark | 0 | 2 (0.2) |
| Cerebral cavernous malformation | 0 | 1 (0.1) |
| Cerebral palsy | 0 | 1 (0.1) |
| Chondrodystrophy | 1 (0.1) | 0 |
| Cleft lip and palate | 1 (0.1) | 0 |
| Cleft palate | 1 (0.1) | 1 (0.1) |
| Colour blindness | 1 (0.1) | 0 |
| Congenital anomaly | 1 (0.1) | 0 |
| Congenital diaphragmatic hernia | 1 (0.1) | 0 |
| Congenital flat feet | 1 (0.1) | 0 |
| Congenital megacolon | 0 | 1 (0.1) |
| Congenital nystagmus | 2 (0.2) | 0 |
| Congenital skin dimples | 0 | 1 (0.1) |
| Cryptorchism | 1 (0.1) | 0 |
| Cystic fibrosis | 0 | 2 (0.2) |
| System Organ Class Preferred Term | Vaccine Group (as Administered) | |
| BNT162b2 (30 µg) (N°=1131) | Placebo (N°=1129) | |
| n° (%) | n° (%) | |
| Developmental hip dysplasia | 0 | 1 (0.1) |
| Ehlers-Danlos syndrome | 0 | 1 (0.1) |
| Factor V Leiden carrier | 1 (0.1) | 1 (0.1) |
| Factor V Leiden mutation | 0 | 1 (0.1) |
| Gilbertś syndrome | 0 | 1 (0.1) |
| Hemivertebra | 0 | 1 (0.1) |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | |
|---|---|---|
| Hereditary motor and sensory neuropathy | 0 | 1 (0.1) |
| Hereditary spherocytosis | 0 | 1 (0.1) |
| Hypoplastic left heart syndrome | 0 | 1 (0.1) |
| Hypospadias | 0 | 1 (0.1) |
| Imperforate hymen | 0 | 1 (0.1) |
| Malformation venous | 1 (0.1) | 0 |
| Metabolic myopathy | 1 (0.1) | 0 |
| Microgenia | 0 | 1 (0.1) |
| Multiple epiphyseal dysplasia | 0 | 1 (0.1) |
| Naevus flammeus | 0 | 1 (0.1) |
| Neurofibromatosis | 0 | 2 (0.2) |
| Oculoauriculovertebral dysplasia | 0 | 1 (0.1) |
| Otospondylomegalyepiphyseal dysplasia | 1 (0.1) | 0 |
| Pectus carinatum | 0 | 1 (0.1) |
| Pectus excavatum | 2 (0.2) | 1 (0.1) |
| Phimosis | 1 (0.1) | 1 (0.1) |
| Polydactyly | 1 (0.1) | 0 |
| Renal dysplasia | 0 | 1 (0.1) |
| Sickle cell anaemia | 1 (0.1) | 0 |
| Sickle cell trait | 0 | 3 (0.3) |
| Spina bifida occulta | 1 (0.1) | 0 |
| Strabismus congenital | 1 (0.1) | 0 |
| Talipes | 1 (0.1) | 0 |
| Thalassaemia beta | 2 (0.2) | 0 |
| Thalassaemia minor | 0 | 1 (0.1) |
| Thyroglossal cyst | 1 (0.1) | 0 |
| Touretteś disorder | 2 (0.2) | 2 (0.2) |
| Transposition of the great vessels | 0 | 1 (0.1) |
| Ventricular septal defect | 0 | 1 (0.1) |
| Von Willebrandś disease | 1 (0.1) | 2 (0.2) |
| Ear and labyrinth disorders | 11 (1.0) | 10 (0.9) |
| Auditory disorder | 1 (0.1) | 0 |
| Deafness | 2 (0.2) | 0 |
| System Organ Class Preferred Term | Vaccine Group (as Administered) | |
| BNT162b2 (30 µg) (N°=1131) | Placebo (N°=1129) | |
| nb (%) | nb (%) | |
| Deafness bilateral | 1 (0.1) | 0 |
| Deafness unilateral | 1 (0.1) | 1 (0.1) |
| Ear pain | 0 | 1 (0.1) |
| Eustachian tube disorder | 1 (0.1) | 0 |
| Eustachian tube dysfunction | 2 (0.2) | 0 |
| Hypoacusis | 1 (0.1) | 2 (0.2) |
| Middle ear adhesions | 0 | 1 (0.1) |
| Motion sickness | 0 | 1 (0.1) |
| Tinnitus | 1 (0.1) | 2 (0.2) |
| Tympanic membrane perforation | 1 (0.1) | 1 (0.1) |
| Vestibular disorder | 0 | 1 (0.1) |
| Endocrine disorders | 7 (0.6) | 7 (0.6) |
| Autoimmune thyroiditis | 0 | 1 (0.1) |
| Growth hormone deficiency | 1 (0.1) | 3 (0.3) |
| Hypopituitarism | 1 (0.1) | 0 |
| Hypothyroidism | 3 (0.3) | 1 (0.1) |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | |
|---|---|---|
| Precocious puberty | 2 (0.2) | 2 (0.2) |
| Eye disorders | 46 (4.1) | 59 (5.2) |
| Amblyopia | 1 (0.1) | 2 (0.2) |
| Amblyopia strabismic | 1 (0.1) | 0 |
| Anisometropia | 1 (0.1) | 0 |
| Astigmatism | 1 (0.1) | 6 (0.5) |
| Blepharitis | 2 (0.2) | 0 |
| Blindness unilateral | 0 | 1 (0.1) |
| Cataract | 1 (0.1) | 0 |
| Chalazion | 0 | 1 (0.1) |
| Conjunctivitis allergic | 1 (0.1) | 0 |
| Dacryostenosis acquired | 0 | 1 (0.1) |
| Eyelid ptosis | 1 (0.1) | 0 |
| Hypermetropia | 5 (0.4) | 7 (0.6) |
| Myopia | 20 (1.8) | 27 (2.4) |
| Optic atrophy | 0 | 1 (0.1) |
| Optic nerve cupping | 0 | 1 (0.1) |
| Presbyopia | 1 (0.1) | 0 |
| Punctate keratitis | 1 (0.1) | 0 |
| Pupils unequal | 0 | 1 (0.1) |
| Recession of chamber angle of eye | 0 | 1 (0.1) |
| Refractive amblyopia | 0 | 1 (0.1) |
| System Organ Class Preferred Term | Vaccine Group (as Administered) | |
| BNT162b2 (30 µg) (N°=1131) | Placebo (N°=1129) | |
| nb (%) | nb (%) | |
| Strabismus | 4 (0.4) | 4 (0.4) |
| Visual acuity reduced | 8 (0.7) | 12 (1.1) |
| Gastrointestinal disorders | 43 (3.8) | 44 (3.9) |
| Abdominal hernia | 1 (0.1) | 0 |
| Abdominal migraine | 2 (0.2) | 1 (0.1) |
| Abdominal pain | 1 (0.1) | 3 (0.3) |
| Abdominal pain upper | 1 (0.1) | 1 (0.1) |
| Coeliac disease | 4 (0.4) | 4 (0.4) |
| Constipation | 10 (0.9) | 11 (1.0) |
| Cyclic vomiting syndrome | 1 (0.1) | 0 |
| Diarrhoea | 1 (0.1) | 2 (0.2) |
| Dyspepsia | 3 (0.3) | 0 |
| Dysphagia | 0 | 1 (0.1) |
| Enterocolitis | 1 (0.1) | 0 |
| Eosinophilic oesophagitis | 1 (0.1) | 3 (0.3) |
| Gastroesophageal reflux disease | 12 (1.1) | 14 (1.2) |
| Hiatus hernia | 0 | 1 (0.1) |
| Inguinal hernia | 1 (0.1) | 1 (0.1) |
| Intussusception | 2 (0.2) | 0 |
| Irritable bowel syndrome | 2 (0.2) | 2 (0.2) |
| Malabsorption | 1 (0.1) | 0 |
| Oesophagitis | 1 (0.1) | 0 |
| Oral pain | 0 | 1 (0.1) |
| Tooth impacted | 0 | 1 (0.1) |
| Toothache | 0 | 1 (0.1) |
| Umbilical hernia | 2 (0.2) | 3 (0.3) |
| General disorders and administration site conditions | 10 (0.9) | 10 (0.9) |
| Adverse food reaction | 1 (0.1) | 0 |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | |
|---|---|---|
| Cyst | 1 (0.1) | 1 (0.1) |
| Developmental delay | 0 | 2 (0.2) |
| Drug intolerance | 1 (0.1) | 2 (0.2) |
| Medical device pain | 1 (0.1) | 1 (0.1) |
| Pain | 5 (0.4) | 3 (0.3) |
| Peripheral swelling | 1 (0.1) | 0 |
| Pyrexia | 0 | 1 (0.1) |
| Hepatobiliary disorders | 4 (0.4) | 0 |
| Cholelithiasis | 2 (0.2) | 0 |
| System Organ Class Preferred Term | Vaccine Group (as Administered) | |
| BNT162b2 (30 µg) (N°=1131) | Placebo (N°=1129) | |
| nb (%) | nb (%) | |
| Hepatic steatosis | 1 (0.1) | 0 |
| Non-alcoholic steatohepatitis | 1 (0.1) | 0 |
| Immune system disorders | 399 (35.3) | 391 (34.6) |
| Allergy to animal | 24 (2.1) | 19 (1.7) |
| Allergy to arthropod bite | 1 (0.1) | 2 (0.2) |
| Allergy to arthropod sting | 2 (0.2) | 4 (0.4) |
| Allergy to chemicals | 1 (0.1) | 0 |
| Allergy to metals | 0 | 2 (0.2) |
| Allergy to plants | 4 (0.4) | 2 (0.2) |
| Anaphylactic reaction | 1 (0.1) | 0 |
| Cockroach allergy | 1 (0.1) | 1 (0.1) |
| Drug hypersensitivity | 130 (11.5) | 97 (8.6) |
| Food allergy | 32 (2.8) | 40 (3.5) |
| Hypersensitivity | 22 (1.9) | 16 (1.4) |
| Milk allergy | 3 (0.3) | 5 (0.4) |
| Mite allergy | 1 (0.1) | 4 (0.4) |
| Multiple allergies | 0 | 1 (0.1) |
| Mycotic allergy | 0 | 1 (0.1) |
| Oral allergy syndrome | 0 | 4 (0.4) |
| Perennial allergy | 2 (0.2) | 3 (0.3) |
| Perfume sensitivity | 1 (0.1) | 0 |
| Reaction to colouring | 1 (0.1) | 0 |
| Reaction to food additive | 0 | 1 (0.1) |
| Rubber sensitivity | 6 (0.5) | 5 (0.4) |
| Seasonal allergy | 241 (21.3) | 247 (21.9) |
| Selective IgA immunodeficiency | 0 | 1 (0.1) |
| Serum sickness | 1 (0.1) | 0 |
| Infections and infestations | 72 (6.4) | 52 (4.6) |
| Abscess limb | 0 | 1 (0.1) |
| Adenoiditis | 11 (1.0) | 6 (0.5) |
| Appendicitis | 6 (0.5) | 9 (0.8) |
| Body tinea | 1 (0.1) | 0 |
| Cellulitis | 1 (0.1) | 0 |
| Chronic sinusitis | 2 (0.2) | 1 (0.1) |
| Chronic tonsillitis | 2 (0.2) | 3 (0.3) |
| Conjunctivitis | 1 (0.1) | 1 (0.1) |
| Croup infectious | 1 (0.1) | 1 (0.1) |
| Dermatophytosis of nail | 1 (0.1) | 0 |
| System Organ Class Preferred Term | Vaccine Group (as Administered) | |
| BNT162b2 (30 µg) (N°=1131) | Placebo (N°=1129) | |
| nb (%) | nb (%) | |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | |
|---|---|---|
| Ear infection | 12 (1.1) | 8 (0.7) |
| Gastrointestinal viral infection | 1 (0.1) | 0 |
| Herpes simplex | 1 (0.1) | 0 |
| Histoplasmosis | 1 (0.1) | 0 |
| Impetigo | 1 (0.1) | 0 |
| Infectious mononucleosis | 0 | 1 (0.1) |
| Kidney infection | 1 (0.1) | 0 |
| Lyme disease | 2 (0.2) | 0 |
| Meningitis | 1 (0.1) | 0 |
| Meningitis viral | 1 (0.1) | 0 |
| Molluscum contagiosum | 0 | 1 (0.1) |
| Nail infection | 1 (0.1) | 0 |
| Oral herpes | 1 (0.1) | 0 |
| Osteomyelitis | 2 (0.2) | 0 |
| Otitis media | 3 (0.3) | 1 (0.1) |
| Otitis media acute | 1 (0.1) | 0 |
| Otitis media chronic | 1 (0.1) | 2 (0.2) |
| Paronychia | 1 (0.1) | 0 |
| Pharyngeal abscess | 0 | 1 (0.1) |
| Pharyngitis | 2 (0.2) | 0 |
| Pharyngitis streptococcal | 1 (0.1) | 3 (0.3) |
| Pneumonia | 6 (0.5) | 4 (0.4) |
| Respiratory syncytial virus bronchiolitis | 0 | 1 (0.1) |
| Respiratory syncytial virus infection | 1 (0.1) | 0 |
| Rhinitis | 1 (0.1) | 1 (0.1) |
| Rotavirus infection | 0 | 1 (0.1) |
| Scarlet fever | 1 (0.1) | 1 (0.1) |
| Sinusitis | 4 (0.4) | 1 (0.1) |
| Staphylococcal scalded skin syndrome | 1 (0.1) | 0 |
| Tinea infection | 0 | 1 (0.1) |
| Tonsillitis | 11 (1.0) | 8 (0.7) |
| Urinary tract infection | 0 | 2 (0.2) |
| Viral infection | 1 (0.1) | 0 |
| Vulvovaginal mycotic infection | 1 (0.1) | 0 |
| Injury, poisoning and procedural complications | 58 (5.1) | 48 (4.3) |
| Ankle fracture | 5 (0.4) | 1 (0.1) |
| Chest injury | 1 (0.1) | 0 |
| Chillblains | 1 (0.1) | 1 (0.1) |
| Clavicle fracture | 2 (0.2) | 3 (0.3) |
| System Organ Class Preferred Term | Vaccine Group (as Administered) | |
| BNT162b2 (30 µg) (N°=1131) | Placebo (N°=1129) | |
| n° (%) | n° (%) | |
| Concussion | 5 (0.4) | 3 (0.3) |
| Contusion | 1 (0.1) | 0 |
| Epiphyseal fracture | 1 (0.1) | 0 |
| Facial bones fracture | 1 (0.1) | 2 (0.2) |
| Fall | 1 (0.1) | 0 |
| Femur fracture | 0 | 1 (0.1) |
| Fibula fracture | 1 (0.1) | 0 |
| Foot fracture | 8 (0.7) | 3 (0.3) |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | |
|---|---|---|
| Foreign body in ear | 0 | 1 (0.1) |
| Hand fracture | 8 (0.7) | 5 (0.4) |
| Humerus fracture | 0 | 1 (0.1) |
| Jaw fracture | 0 | 1 (0.1) |
| Joint dislocation | 0 | 1 (0.1) |
| Joint injury | 0 | 2 (0.2) |
| Ligament injury | 1 (0.1) | 0 |
| Ligament rupture | 1 (0.1) | 0 |
| Ligament sprain | 0 | 2 (0.2) |
| Limb fracture | 2 (0.2) | 0 |
| Limb injury | 1 (0.1) | 0 |
| Lower limb fracture | 1 (0.1) | 1 (0.1) |
| Meniscus injury | 2 (0.2) | 0 |
| Muscle strain | 4 (0.4) | 1 (0.1) |
| Nasal injury | 0 | 1 (0.1) |
| Post concussion syndrome | 0 | 1 (0.1) |
| Radius fracture | 5 (0.4) | 1 (0.1) |
| Skin laceration | 0 | 1 (0.1) |
| Stress fracture | 1 (0.1) | 2 (0.2) |
| Tibia fracture | 3 (0.3) | 5 (0.4) |
| Torus fracture | 0 | 2 (0.2) |
| Upper limb fracture | 12 (1.1) | 11 (1.0) |
| VIIth nerve injury | 1 (0.1) | 0 |
| Wrist fracture | 10 (0.9) | 6 (0.5) |
| Investigations | 14 (1.2) | 8 (0.7) |
| Blood pressure increased | 1 (0.1) | 2 (0.2) |
| Body height decreased | 1 (0.1) | 0 |
| Cardiac murmur | 9 (0.8) | 4 (0.4) |
| Endoscopy | 1 (0.1) | 0 |
| Endoscopy upper gastrointestinal tract | 0 | 1 (0.1) |
| Menstruation normal | 1 (0.1) | 1 (0.1) |
| System Organ Class Preferred Term | Vaccine Group (as Administered) | |
| BNT162b2 (30 µg) (N°=1131) | Placebo (N°=1129) | |
| nb (%) | nb (%) | |
| Serum ferritin decreased | 1 (0.1) | 0 |
| Weight decreased | 1 (0.1) | 0 |
| Metabolism and nutrition disorders | 39 (3.4) | 51 (4.5) |
| Calcium deficiency | 0 | 1 (0.1) |
| Dairy intolerance | 1 (0.1) | 1 (0.1) |
| Decreased appetite | 2 (0.2) | 1 (0.1) |
| Dehydration | 1 (0.1) | 0 |
| Dyslipidaemia | 1 (0.1) | 2 (0.2) |
| Food intolerance | 2 (0.2) | 2 (0.2) |
| Fructose intolerance | 1 (0.1) | 1 (0.1) |
| Glucose tolerance impaired | 1 (0.1) | 0 |
| Gluten sensitivity | 0 | 2 (0.2) |
| Hyperglycaemia | 1 (0.1) | 0 |
| Hyperlipidaemia | 4 (0.4) | 0 |
| Hypertriglyceridaemia | 1 (0.1) | 1 (0.1) |
| Iron deficiency | 0 | 2 (0.2) |
| Lactose intolerance | 5 (0.4) | 7 (0.6) |
| Obesity | 15 (1.3) | 20 (1.8) |
| Overweight | 1 (0.1) | 4 (0.4) |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | |
|---|---|---|
| Type 1 diabetes mellitus | 2 (0.2) | 5 (0.4) |
| Underweight | 1 (0.1) | 0 |
| Vitamin D deficiency | 4 (0.4) | 6 (0.5) |
| Musculoskeletal and connective tissue disorders | 58 (5.1) | 49 (4.3) |
| Arthralgia | 12 (1.1) | 8 (0.7) |
| Back pain | 1 (0.1) | 1 (0.1) |
| Discoid meniscus | 0 | 1 (0.1) |
| Exostosis | 1 (0.1) | 0 |
| Foot deformity | 2 (0.2) | 2 (0.2) |
| Growing pains | 1 (0.1) | 0 |
| Growth retardation | 1 (0.1) | 0 |
| Hypermobility syndrome | 0 | 1 (0.1) |
| Joint instability | 1 (0.1) | 0 |
| Juvenile idiopathic arthritis | 0 | 2 (0.2) |
| Knee deformity | 2 (0.2) | 0 |
| Kyphosis | 2 (0.2) | 1 (0.1) |
| Lordosis | 1 (0.1) | 0 |
| Myalgia | 1 (0.1) | 5 (0.4) |
| Neck pain | 1 (0.1) | 0 |
| System Organ Class Preferred Term | Vaccine Group (as Administered) | |
| BNT162b2 (30 µg) (N°=1131) | Placebo (N°=1129) | |
| nb (%) | nb (%) | |
| Osteitis | 1 (0.1) | 1 (0.1) |
| Osteochondrosis | 3 (0.3) | 4 (0.4) |
| Pain in extremity | 2 (0.2) | 3 (0.3) |
| Pain in jaw | 1 (0.1) | 0 |
| Patellofemoral pain syndrome | 0 | 2 (0.2) |
| Plantar fascial fibromatosis | 0 | 1 (0.1) |
| Rotator cuff syndrome | 1 (0.1) | 0 |
| Scoliosis | 21 (1.9) | 12 (1.1) |
| Short stature | 6 (0.5) | 2 (0.2) |
| Shoulder deformity | 0 | 1 (0.1) |
| Synovial cyst | 0 | 1 (0.1) |
| Temporomandibular joint syndrome | 2 (0.2) | 1 (0.1) |
| Tendon disorder | 0 | 1 (0.1) |
| Tendonitis | 1 (0.1) | 3 (0.3) |
| Toe walking | 1 (0.1) | 0 |
| Neoplasms benign, malignant and unspecified (incl cysts and polyps) | 10 (0.9) | 14 (1.2) |
| Benign ear neoplasm | 1 (0.1) | 0 |
| Cholesteatoma | 0 | 1 (0.1) |
| Eyelid haemangioma | 1 (0.1) | 0 |
| Fibroadenoma of breast | 0 | 1 (0.1) |
| Fibroma | 0 | 1 (0.1) |
| Haemangioma | 1 (0.1) | 0 |
| Lipoma | 0 | 1 (0.1) |
| Melanocytic naevus | 1 (0.1) | 3 (0.3) |
| Nephroblastoma | 0 | 1 (0.1) |
| Skin papilloma | 6 (0.5) | 6 (0.5) |
| Nervous system disorders | 94 (8.3) | 68 (6.0) |
| Apraxia | 1 (0.1) | 0 |
| Arachnoid cyst | 1 (0.1) | 0 |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | |
|---|---|---|
| Benign rolandic epilepsy | 1 (0.1) | 0 |
| Convulsion in childhood | 1 (0.1) | 0 |
| Disturbance in attention | 4 (0.4) | 0 |
| Dizziness | 2 (0.2) | 0 |
| Dysgraphia | 0 | 2 (0.2) |
| Dyslexia | 1 (0.1) | 6 (0.5) |
| Epilepsy | 4 (0.4) | 1 (0.1) |
| Febrile convulsion | 3 (0.3) | 0 |
| Headache | 38 (3.4) | 30 (2.7) |
| System Organ Class Preferred Term | Vaccine Group (as Administered) | |
| BNT162b2 (30 µg) (N°=1131) n° (%) | Placebo (N°=1129) n° (%) | |
| Hydrocephalus | 0 | 1 (0.1) |
| Idiopathic intracranial hypertension | 0 | 1 (0.1) |
| Mental impairment | 2 (0.2) | 0 |
| Migraine | 30 (2.7) | 25 (2.2) |
| Migraine with aura | 2 (0.2) | 1 (0.1) |
| Migraine without aura | 0 | 2 (0.2) |
| Nystagmus | 1 (0.1) | 0 |
| Ophthalmic migraine | 0 | 1 (0.1) |
| Retinal migraine | 1 (0.1) | 0 |
| Seizure | 1 (0.1) | 1 (0.1) |
| Sensory disturbance | 0 | 1 (0.1) |
| Sensory processing disorder | 1 (0.1) | 1 (0.1) |
| Speech disorder | 2 (0.2) | 0 |
| Syncope | 1 (0.1) | 1 (0.1) |
| Tension headache | 1 (0.1) | 0 |
| Tethered cord syndrome | 1 (0.1) | 0 |
| Pregnancy, puerperium and perinatal conditions | 2 (0.2) | 1 (0.1) |
| Premature baby | 2 (0.2) | 1 (0.1) |
| Product issues | 0 | 1 (0.1) |
| Device breakage | 0 | 1 (0.1) |
| Psychiatric disorders | 293 (25.9) | 285 (25.2) |
| Adjustment disorder with depressed mood | 1 (0.1) | 1 (0.1) |
| Adjustment disorder with mixed anxiety and depressed mood | 0 | 1 (0.1) |
| Aggression | 1 (0.1) | 1 (0.1) |
| Anger | 0 | 1 (0.1) |
| Anxiety | 107 (9.5) | 96 (8.5) |
| Anxiety disorder | 4 (0.4) | 3 (0.3) |
| Attention deficit hyperactivity disorder | 182 (16.1) | 166 (14.7) |
| Autism spectrum disorder | 10 (0.9) | 10 (0.9) |
| Behaviour disorder | 2 (0.2) | 1 (0.1) |
| Bipolar disorder | 2 (0.2) | 0 |
| Childhood depression | 1 (0.1) | 0 |
| Chronic tic disorder | 0 | 2 (0.2) |
| Depression | 51 (4.5) | 46 (4.1) |
| Depressive symptom | 0 | 1 (0.1) |
| Disruptive mood dysregulation disorder | 3 (0.3) | 2 (0.2) |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | |
|---|---|---|
| Eating disorder | 1 (0.1) | 2 (0.2) |
| System Organ Class Preferred Term | Vaccine Group (as Administered) | |
| BNT162b2 (30 µg) (N°=1131) | Placebo (N°=1129) | |
| nb (%) | nb (%) | |
| Enuresis | 3 (0.3) | 2 (0.2) |
| Gender dysphoria | 2 (0.2) | 0 |
| Generalised anxiety disorder | 8 (0.7) | 14 (1.2) |
| Hallucination, auditory | 1 (0.1) | 0 |
| Impulse-control disorder | 2 (0.2) | 0 |
| Impulsive behaviour | 0 | 1 (0.1) |
| Insomnia | 27 (2.4) | 29 (2.6) |
| Learning disorder | 0 | 1 (0.1) |
| Major depression | 5 (0.4) | 6 (0.5) |
| Neurodevelopmental disorder | 1 (0.1) | 0 |
| Nightmare | 1 (0.1) | 0 |
| Obsessive-compulsive disorder | 5 (0.4) | 9 (0.8) |
| Oppositional defiant disorder | 2 (0.2) | 3 (0.3) |
| Paediatric autoimmune neuropsychiatric disorders associated with streptococcal infection | 0 | 1 (0.1) |
| Persistent depressive disorder | 1 (0.1) | 0 |
| Post-traumatic stress disorder | 4 (0.4) | 5 (0.4) |
| Reactive attachment disorder of infancy or early childhood | 0 | 1 (0.1) |
| Reading disorder | 0 | 1 (0.1) |
| Separation anxiety disorder | 1 (0.1) | 0 |
| Sleep disorder | 0 | 3 (0.3) |
| Social anxiety disorder | 0 | 1 (0.1) |
| Speech sound disorder | 1 (0.1) | 0 |
| Suicidal ideation | 1 (0.1) | 1 (0.1) |
| Tic | 3 (0.3) | 2 (0.2) |
| Renal and urinary disorders | 5 (0.4) | 6 (0.5) |
| Dysuria | 0 | 1 (0.1) |
| Haematuria | 1 (0.1) | 1 (0.1) |
| Hydronephrosis | 1 (0.1) | 0 |
| Nephrolithiasis | 1 (0.1) | 0 |
| Renal cyst | 0 | 1 (0.1) |
| Renal disorder | 1 (0.1) | 0 |
| Single functional kidney | 0 | 1 (0.1) |
| Urinary retention | 0 | 1 (0.1) |
| Urinary tract disorder | 1 (0.1) | 0 |
| Vesicoureteric reflux | 1 (0.1) | 1 (0.1) |
| Reproductive system and breast disorders | 32 (2.8) | 35 (3.1) |
| Abnormal uterine bleeding | 2 (0.2) | 1 (0.1) |
| System Organ Class Preferred Term | Vaccine Group (as Administered) | |
| BNT162b2 (30 µg) (N°=1131) | Placebo (N°=1129) | |
| nh (%) | nh (%) | |
| Breast cyst | 1 (0.1) | 0 |
| Dysmenorrhoea | 11 (1.0) | 15 (1.3) |
| Epididymal cyst | 1 (0.1) | 0 |
| Gynaecomastia | 0 | 1 (0.1) |
| Heavy menstrual bleeding | 7 (0.6) | 9 (0.8) |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | |
|---|---|---|
| Intermenstrual bleeding | 0 | 1 (0.1) |
| Menstruation irregular | 4 (0.4) | 6 (0.5) |
| Ovulation disorder | 1 (0.1) | 0 |
| Polycystic ovaries | 3 (0.3) | 1 (0.1) |
| Premenstrual dysphoric disorder | 0 | 2 (0.2) |
| Testicular torsion | 1 (0.1) | 0 |
| Vaginal disorder | 1 (0.1) | 0 |
| Varicocele | 1 (0.1) | 1 (0.1) |
| Respiratory, thoracic and mediastinal disorders | 179 (15.8) | 179 (15.9) |
| Adenoidal hypertrophy | 1 (0.1) | 0 |
| Asthma | 108 (9.5) | 110 (9.7) |
| Asthma exercise induced | 11 (1.0) | 16 (1.4) |
| Bronchial hyperreactivity | 6 (0.5) | 4 (0.4) |
| Bronchitis chronic | 0 | 1 (0.1) |
| Bronchospasm | 0 | 1 (0.1) |
| Epistaxis | 6 (0.5) | 5 (0.4) |
| Nasal inflammation | 1 (0.1) | 0 |
| Nasal polyps | 0 | 1 (0.1) |
| Nasal septum deviation | 1 (0.1) | 1 (0.1) |
| Nasal turbinate hypertrophy | 3 (0.3) | 0 |
| Oropharyngeal pain | 1 (0.1) | 1 (0.1) |
| Rhinitis allergic | 41 (3.6) | 46 (4.1) |
| Rhinitis perennial | 1 (0.1) | 0 |
| Sleep apnoea syndrome | 5 (0.4) | 3 (0.3) |
| Snoring | 2 (0.2) | 0 |
| Tonsillar hypertrophy | 2 (0.2) | 1 (0.1) |
| Tonsillolith | 1 (0.1) | 0 |
| Tracheomalacia | 0 | 1 (0.1) |
| Vasomotor rhinitis | 1 (0.1) | 0 |
| Vocal cord disorder | 1 (0.1) | 0 |
| Vocal cord thickening | 1 (0.1) | 0 |
| Wheezing | 2 (0.2) | 0 |
| Skin and subcutaneous tissue disorders | 170 (15.0) | 182 (16.1) |
| System Organ Class Preferred Term | Vaccine Group (as Administered) | |
| BNT162b2 (30 µg) (N°=1131) | Placebo (N°=1129) | |
| nb (%) | nb (%) | |
| Acanthosis nigricans | 0 | 1 (0.1) |
| Acne | 96 (8.5) | 99 (8.8) |
| Acne cosmetica | 1 (0.1) | 0 |
| Acne cystic | 0 | 1 (0.1) |
| Actinic keratosis | 1 (0.1) | 1 (0.1) |
| Alopecia | 1 (0.1) | 0 |
| Alopecia areata | 1 (0.1) | 0 |
| Blister | 1 (0.1) | 0 |
| Dermatitis | 4 (0.4) | 1 (0.1) |
| Dermatitis allergic | 0 | 1 (0.1) |
| Dermatitis atopic | 10 (0.9) | 12 (1.1) |
| Dermatitis contact | 5 (0.4) | 4 (0.4) |
| Drug eruption | 0 | 3 (0.3) |
| Dry skin | 0 | 2 (0.2) |
| Eczema | 35 (3.1) | 44 (3.9) |
| Hand dermatitis | 8 (0.7) | 2 (0.2) |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | |
|---|---|---|
| Hirsutism | 0 | 1 (0.1) |
| Hyperhidrosis | 1 (0.1) | 3 (0.3) |
| Hyperkeratosis | 1 (0.1) | 0 |
| Idiopathic urticaria | 2 (0.2) | 0 |
| Ingrowing nail | 1 (0.1) | 1 (0.1) |
| Keratosis pilaris | 1 (0.1) | 2 (0.2) |
| Miliaria | 1 (0.1) | 0 |
| Nail psoriasis | 1 (0.1) | 0 |
| Pityriasis alba | 0 | 1 (0.1) |
| Pityriasis rosea | 0 | 1 (0.1) |
| Psoriasis | 6 (0.5) | 7 (0.6) |
| Rash | 0 | 2 (0.2) |
| Rosacea | 1 (0.1) | 1 (0.1) |
| Seborrhoea | 1 (0.1) | 0 |
| Spider naevus | 0 | 1 (0.1) |
| Urticaria | 8 (0.7) | 2 (0.2) |
| Vitiligo | 0 | 2 (0.2) |
| Social circumstances | 104 (9.2) | 95 (8.4) |
| Corrective lens user | 8 (0.7) | 10 (0.9) |
| Menarche | 11 (1.0) | 16 (1.4) |
| Premenarche | 88 (7.8) | 69 (6.1) |
| Vegan | 1 (0.1) | 0 |
| Vegetarian | 2 (0.2) | 6 (0.5) |
| System Organ Class Preferred Term | Vaccine Group (as Administered) | |
| BNT162b2 (30 µg) (N°=1131) | Placebo (N°=1129) | |
| n° (%) | n° (%) | |
| Woman of childbearing potential | 0 | 1 (0.1) |
| Surgical and medical procedures | 111 (9.8) | 117 (10.4) |
| Abdominal hernia repair | 1 (0.1) | 0 |
| Abscess drainage | 0 | 2 (0.2) |
| Adenoidectomy | 33 (2.9) | 22 (1.9) |
| Adenotonsillectomy | 1 (0.1) | 2 (0.2) |
| Ankle operation | 0 | 1 (0.1) |
| Anorectal operation | 0 | 1 (0.1) |
| Appendicectomy | 9 (0.8) | 10 (0.9) |
| Arterial switch operation | 0 | 1 (0.1) |
| Atrial septal defect repair | 1 (0.1) | 1 (0.1) |
| Bone operation | 1 (0.1) | 1 (0.1) |
| Cardiac ablation | 0 | 1 (0.1) |
| Cardiac operation | 0 | 1 (0.1) |
| Cataract operation | 0 | 1 (0.1) |
| Cautery to nose | 1 (0.1) | 1 (0.1) |
| Central venous catheterisation | 1 (0.1) | 0 |
| Cerebral cyst excision | 1 (0.1) | 0 |
| Cholecystectomy | 1 (0.1) | 0 |
| Chondroplasty | 1 (0.1) | 0 |
| Circumcision | 2 (0.2) | 4 (0.4) |
| Colon operation | 0 | 1 (0.1) |
| Dacryocystorhinostomy | 0 | 1 (0.1) |
| Ear operation | 0 | 1 (0.1) |
| Ear tube insertion | 12 (1.1) | 10 (0.9) |
| Ear tube removal | 1 (0.1) | 1 (0.1) |
| Elbow operation | 0 | 1 (0.1) |
| Epiphyseal surgery | 1 (0.1) | 0 |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | |
|---|---|---|
| Epiphyisodesis | 1 (0.1) | 0 |
| Eye operation | 2 (0.2) | 4 (0.4) |
| Facial lesion excision | 1 (0.1) | 1 (0.1) |
| Finger amputation | 1 (0.1) | 1 (0.1) |
| Foot operation | 0 | 1 (0.1) |
| Fracture treatment | 3 (0.3) | 5 (0.4) |
| Hernia diaphragmatic repair | 1 (0.1) | 1 (0.1) |
| Hernia repair | 2 (0.2) | 0 |
| Hip surgery | 0 | 1 (0.1) |
| Hydrocele operation | 1 (0.1) | 0 |
| Hymenectomy | 1 (0.1) | 0 |
| System Organ Class Preferred Term | Vaccine Group (as Administered) | |
| BNT162b2 (30 µg) (N°=1131) | Placebo (N°=1129) | |
| nb (%) | nb (%) | |
| Inguinal hernia repair | 1 (0.1) | 1 (0.1) |
| Intestinal operation | 1 (0.1) | 0 |
| Intrauterine contraception | 2 (0.2) | 0 |
| Jaw operation | 1 (0.1) | 0 |
| Joint stabilisation | 0 | 1 (0.1) |
| Knee operation | 2 (0.2) | 0 |
| Lacrimal duct procedure | 0 | 1 (0.1) |
| Ligament operation | 1 (0.1) | 0 |
| Limb operation | 3 (0.3) | 1 (0.1) |
| Limb reconstructive surgery | 0 | 1 (0.1) |
| Lipoma excision | 0 | 1 (0.1) |
| Liposuction | 0 | 1 (0.1) |
| Lymphadenectomy | 0 | 1 (0.1) |
| Mass excision | 1 (0.1) | 0 |
| Mastoidectomy | 1 (0.1) | 0 |
| Medical device change | 0 | 1 (0.1) |
| Medical diet | 1 (0.1) | 0 |
| Meniscus operation | 1 (0.1) | 0 |
| Middle ear operation | 1 (0.1) | 0 |
| Mole excision | 0 | 1 (0.1) |
| Myringotomy | 11 (1.0) | 8 (0.7) |
| Nail operation | 1 (0.1) | 0 |
| Nephrectomy | 0 | 1 (0.1) |
| Oesophagogastric fundoplasty | 0 | 1 (0.1) |
| Open reduction of fracture | 2 (0.2) | 1 (0.1) |
| Orchidopexy | 1 (0.1) | 0 |
| Ostectomy | 1 (0.1) | 0 |
| Pharyngeal reconstruction | 0 | 1 (0.1) |
| Pilonidal sinus repair | 0 | 1 (0.1) |
| Portal shunt procedure | 0 | 1 (0.1) |
| Removal of foreign body from external ear | 0 | 1 (0.1) |
| Rhinoplasty | 0 | 1 (0.1) |
| Scoliosis surgery | 1 (0.1) | 1 (0.1) |
| Scrotal cystectomy | 0 | 1 (0.1) |
| Sinuplasty | 0 | 1 (0.1) |
| Skin lesion removal | 1 (0.1) | 0 |
| Spinal fusion surgery | 2 (0.2) | 1 (0.1) |
| Strabismus correction | 2 (0.2) | 2 (0.2) |
| Suture insertion | 1 (0.1) | 0 |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | |
|---|---|---|
| Temporomandibular joint surgery | 0 | 1 (0.1) |
| System Organ Class Preferred Term | Vaccine Group (as Administered) | |
| BNT162b2 (30 µg) (N°=1131) | Placebo (N°=1129) | |
| n° (%) | n° (%) | |
| Testes exploration | 1 (0.1) | 0 |
| Testicular operation | 1 (0.1) | 0 |
| Thyroglossal cyst excision | 1 (0.1) | 0 |
| Toe operation | 1 (0.1) | 0 |
| Tonsillectomy | 33 (2.9) | 31 (2.7) |
| Tooth extraction | 1 (0.1) | 1 (0.1) |
| Transgender hormonal therapy | 0 | 1 (0.1) |
| Turbinectomy | 1 (0.1) | 0 |
| Turbinoplasty | 1 (0.1) | 0 |
| Tympanoplasty | 1 (0.1) | 0 |
| Umbilical hernia repair | 2 (0.2) | 3 (0.3) |
| Urethral repair | 1 (0.1) | 1 (0.1) |
| Urinary tract operation | 0 | 1 (0.1) |
| Vitrectomy | 0 | 1 (0.1) |
| Wisdom teeth removal | 2 (0.2) | 5 (0.4) |
| Vascular disorders | 3 (0.3) | 2 (0.2) |
| Hypertension | 0 | 1 (0.1) |
| Hypotension | 1 (0.1) | 1 (0.1) |
| Peripheral venous disease | 1 (0.1) | 0 |
| Raynaudś phenomenon | 1 (0.1) | 0 |

Note: MedDRA (v24.0) coding dictionary applied. a. N = number of subjects in the specified group. This value is the denominator for the percentage calculations. b. n = Number of subjects with the specified characteristic. Subjects with multiple occur...

## 14.4 Baseline Charlson Comorbidities – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population

**Table 14.4.** Baseline Charlson Comorbidities – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population

| Charlson Comorbidity Index Category | Vaccine Group (as Administered) | |
|---|---|---|
| BNT162b2 (30 µg) (N°=1131) nb (%) | Placebo (N°=1129) nb (%) | |
| Subjects with any Charlson comorbidity | 123 (10.9) | 136 (12.0) |
| Any malignancy | 0 | 1 (0.1) |
| Chronic pulmonary disease | 119 (10.5) | 127 (11.2) |
| Diabetes without chronic complication | 2 (0.2) | 5 (0.4) |
| Hemiplegia or paraplegia | 0 | 1 (0.1) |
| Mild liver disease | 2 (0.2) | 0 |
| Rheumatic disease | 0 | 2 (0.2) |

Note: MedDRA (v24.0) coding dictionary applied.a. N = number of subjects in the specified group. This value is the denominator for the percentage calculations.b. n = Number of subjects with the specified characteristic. Subjects with multiple occurre...

## 14.5 Demographic Characteristics – Subjects With at Least 6 Months of Follow-up Time After Dose 2 – Phase 2/3 Subjects 12 Through 15 Years of Age (Subjects Who Originally Received BNT162b2) – Safety Population

**Table 14.5.** Demographic Characteristics – Subjects With at Least 6 Months of Follow-up Time After Dose 2 – Phase 2/3 Subjects 12 Through 15 Years of Age (Subjects Who Originally Received BNT162b2) – Safety Population

| | Vaccine Group (as Administered) |
|---|---|
| | BNT162b2 (30 µg)(N=1113)n° (%) |
| Sex | |
| Male | 553 (49.7) |
| Female | 560 (50.3) |
| Race | |
| White | 955 (85.8) |
| Black or African American | 51 (4.6) |
| All others | 107 (9.6) |
| American Indian or Alaska Native | 4 (0.4) |
| Asian | 70 (6.3) |
| Native Hawaiian or other Pacific Islander | 3 (0.3) |
| Multiracial | 24 (2.2) |
| Not reported | 6 (0.5) |
| Racial designation | |
| Japanese | 5 (0.4) |
| Ethnicity | |
| Hispanic/Latino | 130 (11.7) |
| Non-Hispanic/non-Latino | 981 (88.1) |
| Not reported | 2 (0.2) |
| Country | |
| USA | 1113 (100.0) |
| Baseline SARS-CoV-2 status | |
| Positivec | 45 (4.0) |
| Negatived | 1066 (95.8) |
| Missing | 2 (0.2) |
| Comorbiditiese | |
| Yes | 243 (21.8) |
| No | 870 (78.2) |
| Obesef | |
| Yes | 140 (12.6) |
| No | 973 (87.4) |
| Age at vaccination (years) | |
| Mean (SD) | 13.6 (1.11) |
| Median | 14.0 |
| Min, max | (12, 15) |
| Vaccine Group (as Administered) | |
| BNT162b2 (30 µg) (N°=1113) n° (%) | |

Abbreviation: SARS-CoV-2 = severe acute respiratory syndrome coronavirus 2. a. N = number of subjects in the specified group. This value is the denominator for the percentage calculations. b. n = Number of subjects with the specified characteristic. ...

## 14.6 Demographic Characteristics – Subjects Who Originally Received Placebo and Then Received BNT162b2 After Unblinding – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population

**Table 14.6.** Demographic Characteristics – Subjects Who Originally Received Placebo and Then Received BNT162b2 After Unblinding – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population

| | Vaccine Group (as Administered) |
|---|---|
| | BNT162b2 (30 µg)(N=1010)n° (%) |
| Sex | |
| Male | 518 (51.3) |
| Female | 492 (48.7) |
| Race | |
| White | 866 (85.7) |
| Black or African American | 48 (4.8) |
| All others | 96 (9.5) |
| American Indian or Alaska Native | 2 (0.2) |
| Asian | 62 (6.1) |
| Multiracial | 26 (2.6) |
| Not reported | 6 (0.6) |
| Racial designation | |
| Japanese | 2 (0.2) |
| Ethnicity | |
| Hispanic/Latino | 115 (11.4) |
| Non-Hispanic/non-Latino | 892 (88.3) |
| Not reported | 3 (0.3) |
| Country | |
| USA | 1010 (100.0) |
| Baseline SARS-CoV-2 status | |
| Positivec | 43 (4.3) |
| Negatived | 966 (95.6) |
| Missing | 1 (0.1) |
| Comorbiditiese | |
| Yes | 215 (21.3) |
| No | 795 (78.7) |
| Obesef | |
| Yes | 116 (11.5) |
| No | 894 (88.5) |
| Age at vaccination (years) | |
| Mean (SD) | 13.6 (1.11) |
| Median | 14.0 |
| Min, max | (12, 15) |
| Vaccine Group (as Administered) | |
| BNT162b2 (30 µg) (Na=1010) nb (%) | |

## 14.7 Demographic Characteristics – Blinded Placebo-Controlled Follow-up Period – Subjects 12 Through 15 Years of Age and Without Evidence of Infection Prior to 7 Days After Dose 2 – Evaluable Efficacy (7 Days) Population

**Table 14.7.** Demographic Characteristics – Blinded Placebo-Controlled Follow-up Period – Subjects 12 Through 15 Years of Age and Without Evidence of Infection Prior to 7 Days After Dose 2 – Evaluable Efficacy (7 Days) Population

| | Vaccine Group (as Randomized) | | Total (N°=2087) nb (%) |
|---|---|---|---|
| BNT162b2 (30 µg) (N°=1057) nb (%) | Placebo (N°=1030) nb (%) | | |
| Sex | | | |
| Male | 530 (50.1) | 533 (51.7) | 1063 (50.9) |
| Female | 527 (49.9) | 497 (48.3) | 1024 (49.1) |
| Race | | | |
| White | 909 (86.0) | 874 (84.9) | 1783 (85.4) |
| Black or African American | 44 (4.2) | 50 (4.9) | 94 (4.5) |
| All others | 104 (9.8) | 106 (10.3) | 210 (10.1) |
| American Indian or Alaska Native | 4 (0.4) | 2 (0.2) | 6 (0.3) |
| Asian | 68 (6.4) | 69 (6.7) | 137 (6.6) |
| Native Hawaiian or other Pacific Islander | 3 (0.3) | 0 | 3 (0.1) |
| Multiracial | 23 (2.2) | 29 (2.8) | 52 (2.5) |
| Not reported | 6 (0.6) | 6 (0.6) | 12 (0.6) |
| Ethnicity | | | |
| Hispanic/Latino | 121 (11.4) | 113 (11.0) | 234 (11.2) |
| Non-Hispanic/non-Latino | 934 (88.4) | 914 (88.7) | 1848 (88.5) |
| Not reported | 2 (0.2) | 3 (0.3) | 5 (0.2) |
| Country | | | |
| USA | 1057 (100.0) | 1030 (100.0) | 2087 (100.0) |
| Comorbiditiesc | | | |
| Yes | 233 (22.0) | 216 (21.0) | 449 (21.5) |
| No | 824 (78.0) | 814 (79.0) | 1638 (78.5) |
| Obesed | | | |
| Yes | 135 (12.8) | 111 (10.8) | 246 (11.8) |
| No | 922 (87.2) | 919 (89.2) | 1841 (88.2) |
| Age at vaccination (years) | | | |
| Mean (SD) | 13.6 (1.10) | 13.6 (1.11) | 13.6 (1.11) |
| Median | 14.0 | 14.0 | 14.0 |
| Min, max | (12, 15) | (12, 15) | (12, 15) |
| | Vaccine Group (as Randomized) | | |
| | BNT162b2 (30 µg) (N°=1057) nb (%) | Placebo (N°=1030) nb (%) | Total (N°=2087) nb (%) |

a. N = number of subjects in the specified group, or the total sample. This value is the denominator for the percentage calculations. b. n = Number of subjects with the specified characteristic. c. Number of subjects who have 1 or more comorbidities ...

## 14.8 Demographic Characteristics – Blinded Placebo-Controlled Follow-up Period – Subjects 12 Through 15 Years of Age – Dose 1 All-Available Efficacy Population

**Table 14.8.** Demographic Characteristics – Blinded Placebo-Controlled Follow-up Period – Subjects 12 Through 15 Years of Age – Dose 1 All-Available Efficacy Population

| | Vaccine Group (as Randomized) | | Total (N*=2260) nb (%) |
|---|---|---|---|
| BNT162b2 (30 µg) (N*=1131) nb (%) | Placebo (N*=1129) nb (%) | | |
| Sex | | | |
| Male | 567 (50.1) | 585 (51.8) | 1152 (51.0) |
| Female | 564 (49.9) | 544 (48.2) | 1108 (49.0) |
| Race | | | |
| White | 970 (85.8) | 962 (85.2) | 1932 (85.5) |
| Black or African American | 52 (4.6) | 57 (5.0) | 109 (4.8) |
| All others | 109 (9.6) | 110 (9.7) | 219 (9.7) |
| American Indian or Alaska Native | 4 (0.4) | 3 (0.3) | 7 (0.3) |
| Asian | 72 (6.4) | 71 (6.3) | 143 (6.3) |
| Native Hawaiian or other Pacific Islander | 3 (0.3) | 0 | 3 (0.1) |
| Multiracial | 24 (2.1) | 29 (2.6) | 53 (2.3) |
| Not reported | 6 (0.5) | 7 (0.6) | 13 (0.6) |
| Ethnicity | | | |
| Hispanic/Latino | 132 (11.7) | 130 (11.5) | 262 (11.6) |
| Non-Hispanic/non-Latino | 997 (88.2) | 996 (88.2) | 1993 (88.2) |
| Not reported | 2 (0.2) | 3 (0.3) | 5 (0.2) |
| Country | | | |
| USA | 1131 (100.0) | 1129 (100.0) | 2260 (100.0) |
| Comorbiditiesc | | | |
| Yes | 249 (22.0) | 242 (21.4) | 491 (21.7) |
| No | 882 (78.0) | 887 (78.6) | 1769 (78.3) |
| Obesed | | | |
| Yes | 143 (12.6) | 128 (11.3) | 271 (12.0) |
| No | 988 (87.4) | 1001 (88.7) | 1989 (88.0) |
| Baseline SARS-CoV-2 status | | | |
| Positivee | 46 (4.1) | 50 (4.4) | 96 (4.2) |
| Negativef | 1083 (95.8) | 1078 (95.5) | 2161 (95.6) |
| Unknown | 2 (0.2) | 1 (0.1) | 3 (0.1) |
| Age at vaccination (years) | | | |
| Mean (SD) | 13.6 (1.11) | 13.6 (1.11) | 13.6 (1.11) |
| Median | 14.0 | 14.0 | 14.0 |
| Min, max | (12, 15) | (12, 15) | (12, 15) |
| | Vaccine Group (as Randomized) | | |
| BNT162b2 (30 µg) (Nα=1131) nb (%) | Placebo (Nα=1129) nb (%) | Total (Nα=2260) nb (%) | |

a. N = number of subjects in the specified group, or the total sample. This value is the denominator for the percentage calculations. b. n = Number of subjects with the specified characteristic. c. Number of subjects who have 1 or more comorbidities ...

## 14.9 Demographic Characteristics – Blinded Placebo-Controlled Follow-up Period – Subjects 12 Through 15 Years of Age and With or Without Evidence of Infection Prior to 7 Days After Dose 2 – Evaluable Efficacy (7 Days) Population

**Table 14.9.** Demographic Characteristics – Blinded Placebo-Controlled Follow-up Period – Subjects 12 Through 15 Years of Age and With or Without Evidence of Infection Prior to 7 Days After Dose 2 – Evaluable Efficacy (7 Days) Population

| | Vaccine Group (as Randomized) | | Total (N |
|---|---|---|---|
| (^a<br>)=2228) n<br>(^b<br>) (%) | | | |
| BNT162b2 (30 µg) (N<br>(^a<br>)=1119) n<br>(^b<br>) (%) | | | |
| | Placebo (N<br>(^a<br>)=1109) n<br>(^b<br>) (%) | | |
| Sex | | | |
| Male | 559 (50.0) | 573 (51.7) | 1132 (50.8) |
| Female | 560 (50.0) | 536 (48.3) | 1096 (49.2) |
| Race | | | |
| White | 961 (85.9) | 943 (85.0) | 1904 (85.5) |
| Black or African American | 50 (4.5) | 57 (5.1) | 107 (4.8) |
| All others | 108 (9.7) | 109 (9.8) | 217 (9.7) |
| American Indian or Alaska Native | 4 (0.4) | 2 (0.2) | 6 (0.3) |
| Asian | 71 (6.3) | 71 (6.4) | 142 (6.4) |
| Native Hawaiian or other Pacific Islander | 3 (0.3) | 0 | 3 (0.1) |
| Multiracial | 24 (2.1) | 29 (2.6) | 53 (2.4) |
| Not reported | 6 (0.5) | 7 (0.6) | 13 (0.6) |
| Ethnicity | | | |
| Hispanic/Latino | 131 (11.7) | 127 (11.5) | 258 (11.6) |
| Non-Hispanic/non-Latino | 986 (88.1) | 979 (88.3) | 1965 (88.2) |
| Not reported | 2 (0.2) | 3 (0.3) | 5 (0.2) |
| Country | | | |
| USA | 1119 (100.0) | 1109 (100.0) | 2228 (100.0) |
| Comorbiditiesc | | | |
| Yes | 244 (21.8) | 236 (21.3) | 480 (21.5) |
| No | 875 (78.2) | 873 (78.7) | 1748 (78.5) |
| Obesed | | | |
| Yes | 141 (12.6) | 125 (11.3) | 266 (11.9) |
| No | 978 (87.4) | 984 (88.7) | 1962 (88.1) |
| Baseline SARS-CoV-2 status | | | |
| Positivee | 46 (4.1) | 49 (4.4) | 95 (4.3) |
| Negativef | 1071 (95.7) | 1059 (95.5) | 2130 (95.6) |
| Unknown | 2 (0.2) | 1 (0.1) | 3 (0.1) |
| Age at vaccination (years) | | | |
| Mean (SD) | 13.6 (1.11) | 13.6 (1.11) | 13.6 (1.11) |
| Median | 14.0 | 14.0 | 14.0 |

| | Vaccine Group (as Randomized) | | Total (N |
|---|---|---|---|
| (^a<br>)=2228) n<br>(^b<br>) (%) | | | |
| Min, max | (12, 15) | (12, 15) | (12, 15) |
| | Vaccine Group (as Randomized)<br>BNT162b2 (30 µg) (N=1119) nb (%) | Placebo (N=1109) nb (%) | Total (N=2228) nb (%) |

a. N = number of subjects in the specified group, or the total sample. This value is the denominator for the percentage calculations. b. n = Number of subjects with the specified characteristic. c. Number of subjects who have 1 or more comorbidities ...

## 14.10 Concomitant Vaccines Received After Dose 1 – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population

**Table 14.10.** Concomitant Vaccines Received After Dose 1 – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population

| Vaccineb | Vaccine Group (as Administered) | |
|---|---|---|
| BNT162b2 (30 µg) (Nα=1131) n° (%) | Placebo (Nα=1129) n° (%) | |
| Any concomitant vaccine | 32 (2.8) | 31 (2.7) |
| DIPHTHERIA VACCINE TOXOID;PERTUSSIS VACCINE ACELLULAR;TETANUS VACCINE TOXOID | 3 (0.3) | 0 |
| HPV VACCINE | 3 (0.3) | 10 (0.9) |
| HPV VACCINE VLP RL1 4V (YEAST) | 4 (0.4) | 2 (0.2) |
| INFLUENZA VACCINE | 12 (1.1) | 5 (0.4) |
| INFLUENZA VACCINE INACT SPLIT 4V | 2 (0.2) | 1 (0.1) |
| MENINGOCOCCAL VACCINE | 3 (0.3) | 7 (0.6) |
| MENINGOCOCCAL VACCINE A/C/Y/W | 1 (0.1) | 1 (0.1) |
| MENINGOCOCCAL VACCINE A/C/Y/W CONJ (DIP TOX) | 3 (0.3) | 4 (0.4) |
| MENINGOCOCCAL VACCINE B | 1 (0.1) | 0 |
| MENINGOCOCCAL VACCINE B RFHBP/NADA/NHBA OMV | 1 (0.1) | 0 |
| MENINGOCOCCAL VACCINE B RFHBPA/FHBPB | 0 | 2 (0.2) |
| MENINGOCOCCAL VACCINE CONJ | 1 (0.1) | 1 (0.1) |
| POLIO VACCINE | 0 | 1 (0.1) |
| RABIES VACCINE | 1 (0.1) | 0 |
| TETANUS VACCINE | 0 | 1 (0.1) |

Note: WHODDG B3 v202103 coding dictionary applied. a. N = number of subjects in the specified group. This value is the denominator for the percentage calculations. b. Subjects are counted only once for each preferred term. c. n = Number of subjects w...

## 14.11 Vaccine Efficacy – First COVID-19 Occurrence From 7 Days After Dose 2 – Blinded Placebo-Controlled Follow-up Period – Subjects 12 Through 15 Years of Age and Without Evidence of Infection Prior to 7 Days After Dose 2 – Dose 2 All-Available Efficacy Population

**Table 14.11.** Vaccine Efficacy – First COVID-19 Occurrence From 7 Days After Dose 2 – Blinded Placebo-Controlled Follow-up Period – Subjects 12 Through 15 Years of Age and Without Evidence of Infection Prior to 7 Days After Dose 2 – Dose 2 All-Available Efficacy Population

| Efficacy Endpoint Subgroup | Vaccine Group (as Randomized) | | | | VE (%) | (95% CIc) |
|---|---|---|---|---|---|---|
| | BNT162b2 (30 µg) (Na=1061) | Placebo (Na=1037) | | | | |
| n1b | Surveillance Timec (n2d) | n1b | Surveillance Timec (n2d) | | | |
| First COVID-19 occurrence from 7 days after Dose 2 | 0 | 0.345 (1047) | 29 | 0.325 (1026) | 100.0 | (87.2, 100.0) |
| ≥7 days after Dose 2 to <2 Months after Dose 2 | 0 | 0.139 (1047) | 15 | 0.134 (1026) | 100.0 | (73.1, 100.0) |
| ≥2 Months after Dose 2 to <4 Months after Dose 2 | 0 | 0.148 (1012) | 10 | 0.140 (964) | 100.0 | (57.9, 100.0) |
| ≥4 Months after Dose 2 | 0 | 0.058 (726) | 4 | 0.051 (688) | 100.0 | (-34.2, 100.0) |

## 14.12 Vaccine Efficacy – First COVID-19 Occurrence From 7 Days After Dose 2, by Subgroup – Blinded Placebo-Controlled Follow-up Period – Subjects 12 Through 15 Years of Age and With or Without Evidence of Infection Prior to 7 Days After Dose 2 – Evaluable Efficacy (7 Days) Population

**Table 14.12.** Vaccine Efficacy – First COVID-19 Occurrence From 7 Days After Dose 2, by Subgroup – Blinded Placebo-Controlled Follow-up Period – Subjects 12 Through 15 Years of Age and With or Without Evidence of Infection Prior to 7 Days After Dose 2 – Evaluable Efficacy (7 Days) Population

| Efficacy Endpoint Subgroup | Vaccine Group (as Randomized) | | | | VE (%) | (95% CI#) |
|---|---|---|---|---|---|---|
| | BNT162b2 (30 µg) (Nα=1119) | | Placebo (Nα=1109) | | | |
| | n1b | Surveillance Timec (n2d) | n1b | Surveillance Timec (n2d) | | |
| First COVID-19 occurrence from 7 days after Dose 2 | | | | | | |
| Overall | 0 | 0.362 (1098) | 30 | 0.345 (1088) | 100.0 | (87.5, 100.0) |
| Sex | | | | | | |
| Male | 0 | 0.183 (550) | 18 | 0.177 (561) | 100.0 | (78.0, 100.0) |
| Female | 0 | 0.179 (548) | 12 | 0.169 (527) | 100.0 | (66.1, 100.0) |
| Race | | | | | | |
| White | 0 | 0.309 (945) | 28 | 0.291 (926) | 100.0 | (86.8, 100.0) |
| Black or African American | 0 | 0.019 (47) | 2 | 0.021 (56) | 100.0 | (-492.9, 100.0) |
| Ethnicity | | | | | | |
| Hispanic/Latino | 0 | 0.045 (127) | 7 | 0.040 (125) | 100.0 | (37.8, 100.0) |
| Non-Hispanic/non-Latino | 0 | 0.317 (969) | 23 | 0.304 (960) | 100.0 | (83.3, 100.0) |
| Country | | | | | | |
| USA | 0 | 0.362 (1098) | 30 | 0.345 (1088) | 100.0 | (87.5, 100.0) |
| Comorbiditiesf | | | | | | |
| Yes | 0 | 0.082 (241) | 11 | 0.073 (228) | 100.0 | (64.6, 100.0) |
| No | 0 | 0.280 (857) | 19 | 0.273 (860) | 100.0 | (79.2, 100.0) |
| Obeseg | | | | | | |
| Yes | 0 | 0.048 (140) | 7 | 0.039 (122) | 100.0 | (43.1, 100.0) |
| No | 0 | 0.314 (958) | 23 | 0.306 (966) | 100.0 | (83.1, 100.0) |
| Prior SARS-CoV-2 Status | | | | | | |
| Negative at baseline but positive prior to 7 days after Dose 2h | 0 | 0.001 (3) | 2 | 0.004 (11) | 100.0 | (-1374.1, 100.0) |
| Negative prior to 7 days after Dose 2i | 0 | 0.343 (1043) | 28 | 0.322 (1019) | 100.0 | (86.8, 100.0) |
| Efficacy Endpoint Subgroup | Vaccine Group (as Randomized) | | | | VE (%) | (95% CI#) |

| Efficacy Endpoint Subgroup | Vaccine Group (as Randomized) | | VE (%) | (95% CI#) |
|---|---|---|---|---|
| BNT162b2 (30 µg) (Na=1119) n1b Surveil-lance Timea (n2d) | Placebo (Na=1109) n1b Surveil-lance Timea (n2d) | VE (%) | | |
| Abbreviations: N-binding = SARS-CoV-2 nucleoprotein–binding; NAAT = nucleic acid amplification test; SARS-CoV-2 = severe acute respiratory syndrome coronavirus 2; VE = vaccine efficacy. a. N = number of subjects in the specified group. b. n1 = Number... | | | | |

## 14.13 Vaccine Efficacy – First COVID-19 Occurrence After Dose 1, by Subgroup – Blinded Placebo-Controlled Follow-up Period – Subjects 12 Through 15 Years of Age – Dose 1 All-Available Efficacy Population

**Table 14.13.** Vaccine Efficacy – First COVID-19 Occurrence After Dose 1, by Subgroup – Blinded Placebo-Controlled Follow-up Period – Subjects 12 Through 15 Years of Age – Dose 1 All-Available Efficacy Population

| Efficacy Endpoint Subgroup | Vaccine Group (as Randomized) | | | | VE (%) | (95% CI#) |
|---|---|---|---|---|---|---|
| | BNT162b2 (30 µg) (N=1131) | | Placebo (N=1129) | | | |
| | n1b | Surveillance Timec (n2d) | n1b | Surveillance Timec (n2d) | | |
| First COVID-19 occurrence after Dose 1 | | | | | | |
| Overall | 3 | 0.450 (1109) | 48 | 0.434 (1114) | 94.0 | (81.3, 98.8) |
| Sex | | | | | | |
| Male | 3 | 0.227 (557) | 26 | 0.223 (575) | 88.7 | (63.0, 97.8) |
| Female | 0 | 0.223 (552) | 22 | 0.211 (539) | 100.0 | (82.7, 100.0) |
| Race | | | | | | |
| White | 2 | 0.384 (953) | 45 | 0.367 (951) | 95.8 | (83.8, 99.5) |
| Black or African American | 0 | 0.024 (49) | 2 | 0.026 (56) | 100.0 | (-479.7, 100.0) |
| All others | 1 | 0.042 (107) | 1 | 0.042 (107) | 1.4 | (-7638.0, 98.7) |
| American Indian or Alaska native | 0 | 0.002 (4) | 1 | 0.001 (3) | 100.0 | (-1965.1, 100.0) |
| Asian | 1 | 0.027 (70) | 0 | 0.027 (69) | UND | (NA, NA) |
| Ethnicity | | | | | | |
| Hispanic/Latino | 1 | 0.055 (128) | 11 | 0.051 (130) | 91.6 | (42.3, 99.8) |
| Non-Hispanic/non-Latino | 2 | 0.394 (979) | 37 | 0.382 (981) | 94.8 | (79.7, 99.4) |
| Country | | | | | | |
| USA | 3 | 0.450 (1109) | 48 | 0.434 (1114) | 94.0 | (81.3, 98.8) |
| Comorbiditiesf | | | | | | |
| Yes | 1 | 0.102 (246) | 18 | 0.092 (238) | 95.0 | (68.3, 99.9) |
| No | 2 | 0.348 (863) | 30 | 0.342 (876) | 93.4 | (74.2, 99.2) |
| Obeseg | | | | | | |
| Yes | 0 | 0.060 (142) | 11 | 0.049 (127) | 100.0 | (67.0, 100.0) |
| No | 3 | 0.391 (967) | 37 | 0.385 (987) | 92.0 | (74.8, 98.4) |
| Baseline SARS-CoV-2 status | | | | | | |
| Positiveh | 0 | 0.019 (45) | 1 | 0.021 (50) | 100.0 | (-4202.3, 100.0) |
| Positive NAAT only | 0 | 0.002 (6) | 1 | 0.004 (10) | 100.0 | (-5725.9, 100.0) |
| Negativel | 3 | 0.431 (1062) | 47 | 0.413 (1063) | 93.9 | (81.0, 98.8) |
| Efficacy Endpoint Subgroup | Vaccine Group (as Randomized) | | | | VE (%) | (95% CI#) |

| Efficacy Endpoint Subgroup | Vaccine Group (as Randomized) | VE (%) | (95% CI#) |
|---|---|---|---|
| BNT162b2 (30 µg) (N=1131) n1b Surveillance Timec (n2d) | Placebo (N=1129) n1b Surveillance Timec (n2d) | | |
| Abbreviations: N-binding = SARS-CoV-2 nucleoprotein–binding; NAAT = nucleic acid amplification test; SARS-CoV-2 = severe acute respiratory syndrome coronavirus 2; UND=Undefined; VE = vaccine efficacy. a. N = number of subjects in the specified group.... | | | |

## 14.14 Incidence Rates of at Least 1 Adverse Event From Dose 1 to Unblinding Date, by Baseline SARS-CoV-2 Status – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Baseline SARS-CoV-2 Status: Positive

**Table 14.14.** Incidence Rates of at Least 1 Adverse Event From Dose 1 to Unblinding Date, by Baseline SARS-CoV-2 Status – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Baseline SARS-CoV-2 Status: Positive

| Adverse Event | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| | BNT162b2 (30 µg) (Nα=46, TEb=0.2) | | | Placebo (Nα=50, TEb=0.2) | | |
| n° (%) | IRd | (95% CIe) | n° (%) | IRd | (95% CIe) | |
| Any event | 4 (8.7) | 20.8 | (5.7, 53.3) | 4 (8.0) | 19.2 | (5.2, 49.1) |
| Relatedf | 1 (2.2) | 5.2 | (0.1, 29.0) | 2 (4.0) | 9.6 | (1.2, 34.6) |
| Severe | 0 | 0.0 | (0.0, 19.2) | 0 | 0.0 | (0.0, 17.7) |
| Life-threatening | 0 | 0.0 | (0.0, 19.2) | 0 | 0.0 | (0.0, 17.7) |
| Any serious adverse event | 0 | 0.0 | (0.0, 19.2) | 0 | 0.0 | (0.0, 17.7) |
| Relatedf | 0 | 0.0 | (0.0, 19.2) | 0 | 0.0 | (0.0, 17.7) |
| Severe | 0 | 0.0 | (0.0, 19.2) | 0 | 0.0 | (0.0, 17.7) |
| Life-threatening | 0 | 0.0 | (0.0, 19.2) | 0 | 0.0 | (0.0, 17.7) |
| Any nonserious adverse event | 4 (8.7) | 20.8 | (5.7, 53.3) | 4 (8.0) | 19.2 | (5.2, 49.1) |
| Relatedf | 1 (2.2) | 5.2 | (0.1, 29.0) | 2 (4.0) | 9.6 | (1.2, 34.6) |
| Severe | 0 | 0.0 | (0.0, 19.2) | 0 | 0.0 | (0.0, 17.7) |
| Life-threatening | 0 | 0.0 | (0.0, 19.2) | 0 | 0.0 | (0.0, 17.7) |
| Any adverse event leading to withdrawal | 0 | 0.0 | (0.0, 19.2) | 0 | 0.0 | (0.0, 17.7) |
| Relatedf | 0 | 0.0 | (0.0, 19.2) | 0 | 0.0 | (0.0, 17.7) |
| Severe | 0 | 0.0 | (0.0, 19.2) | 0 | 0.0 | (0.0, 17.7) |
| Life-threatening | 0 | 0.0 | (0.0, 19.2) | 0 | 0.0 | (0.0, 17.7) |
| Death | 0 | 0.0 | (0.0, 19.2) | 0 | 0.0 | (0.0, 17.7) |

## 14.15 Incidence Rates of at Least 1 Adverse Event From Dose 1 to Unblinding Date, by Baseline SARS-CoV-2 Status – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Baseline SARS-CoV-2 Status: Negative

**Table 14.15.** Incidence Rates of at Least 1 Adverse Event From Dose 1 to Unblinding Date, by Baseline SARS-CoV-2 Status – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Baseline SARS-CoV-2 Status: Negative

| Adverse Event | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| | BNT162b2 (30 µg) (Nα=1083, TEb=4.4) | | | Placebo (Nα=1078, TEb=4.3) | | |
| nc(%) | IRd | (95% CIe) | nc(%) | IRd | (95% CIe) | |
| Any event | 91 (8.4) | 20.8 | (16.8, 25.6) | 109 (10.1) | 25.4 | (20.8, 30.6) |
| Relatedf | 35 (3.2) | 8.0 | (5.6, 11.1) | 22 (2.0) | 5.1 | (3.2, 7.8) |
| Severe | 13 (1.2) | 3.0 | (1.6, 5.1) | 5 (0.5) | 1.2 | (0.4, 2.7) |
| Life-threatening | 2 (0.2) | 0.5 | (0.1, 1.7) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Any serious adverse event | 10 (0.9) | 2.3 | (1.1, 4.2) | 2 (0.2) | 0.5 | (0.1, 1.7) |
| Relatedf | 0 | 0.0 | (0.0, 0.8) | 0 | 0.0 | (0.0, 0.9) |
| Severe | 7 (0.6) | 1.6 | (0.6, 3.3) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Life-threatening | 1 (0.1) | 0.2 | (0.0, 1.3) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Any nonserious adverse event | 85 (7.8) | 19.4 | (15.5, 24.0) | 107 (9.9) | 24.9 | (20.4, 30.1) |
| Relatedf | 35 (3.2) | 8.0 | (5.6, 11.1) | 22 (2.0) | 5.1 | (3.2, 7.8) |
| Severe | 6 (0.6) | 1.4 | (0.5, 3.0) | 4 (0.4) | 0.9 | (0.3, 2.4) |
| Life-threatening | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| Any adverse event leading to withdrawal | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| Relatedf | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| Severe | 0 | 0.0 | (0.0, 0.8) | 0 | 0.0 | (0.0, 0.9) |
| Life-threatening | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| Death | 0 | 0.0 | (0.0, 0.8) | 0 | 0.0 | (0.0, 0.9) |

## 14.16 Incidence Rates of at Least 1 Adverse Event From Dose 1 to Unblinding Date, by Ethnicity – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Ethnicity: Hispanic/Latino

**Table 14.16.** Incidence Rates of at Least 1 Adverse Event From Dose 1 to Unblinding Date, by Ethnicity – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Ethnicity: Hispanic/Latino

| Adverse Event | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| | BNT162b2 (30 µg) (Na=132, TEb=0.6) | | Placebo (Na=130, TEb=0.5) | | | |
| nc (%) | IRd | (95% CIe) | nc (%) | IRd | (95% CIe) | |
| Any event | 9 (6.8) | 16.0 | (7.3, 30.3) | 16 (12.3) | 29.8 | (17.0, 48.4) |
| Relatedf | 3 (2.3) | 5.3 | (1.1, 15.6) | 4 (3.1) | 7.5 | (2.0, 19.1) |
| Severe | 2 (1.5) | 3.6 | (0.4, 12.8) | 0 | 0.0 | (0.0, 6.9) |
| Life-threatening | 0 | 0.0 | (0.0, 6.6) | 0 | 0.0 | (0.0, 6.9) |
| Any serious adverse event | 2 (1.5) | 3.6 | (0.4, 12.8) | 0 | 0.0 | (0.0, 6.9) |
| Relatedf | 0 | 0.0 | (0.0, 6.6) | 0 | 0.0 | (0.0, 6.9) |
| Severe | 1 (0.8) | 1.8 | (0.0, 9.9) | 0 | 0.0 | (0.0, 6.9) |
| Life-threatening | 0 | 0.0 | (0.0, 6.6) | 0 | 0.0 | (0.0, 6.9) |
| Any nonserious adverse event | 8 (6.1) | 14.2 | (6.1, 28.0) | 16 (12.3) | 29.8 | (17.0, 48.4) |
| Relatedf | 3 (2.3) | 5.3 | (1.1, 15.6) | 4 (3.1) | 7.5 | (2.0, 19.1) |
| Severe | 1 (0.8) | 1.8 | (0.0, 9.9) | 0 | 0.0 | (0.0, 6.9) |
| Life-threatening | 0 | 0.0 | (0.0, 6.6) | 0 | 0.0 | (0.0, 6.9) |
| Any adverse event leading to withdrawal | 0 | 0.0 | (0.0, 6.6) | 0 | 0.0 | (0.0, 6.9) |
| Relatedf | 0 | 0.0 | (0.0, 6.6) | 0 | 0.0 | (0.0, 6.9) |
| Severe | 0 | 0.0 | (0.0, 6.6) | 0 | 0.0 | (0.0, 6.9) |
| Life-threatening | 0 | 0.0 | (0.0, 6.6) | 0 | 0.0 | (0.0, 6.9) |
| Death | 0 | 0.0 | (0.0, 6.6) | 0 | 0.0 | (0.0, 6.9) |

## 14.17 Incidence Rates of at Least 1 Adverse Event From Dose 1 to Unblinding Date, by Ethnicity – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Ethnicity: Non-Hispanic/Non-Latino

**Table 14.17.** Incidence Rates of at Least 1 Adverse Event From Dose 1 to Unblinding Date, by Ethnicity – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Ethnicity: Non-Hispanic/Non-Latino

| Adverse Event | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| | BNT162b2 (30 µg) (Nα=997, TEb=4.0) | | | Placebo (Nα=996, TEb=4.0) | | |
| n° (%) | IRd | (95% CIe) | n° (%) | IRd | (95% CIe) | |
| Any event | 86 (8.6) | 21.5 | (17.2, 26.5) | 96 (9.6) | 24.2 | (19.6, 29.6) |
| Relatedf | 33 (3.3) | 8.2 | (5.7, 11.6) | 20 (2.0) | 5.1 | (3.1, 7.8) |
| Severe | 11 (1.1) | 2.7 | (1.4, 4.9) | 5 (0.5) | 1.3 | (0.4, 2.9) |
| Life-threatening | 2 (0.2) | 0.5 | (0.1, 1.8) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| Any serious adverse event | 8 (0.8) | 2.0 | (0.9, 3.9) | 2 (0.2) | 0.5 | (0.1, 1.8) |
| Relatedf | 0 | 0.0 | (0.0, 0.9) | 0 | 0.0 | (0.0, 0.9) |
| Severe | 6 (0.6) | 1.5 | (0.6, 3.3) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| Life-threatening | 1 (0.1) | 0.2 | (0.0, 1.4) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| Any nonserious adverse event | 81 (8.1) | 20.2 | (16.1, 25.2) | 94 (9.4) | 23.7 | (19.2, 29.0) |
| Relatedf | 33 (3.3) | 8.2 | (5.7, 11.6) | 20 (2.0) | 5.1 | (3.1, 7.8) |
| Severe | 5 (0.5) | 1.2 | (0.4, 2.9) | 4 (0.4) | 1.0 | (0.3, 2.6) |
| Life-threatening | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |
| Any adverse event leading to withdrawal | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |
| Relatedf | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |
| Severe | 0 | 0.0 | (0.0, 0.9) | 0 | 0.0 | (0.0, 0.9) |
| Life-threatening | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |
| Death | 0 | 0.0 | (0.0, 0.9) | 0 | 0.0 | (0.0, 0.9) |

## 14.18 Incidence Rates of at Least 1 Adverse Event From Dose 1 to Unblinding Date, by Race – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Race: White

**Table 14.18.** Incidence Rates of at Least 1 Adverse Event From Dose 1 to Unblinding Date, by Race – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Race: White

| Adverse Event | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| | BNT162b2 (30 µg) (Na=970, TEb=3.9) | | | Placebo (Na=962, TEb=3.8) | | |
| nc (%) | IRd | (95% CIe) | nc (%) | IRd | (95% CIe) | |
| Any event | 83 (8.6) | 21.3 | (17.0, 26.4) | 100 (10.4) | 26.2 | (21.3, 31.9) |
| Relatedf | 29 (3.0) | 7.5 | (5.0, 10.7) | 19 (2.0) | 5.0 | (3.0, 7.8) |
| Severe | 10 (1.0) | 2.6 | (1.2, 4.7) | 5 (0.5) | 1.3 | (0.4, 3.1) |
| Life-threatening | 2 (0.2) | 0.5 | (0.1, 1.9) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| Any serious adverse event | 7 (0.7) | 1.8 | (0.7, 3.7) | 2 (0.2) | 0.5 | (0.1, 1.9) |
| Relatedf | 0 | 0.0 | (0.0, 0.9) | 0 | 0.0 | (0.0, 1.0) |
| Severe | 5 (0.5) | 1.3 | (0.4, 3.0) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| Life-threatening | 1 (0.1) | 0.3 | (0.0, 1.4) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| Any nonserious adverse event | 79 (8.1) | 20.3 | (16.1, 25.3) | 98 (10.2) | 25.7 | (20.8, 31.3) |
| Relatedf | 29 (3.0) | 7.5 | (5.0, 10.7) | 19 (2.0) | 5.0 | (3.0, 7.8) |
| Severe | 5 (0.5) | 1.3 | (0.4, 3.0) | 4 (0.4) | 1.0 | (0.3, 2.7) |
| Life-threatening | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| Any adverse event leading to withdrawal | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| Relatedf | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| Severe | 0 | 0.0 | (0.0, 0.9) | 0 | 0.0 | (0.0, 1.0) |
| Life-threatening | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| Death | 0 | 0.0 | (0.0, 0.9) | 0 | 0.0 | (0.0, 1.0) |

## 14.19 Incidence Rates of at Least 1 Adverse Event From Dose 1 to Unblinding Date, by Race – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Race: Black or African American

**Table 14.19.** Incidence Rates of at Least 1 Adverse Event From Dose 1 to Unblinding Date, by Race – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Race: Black or African American

| Adverse Event | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| | BNT162b2 (30 µg) (Na=52, TEb=0.2) | | | Placebo (Na=57, TEb=0.3) | | |
| nc(%) | IRd | (95% CIe) | nc(%) | IRd | (95% CIe) | |
| Any event | 3 (5.8) | 12.1 | (2.5, 35.4) | 3 (5.3) | 11.5 | (2.4, 33.6) |
| Relatedf | 1 (1.9) | 4.0 | (0.1, 22.5) | 3 (5.3) | 11.5 | (2.4, 33.6) |
| Severe | 1 (1.9) | 4.0 | (0.1, 22.5) | 0 | 0.0 | (0.0, 14.1) |
| Life-threatening | 0 | 0.0 | (0.0, 14.9) | 0 | 0.0 | (0.0, 14.1) |
| Any serious adverse event | 1 (1.9) | 4.0 | (0.1, 22.5) | 0 | 0.0 | (0.0, 14.1) |
| Relatedf | 0 | 0.0 | (0.0, 14.9) | 0 | 0.0 | (0.0, 14.1) |
| Severe | 1 (1.9) | 4.0 | (0.1, 22.5) | 0 | 0.0 | (0.0, 14.1) |
| Life-threatening | 0 | 0.0 | (0.0, 14.9) | 0 | 0.0 | (0.0, 14.1) |
| Any nonserious adverse event | 3 (5.8) | 12.1 | (2.5, 35.4) | 3 (5.3) | 11.5 | (2.4, 33.6) |
| Relatedf | 1 (1.9) | 4.0 | (0.1, 22.5) | 3 (5.3) | 11.5 | (2.4, 33.6) |
| Severe | 0 | 0.0 | (0.0, 14.9) | 0 | 0.0 | (0.0, 14.1) |
| Life-threatening | 0 | 0.0 | (0.0, 14.9) | 0 | 0.0 | (0.0, 14.1) |
| Any adverse event leading to withdrawal | 0 | 0.0 | (0.0, 14.9) | 0 | 0.0 | (0.0, 14.1) |
| Relatedf | 0 | 0.0 | (0.0, 14.9) | 0 | 0.0 | (0.0, 14.1) |
| Severe | 0 | 0.0 | (0.0, 14.9) | 0 | 0.0 | (0.0, 14.1) |
| Life-threatening | 0 | 0.0 | (0.0, 14.9) | 0 | 0.0 | (0.0, 14.1) |
| Death | 0 | 0.0 | (0.0, 14.9) | 0 | 0.0 | (0.0, 14.1) |

## 14.20 Incidence Rates of at Least 1 Adverse Event From Dose 1 to Unblinding Date, by Race – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Race: All Others

**Table 14.20.** Incidence Rates of at Least 1 Adverse Event From Dose 1 to Unblinding Date, by Race – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Race: All Others

| Adverse Event | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| | BNT162b2 (30 µg) (Nα=109, TEb=0.4) | | | Placebo (Nα=110, TEb=0.4) | | |
| n^c (%) | IR^d | (95% CI^c) | n^c (%) | IR^d | (95% CI^c) | |
| Any event | 9 (8.3) | 20.8 | (9.5, 39.4) | 10 (9.1) | 23.3 | (11.2, 42.9) |
| Relatedf | 6 (5.5) | 13.8 | (5.1, 30.1) | 2 (1.8) | 4.7 | (0.6, 16.9) |
| Severe | 2 (1.8) | 4.6 | (0.6, 16.7) | 0 | 0.0 | (0.0, 8.6) |
| Life-threatening | 0 | 0.0 | (0.0, 8.5) | 0 | 0.0 | (0.0, 8.6) |
| Any serious adverse event | 2 (1.8) | 4.6 | (0.6, 16.7) | 0 | 0.0 | (0.0, 8.6) |
| Relatedf | 0 | 0.0 | (0.0, 8.5) | 0 | 0.0 | (0.0, 8.6) |
| Severe | 1 (0.9) | 2.3 | (0.1, 12.8) | 0 | 0.0 | (0.0, 8.6) |
| Life-threatening | 0 | 0.0 | (0.0, 8.5) | 0 | 0.0 | (0.0, 8.6) |
| Any nonserious adverse event | 7 (6.4) | 16.1 | (6.5, 33.3) | 10 (9.1) | 23.3 | (11.2, 42.9) |
| Relatedf | 6 (5.5) | 13.8 | (5.1, 30.1) | 2 (1.8) | 4.7 | (0.6, 16.9) |
| Severe | 1 (0.9) | 2.3 | (0.1, 12.8) | 0 | 0.0 | (0.0, 8.6) |
| Life-threatening | 0 | 0.0 | (0.0, 8.5) | 0 | 0.0 | (0.0, 8.6) |
| Any adverse event leading to withdrawal | 0 | 0.0 | (0.0, 8.5) | 0 | 0.0 | (0.0, 8.6) |
| Relatedf | 0 | 0.0 | (0.0, 8.5) | 0 | 0.0 | (0.0, 8.6) |
| Severe | 0 | 0.0 | (0.0, 8.5) | 0 | 0.0 | (0.0, 8.6) |
| Life-threatening | 0 | 0.0 | (0.0, 8.5) | 0 | 0.0 | (0.0, 8.6) |
| Death | 0 | 0.0 | (0.0, 8.5) | 0 | 0.0 | (0.0, 8.6) |

## 14.21 Incidence Rates of at Least 1 Adverse Event From Dose 1 to Unblinding Date, by Sex – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Sex: Male

**Table 14.21.** Incidence Rates of at Least 1 Adverse Event From Dose 1 to Unblinding Date, by Sex – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Sex: Male

| Adverse Event | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| | BNT162b2 (30 µg)(Nα=567, TEb=2.3) | | | Placebo(Nα=585, TEb=2.3) | | |
| n c (%) | IRd | (95% CIe) | n c (%) | IRd | (95% CIe) | |
| Any event | 42 (7.4) | 18.2 | (13.1, 24.5) | 57 (9.7) | 24.4 | (18.5, 31.7) |
| Relatedf | 21 (3.7) | 9.1 | (5.6, 13.9) | 12 (2.1) | 5.1 | (2.7, 9.0) |
| Severe | 5 (0.9) | 2.2 | (0.7, 5.0) | 4 (0.7) | 1.7 | (0.5, 4.4) |
| Life-threatening | 1 (0.2) | 0.4 | (0.0, 2.4) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| Any serious adverse event | 3 (0.5) | 1.3 | (0.3, 3.8) | 2 (0.3) | 0.9 | (0.1, 3.1) |
| Relatedf | 0 | 0.0 | (0.0, 1.6) | 0 | 0.0 | (0.0, 1.6) |
| Severe | 3 (0.5) | 1.3 | (0.3, 3.8) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| Life-threatening | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| Any nonserious adverse event | 39 (6.9) | 16.9 | (12.0, 23.1) | 55 (9.4) | 23.6 | (17.8, 30.7) |
| Relatedf | 21 (3.7) | 9.1 | (5.6, 13.9) | 12 (2.1) | 5.1 | (2.7, 9.0) |
| Severe | 2 (0.4) | 0.9 | (0.1, 3.1) | 3 (0.5) | 1.3 | (0.3, 3.8) |
| Life-threatening | 1 (0.2) | 0.4 | (0.0, 2.4) | 0 | 0.0 | (0.0, 1.6) |
| Any adverse event leading to withdrawal | 1 (0.2) | 0.4 | (0.0, 2.4) | 0 | 0.0 | (0.0, 1.6) |
| Relatedf | 1 (0.2) | 0.4 | (0.0, 2.4) | 0 | 0.0 | (0.0, 1.6) |
| Severe | 0 | 0.0 | (0.0, 1.6) | 0 | 0.0 | (0.0, 1.6) |
| Life-threatening | 1 (0.2) | 0.4 | (0.0, 2.4) | 0 | 0.0 | (0.0, 1.6) |
| Death | 0 | 0.0 | (0.0, 1.6) | 0 | 0.0 | (0.0, 1.6) |

## 14.22 Incidence Rates of at Least 1 Adverse Event From Dose 1 to Unblinding Date, by Sex – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Sex: Female

**Table 14.22.** Incidence Rates of at Least 1 Adverse Event From Dose 1 to Unblinding Date, by Sex – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Sex: Female

| Adverse Event | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| | BNT162b2 (30 µg) (Nα=564, TEb=2.3) | | | Placebo (Nα=544, TEb=2.2) | | |
| nc(%) | IRd | (95% CIc) | nc(%) | IRd | (95% CIc) | |
| Any event | 53 (9.4) | 23.5 | (17.6, 30.7) | 56 (10.3) | 25.7 | (19.4, 33.4) |
| Relatedf | 15 (2.7) | 6.6 | (3.7, 11.0) | 12 (2.2) | 5.5 | (2.8, 9.6) |
| Severe | 8 (1.4) | 3.5 | (1.5, 7.0) | 1 (0.2) | 0.5 | (0.0, 2.6) |
| Life-threatening | 1 (0.2) | 0.4 | (0.0, 2.5) | 0 | 0.0 | (0.0, 1.7) |
| Any serious adverse event | 7 (1.2) | 3.1 | (1.2, 6.4) | 0 | 0.0 | (0.0, 1.7) |
| Relatedf | 0 | 0.0 | (0.0, 1.6) | 0 | 0.0 | (0.0, 1.7) |
| Severe | 4 (0.7) | 1.8 | (0.5, 4.5) | 0 | 0.0 | (0.0, 1.7) |
| Life-threatening | 1 (0.2) | 0.4 | (0.0, 2.5) | 0 | 0.0 | (0.0, 1.7) |
| Any nonserious adverse event | 50 (8.9) | 22.1 | (16.4, 29.2) | 56 (10.3) | 25.7 | (19.4, 33.4) |
| Relatedf | 15 (2.7) | 6.6 | (3.7, 11.0) | 12 (2.2) | 5.5 | (2.8, 9.6) |
| Severe | 4 (0.7) | 1.8 | (0.5, 4.5) | 1 (0.2) | 0.5 | (0.0, 2.6) |
| Life-threatening | 0 | 0.0 | (0.0, 1.6) | 0 | 0.0 | (0.0, 1.7) |
| Any adverse event leading to withdrawal | 0 | 0.0 | (0.0, 1.6) | 0 | 0.0 | (0.0, 1.7) |
| Relatedf | 0 | 0.0 | (0.0, 1.6) | 0 | 0.0 | (0.0, 1.7) |
| Severe | 0 | 0.0 | (0.0, 1.6) | 0 | 0.0 | (0.0, 1.7) |
| Life-threatening | 0 | 0.0 | (0.0, 1.6) | 0 | 0.0 | (0.0, 1.7) |
| Death | 0 | 0.0 | (0.0, 1.6) | 0 | 0.0 | (0.0, 1.7) |

## 14.23 Incidence Rates of at Least 1 Adverse Event From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term, by Baseline SARS-CoV-2 Status – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Baseline SARS-CoV-2 Status: Positive

**Table 14.23.** Incidence Rates of at Least 1 Adverse Event From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term, by Baseline SARS-CoV-2 Status – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Baseline SARS-CoV-2 Status: Positive

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| | BNT162b2 (30 µg) (N*=46, TEb=0.2) | | | Placebo (N*=50, TEb=0.2) | | |
| n° (%) | IRd | (95% CIe) | n° (%) | IRd | (95% CIe) | |
| Any event | 4 (8.7) | 20.8 | (5.7, 53.3) | 4 (8.0) | 19.2 | (5.2, 49.1) |
| GASTROINTESTINAL DISORDERS | 2 (4.3) | 10.4 | (1.3, 37.6) | 0 | 0.0 | (0.0, 17.7) |
| Diarrhoea | 1 (2.2) | 5.2 | (0.1, 29.0) | 0 | 0.0 | (0.0, 17.7) |
| Nausea | 1 (2.2) | 5.2 | (0.1, 29.0) | 0 | 0.0 | (0.0, 17.7) |
| GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS | 0 | 0.0 | (0.0, 19.2) | 1 (2.0) | 4.8 | (0.1, 26.7) |
| Fatigue | 0 | 0.0 | (0.0, 19.2) | 1 (2.0) | 4.8 | (0.1, 26.7) |
| INFECTIONS AND INFESTATIONS | 1 (2.2) | 5.2 | (0.1, 29.0) | 0 | 0.0 | (0.0, 17.7) |
| Ear infection | 1 (2.2) | 5.2 | (0.1, 29.0) | 0 | 0.0 | (0.0, 17.7) |
| Otitis externa | 1 (2.2) | 5.2 | (0.1, 29.0) | 0 | 0.0 | (0.0, 17.7) |
| Otitis media | 1 (2.2) | 5.2 | (0.1, 29.0) | 0 | 0.0 | (0.0, 17.7) |
| MUSCULOSKELETAL AND CONNECTIVE TISSUE DISORDERS | 0 | 0.0 | (0.0, 19.2) | 2 (4.0) | 9.6 | (1.2, 34.6) |
| Musculoskeletal chest pain | 0 | 0.0 | (0.0, 19.2) | 1 (2.0) | 4.8 | (0.1, 26.7) |
| Myalgia | 0 | 0.0 | (0.0, 19.2) | 1 (2.0) | 4.8 | (0.1, 26.7) |
| NERVOUS SYSTEM DISORDERS | 2 (4.3) | 10.4 | (1.3, 37.6) | 1 (2.0) | 4.8 | (0.1, 26.7) |
| Dizziness | 1 (2.2) | 5.2 | (0.1, 29.0) | 1 (2.0) | 4.8 | (0.1, 26.7) |
| Syncope | 1 (2.2) | 5.2 | (0.1, 29.0) | 0 | 0.0 | (0.0, 17.7) |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| SKIN AND SUBCUTANEOUS TISSUE DISORDERS | 0 | 0.0 | (0.0, 19.2) | 1 (2.0) | 4.8 | (0.1, 26.7) |
| Rash | 0 | 0.0 | (0.0, 19.2) | 1 (2.0) | 4.8 | (0.1, 26.7) |
| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
| | BNT162b2 (30 µg) (Na=46, TEb=0.2) | | Placebo (Na=50, TEb=0.2) | | | |
| nc (%) | IRd | (95% CIe) | nc (%) | IRd | (95% CIe) | |
| Abbreviation: SARS-CoV-2 = severe acute respiratory syndrome coronavirus 2.Note: MedDRA (v24.0) coding dictionary applied.Note: Subjects whose baseline SARS-CoV-2 status cannot be determined because of missing N-binding antibody or NAAT at Visit 1 we... | | | | | | |

## 14.24 Incidence Rates of at Least 1 Adverse Event From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term, by Baseline SARS-CoV-2 Status – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Baseline SARS-CoV-2 Status: Negative

**Table 14.24.** Incidence Rates of at Least 1 Adverse Event From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term, by Baseline SARS-CoV-2 Status – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Baseline SARS-CoV-2 Status: Negative

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| | BNT162b2 (30 µg) (Nα=1083, TEb=4.4) | | | Placebo (Nα=1078, TEb=4.3) | | |
| nc(%) | IRd | (95% CIc) | nc(%) | IRd | (95% CIc) | |
| Any event | 91 (8.4) | 20.8 | (16.8, 25.6) | 109 (10.1) | 25.4 | (20.8, 30.6) |
| BLOOD AND LYMPHATIC SYSTEM DISORDERS | 9 (0.8) | 2.1 | (0.9, 3.9) | 2 (0.2) | 0.5 | (0.1, 1.7) |
| Lymphadenopathy | 9 (0.8) | 2.1 | (0.9, 3.9) | 2 (0.2) | 0.5 | (0.1, 1.7) |
| CONGENITAL, FAMILIAL AND GENETIC DISORDERS | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Spine malformation | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| EAR AND LABYRINTH DISORDERS | 1 (0.1) | 0.2 | (0.0, 1.3) | 3 (0.3) | 0.7 | (0.1, 2.0) |
| Cerumen impaction | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Conductive deafness | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Ear pain | 1 (0.1) | 0.2 | (0.0, 1.3) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| EYE DISORDERS | 2 (0.2) | 0.5 | (0.1, 1.7) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Eye pain | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| Eyelid rash | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| Retinal haemorrhage | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| GASTROINTESTINAL DISORDERS | 12 (1.1) | 2.7 | (1.4, 4.8) | 8 (0.7) | 1.9 | (0.8, 3.7) |
| Abdominal pain | 2 (0.2) | 0.5 | (0.1, 1.7) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Aphthous ulcer | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| Constipation | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| Diarrhoea | 2 (0.2) | 0.5 | (0.1, 1.7) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Gastritis | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| Lip swelling | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| Mouth swelling | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| Mouth ulceration | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Nausea | 4 (0.4) | 0.9 | (0.2, 2.3) | 3 (0.3) | 0.7 | (0.1, 2.0) |
| Oral mucosal blistering | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| Rectal prolapse | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| Tooth impacted | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Toothache | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Vomiting | 1 (0.1) | 0.2 | (0.0, 1.3) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS | 17 (1.6) | 3.9 | (2.3, 6.2) | 11 (1.0) | 2.6 | (1.3, 4.6) |
| Chills | 2 (0.2) | 0.5 | (0.1, 1.7) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Fatigue | 8 (0.7) | 1.8 | (0.8, 3.6) | 3 (0.3) | 0.7 | (0.1, 2.0) |
| Injection site pain | 8 (0.7) | 1.8 | (0.8, 3.6) | 8 (0.7) | 1.9 | (0.8, 3.7) |
| Injection site swelling | 2 (0.2) | 0.5 | (0.1, 1.7) | 0 | 0.0 | (0.0, 0.9) |
| Nodule | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
| BNT162b2 (30 µg) (Na=1083, TEb=4.4) | | | Placebo (Na=1078, TEb=4.3) | | | |
| nc(%) | IRd | (95% CIc) | nc(%) | IRd | (95% CIc) | |
| Oedema peripheral | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Peripheral swelling | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| Pyrexia | 6 (0.6) | 1.4 | (0.5, 3.0) | 0 | 0.0 | (0.0, 0.9) |
| Vessel puncture site pain | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| IMMUNE SYSTEM DISORDERS | 1 (0.1) | 0.2 | (0.0, 1.3) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Food allergy | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Seasonal allergy | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| INFECTIONS AND INFESTATIONS | 9 (0.8) | 2.1 | (0.9, 3.9) | 9 (0.8) | 2.1 | (1.0, 4.0) |
| Anal abscess | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| Appendicitis | 0 | 0.0 | (0.0, 0.8) | 2 (0.2) | 0.5 | (0.1, 1.7) |
| Body tinea | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| Candida infection | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Cellulitis | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Conjunctivitis | 0 | 0.0 | (0.0, 0.8) | 2 (0.2) | 0.5 | (0.1, 1.7) |
| Ear infection | 2 (0.2) | 0.5 | (0.1, 1.7) | 0 | 0.0 | (0.0, 0.9) |
| Focal peritonitis | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Infectious mononucleosis | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Paronychia | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| Pilonidal cyst | 1 (0.1) | 0.2 | (0.0, 1.3) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Subcutaneous abscess | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Tinea capitis | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| Vulval abscess | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| Vulvovaginal mycotic infection | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| INJURY, POISONING AND PROCEDURAL COMPLICATIONS | 15 (1.4) | 3.4 | (1.9, 5.7) | 25 (2.3) | 5.8 | (3.8, 8.6) |
| Accident | 1 (0.1) | 0.2 | (0.0, 1.3) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Ankle fracture | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Bone contusion | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| Clavicle fracture | 1 (0.1) | 0.2 | (0.0, 1.3) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Concussion | 3 (0.3) | 0.7 | (0.1, 2.0) | 4 (0.4) | 0.9 | (0.3, 2.4) |
| Contusion | 2 (0.2) | 0.5 | (0.1, 1.7) | 2 (0.2) | 0.5 | (0.1, 1.7) |
| Fall | 2 (0.2) | 0.5 | (0.1, 1.7) | 5 (0.5) | 1.2 | (0.4, 2.7) |
| Femur fracture | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| Foot fracture | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Hand fracture | 1 (0.1) | 0.2 | (0.0, 1.3) | 4 (0.4) | 0.9 | (0.3, 2.4) |
| Humerus fracture | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Ligament sprain | 1 (0.1) | 0.2 | (0.0, 1.3) | 4 (0.4) | 0.9 | (0.3, 2.4) |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
| | BNT162b2 (30 µg) (Nα=1083, TEb=4.4) | | | Placebo (Nα=1078, TEb=4.3) | | |
| nc(%) | IRd | (95% CIc) | nc(%) | IRd | (95% CIc) | |
| Lip injury | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Meniscus injury | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| Muscle strain | 1 (0.1) | 0.2 | (0.0, 1.3) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Patella fracture | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Procedural pain | 2 (0.2) | 0.5 | (0.1, 1.7) | 3 (0.3) | 0.7 | (0.1, 2.0) |
| Radius fracture | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| Skin laceration | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Tibia fracture | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Tooth fracture | 0 | 0.0 | (0.0, 0.8) | 2 (0.2) | 0.5 | (0.1, 1.7) |
| Upper limb fracture | 1 (0.1) | 0.2 | (0.0, 1.3) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| INVESTIGATIONS | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| SARS-CoV-2 antibody test positive | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| MUSCULOSKELETAL AND CONNECTIVE TISSUE DISORDERS | 8 (0.7) | 1.8 | (0.8, 3.6) | 12 (1.1) | 2.8 | (1.4, 4.9) |
| Arthralgia | 2 (0.2) | 0.5 | (0.1, 1.7) | 4 (0.4) | 0.9 | (0.3, 2.4) |
| Back pain | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Joint swelling | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Musculoskeletal chest pain | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| Myalgia | 3 (0.3) | 0.7 | (0.1, 2.0) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Neck pain | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Osteochondrosis | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| Pain in extremity | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| Tendonitis | 0 | 0.0 | (0.0, 0.8) | 4 (0.4) | 0.9 | (0.3, 2.4) |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| NEOPLASMS BENIGN, MALIGNANT AND UNSPECIFIED (INCL CYSTS AND POLYPS) | 1 (0.1) | 0.2 | (0.0, 1.3) | 3 (0.3) | 0.7 | (0.1, 2.0) |
| Fibroadenoma of breast | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Hair follicle tumour benign | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| Melanocytic naevus | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Skin papilloma | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| NERVOUS SYSTEM DISORDERS | 11 (1.0) | 2.5 | (1.3, 4.5) | 12 (1.1) | 2.8 | (1.4, 4.9) |
| Dizziness | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| Headache | 5 (0.5) | 1.1 | (0.4, 2.7) | 7 (0.6) | 1.6 | (0.7, 3.4) |
| Migraine | 3 (0.3) | 0.7 | (0.1, 2.0) | 0 | 0.0 | (0.0, 0.9) |
| Paraesthesia | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| Presyncope | 1 (0.1) | 0.2 | (0.0, 1.3) | 4 (0.4) | 0.9 | (0.3, 2.4) |
| Syncope | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| PSYCHIATRIC DISORDERS | 17 (1.6) | 3.9 | (2.3, 6.2) | 13 (1.2) | 3.0 | (1.6, 5.2) |
| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
| | BNT162b2 (30 µg) (Nα=1083, TEb=4.4) | | | Placebo (Nα=1078, TEb=4.3) | | |
| nc(%) | IRd | (95% CIc) | nc(%) | IRd | (95% CIc) | |
| Anxiety | 4 (0.4) | 0.9 | (0.2, 2.3) | 6 (0.6) | 1.4 | (0.5, 3.0) |
| Attention deficit hyperactivity disorder | 2 (0.2) | 0.5 | (0.1, 1.7) | 4 (0.4) | 0.9 | (0.3, 2.4) |
| Conversion disorder | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| Depression | 6 (0.6) | 1.4 | (0.5, 3.0) | 3 (0.3) | 0.7 | (0.1, 2.0) |
| Disorientation | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| Generalised anxiety disorder | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| Obsessive-compulsive disorder | 0 | 0.0 | (0.0, 0.8) | 2 (0.2) | 0.5 | (0.1, 1.7) |
| Panic attack | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| Sleep terror | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| Suicidal ideation | 4 (0.4) | 0.9 | (0.2, 2.3) | 0 | 0.0 | (0.0, 0.9) |
| Tic | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| RENAL AND URINARY DISORDERS | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Dysuria | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| REPRODUCTIVE SYSTEM AND BREAST DISORDERS | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| Amenorrhoea | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| RESPIRATORY, THORACIC AND MEDIASTINAL DISORDERS | 3 (0.3) | 0.7 | (0.1, 2.0) | 8 (0.7) | 1.9 | (0.8, 3.7) |
| Epistaxis | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Nasal congestion | 2 (0.2) | 0.5 | (0.1, 1.7) | 3 (0.3) | 0.7 | (0.1, 2.0) |
| Rhinorrhoea | 2 (0.2) | 0.5 | (0.1, 1.7) | 4 (0.4) | 0.9 | (0.3, 2.4) |
| Sneezing | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| SKIN AND SUBCUTANEOUS TISSUE DISORDERS | 9 (0.8) | 2.1 | (0.9, 3.9) | 15 (1.4) | 3.5 | (2.0, 5.8) |
| Acne | 2 (0.2) | 0.5 | (0.1, 1.7) | 3 (0.3) | 0.7 | (0.1, 2.0) |
| Dermatitis contact | 2 (0.2) | 0.5 | (0.1, 1.7) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Eczema | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Pityriasis rosea | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Rash | 3 (0.3) | 0.7 | (0.1, 2.0) | 4 (0.4) | 0.9 | (0.3, 2.4) |
| Rash maculopapular | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Seborrhoeic dermatitis | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Urticaria | 2 (0.2) | 0.5 | (0.1, 1.7) | 5 (0.5) | 1.2 | (0.4, 2.7) |
| SURGICAL AND MEDICAL PROCEDURES | 1 (0.1) | 0.2 | (0.0, 1.3) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| Wisdom teeth removal | 1 (0.1) | 0.2 | (0.0, 1.3) | 1 (0.1) | 0.2 | (0.0, 1.3) |

| System Organ Class Preferred Term | Vaccine Group (as Administered) |
|---|---|

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | |
|---|---|---|---|---|---|
| | BNT162b2 (30 µg) (Na=1083, TEb=4.4) | | Placebo (Na=1078, TEb=4.3) | | |
| nc (%) | IRd | (95% CIe) | nc (%) | IRd | (95% CIe) |

Abbreviation: SARS-CoV-2 = severe acute respiratory syndrome coronavirus 2.Note: MedDRA (v24.0) coding dictionary applied.Note: Subjects whose baseline SARS-CoV-2 status cannot be determined because of missing N-binding antibody or NAAT at Visit 1 we...

## 14.25 Incidence Rates of at Least 1 Adverse Event From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term, by Ethnicity – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Ethnicity: Hispanic/Latino

**Table 14.25.** Incidence Rates of at Least 1 Adverse Event From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term, by Ethnicity – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Ethnicity: Hispanic/Latino

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| | BNT162b2 (30 µg) (N=132, TEb=0.6) | | | Placebo (N=130, TEb=0.5) | | |
| n° (%) | IRd | (95% CIe) | n° (%) | IRd | (95% CIe) | |
| Any event | 9 (6.8) | 16.0 | (7.3, 30.3) | 16 (12.3) | 29.8 | (17.0, 48.4) |
| BLOOD AND LYMPHATIC SYSTEM DISORDERS | 1 (0.8) | 1.8 | (0.0, 9.9) | 0 | 0.0 | (0.0, 6.9) |
| Lymphadenopathy | 1 (0.8) | 1.8 | (0.0, 9.9) | 0 | 0.0 | (0.0, 6.9) |
| EAR AND LABYRINTH DISORDERS | 0 | 0.0 | (0.0, 6.6) | 1 (0.8) | 1.9 | (0.0, 10.4) |
| Conductive deafness | 0 | 0.0 | (0.0, 6.6) | 1 (0.8) | 1.9 | (0.0, 10.4) |
| GASTROINTESTINAL DISORDERS | 2 (1.5) | 3.6 | (0.4, 12.8) | 1 (0.8) | 1.9 | (0.0, 10.4) |
| Abdominal pain | 1 (0.8) | 1.8 | (0.0, 9.9) | 0 | 0.0 | (0.0, 6.9) |
| Constipation | 1 (0.8) | 1.8 | (0.0, 9.9) | 0 | 0.0 | (0.0, 6.9) |
| Diarrhoea | 1 (0.8) | 1.8 | (0.0, 9.9) | 0 | 0.0 | (0.0, 6.9) |
| Gastritis | 1 (0.8) | 1.8 | (0.0, 9.9) | 0 | 0.0 | (0.0, 6.9) |
| Nausea | 0 | 0.0 | (0.0, 6.6) | 1 (0.8) | 1.9 | (0.0, 10.4) |
| Vomiting | 0 | 0.0 | (0.0, 6.6) | 1 (0.8) | 1.9 | (0.0, 10.4) |
| GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS | 1 (0.8) | 1.8 | (0.0, 9.9) | 1 (0.8) | 1.9 | (0.0, 10.4) |
| Injection site pain | 1 (0.8) | 1.8 | (0.0, 9.9) | 1 (0.8) | 1.9 | (0.0, 10.4) |
| Vessel puncture site pain | 0 | 0.0 | (0.0, 6.6) | 1 (0.8) | 1.9 | (0.0, 10.4) |
| INFECTIONS AND INFESTATIONS | 3 (2.3) | 5.3 | (1.1, 15.6) | 0 | 0.0 | (0.0, 6.9) |
| Body tinea | 1 (0.8) | 1.8 | (0.0, 9.9) | 0 | 0.0 | (0.0, 6.9) |
| Paronychia | 1 (0.8) | 1.8 | (0.0, 9.9) | 0 | 0.0 | (0.0, 6.9) |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| Vulval abscess | 1 (0.8) | 1.8 | (0.0, 9.9) | 0 | 0.0 | (0.0, 6.9) |
| INJURY, POISONING AND PROCEDURAL COMPLICATIONS | 0 | 0.0 | (0.0, 6.6) | 3 (2.3) | 5.6 | (1.2, 16.3) |
| Accident | 0 | 0.0 | (0.0, 6.6) | 1 (0.8) | 1.9 | (0.0, 10.4) |
| Ankle fracture | 0 | 0.0 | (0.0, 6.6) | 1 (0.8) | 1.9 | (0.0, 10.4) |
| Contusion | 0 | 0.0 | (0.0, 6.6) | 1 (0.8) | 1.9 | (0.0, 10.4) |
| Fall | 0 | 0.0 | (0.0, 6.6) | 1 (0.8) | 1.9 | (0.0, 10.4) |
| Hand fracture | 0 | 0.0 | (0.0, 6.6) | 1 (0.8) | 1.9 | (0.0, 10.4) |
| Lip injury | 0 | 0.0 | (0.0, 6.6) | 1 (0.8) | 1.9 | (0.0, 10.4) |
| Tooth fracture | 0 | 0.0 | (0.0, 6.6) | 1 (0.8) | 1.9 | (0.0, 10.4) |
| MUSCULOSKELETAL AND CONNECTIVE TISSUE DISORDERS | 0 | 0.0 | (0.0, 6.6) | 3 (2.3) | 5.6 | (1.2, 16.3) |
| Arthralgia | 0 | 0.0 | (0.0, 6.6) | 2 (1.5) | 3.7 | (0.5, 13.5) |
| Musculoskeletal chest pain | 0 | 0.0 | (0.0, 6.6) | 1 (0.8) | 1.9 | (0.0, 10.4) |
| NEOPLASMS BENIGN, MALIGNANT AND UNSPECIFIED (INCL CYSTS AND POLYPS) | 0 | 0.0 | (0.0, 6.6) | 1 (0.8) | 1.9 | (0.0, 10.4) |
| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
| | BNT162b2 (30 µg) (Na=132, TEb=0.6) | | Placebo (Na=130, TEb=0.5) | | | |
| nc (%) | IRd | (95% CIe) | nc (%) | IRd | (95% CIe) | |
| Melanocytic naevus | 0 | 0.0 | (0.0, 6.6) | 1 (0.8) | 1.9 | (0.0, 10.4) |
| NERVOUS SYSTEM DISORDERS | 2 (1.5) | 3.6 | (0.4, 12.8) | 3 (2.3) | 5.6 | (1.2, 16.3) |
| Headache | 1 (0.8) | 1.8 | (0.0, 9.9) | 3 (2.3) | 5.6 | (1.2, 16.3) |
| Paraesthesia | 1 (0.8) | 1.8 | (0.0, 9.9) | 0 | 0.0 | (0.0, 6.9) |
| PSYCHIATRIC DISORDERS | 2 (1.5) | 3.6 | (0.4, 12.8) | 1 (0.8) | 1.9 | (0.0, 10.4) |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| Anxiety | 1 (0.8) | 1.8 | (0.0, 9.9) | 0 | 0.0 | (0.0, 6.9) |
| Conversion disorder | 1 (0.8) | 1.8 | (0.0, 9.9) | 0 | 0.0 | (0.0, 6.9) |
| Obsessive-compulsive disorder | 0 | 0.0 | (0.0, 6.6) | 1 (0.8) | 1.9 | (0.0, 10.4) |
| Suicidal ideation | 1 (0.8) | 1.8 | (0.0, 9.9) | 0 | 0.0 | (0.0, 6.9) |
| SKIN AND SUBCUTANEOUS TISSUE DISORDERS | 1 (0.8) | 1.8 | (0.0, 9.9) | 6 (4.6) | 11.2 | (4.1, 24.3) |
| Acne | 0 | 0.0 | (0.0, 6.6) | 2 (1.5) | 3.7 | (0.5, 13.5) |
| Dermatitis contact | 1 (0.8) | 1.8 | (0.0, 9.9) | 0 | 0.0 | (0.0, 6.9) |
| Rash | 0 | 0.0 | (0.0, 6.6) | 3 (2.3) | 5.6 | (1.2, 16.3) |
| Seborrhoeic dermatitis | 0 | 0.0 | (0.0, 6.6) | 1 (0.8) | 1.9 | (0.0, 10.4) |
| Urticaria | 0 | 0.0 | (0.0, 6.6) | 1 (0.8) | 1.9 | (0.0, 10.4) |

## 14.26 Incidence Rates of at Least 1 Adverse Event From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term, by Ethnicity – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Ethnicity: Non-Hispanic/Non-Latino

**Table 14.26.** Incidence Rates of at Least 1 Adverse Event From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term, by Ethnicity – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Ethnicity: Non-Hispanic/Non-Latino

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| | BNT162b2 (30 µg) (N°=997, TEb=4.0) | | | Placebo (N°=996, TEb=4.0) | | |
| | n^c (%) | IRd | (95% CIe) | n^c (%) | IRd | (95% CIe) |
| Any event | 86 (8.6) | 21.5 | (17.2, 26.5) | 96 (9.6) | 24.2 | (19.6, 29.6) |
| BLOOD AND LYMPHATIC SYSTEM DISORDERS | 8 (0.8) | 2.0 | (0.9, 3.9) | 2 (0.2) | 0.5 | (0.1, 1.8) |
| Lymphadenopathy | 8 (0.8) | 2.0 | (0.9, 3.9) | 2 (0.2) | 0.5 | (0.1, 1.8) |
| CONGENITAL, FAMILIAL AND GENETIC DISORDERS | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| Spine malformation | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| EAR AND LABYRINTH DISORDERS | 1 (0.1) | 0.2 | (0.0, 1.4) | 2 (0.2) | 0.5 | (0.1, 1.8) |
| Cerumen impaction | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| Ear pain | 1 (0.1) | 0.2 | (0.0, 1.4) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| EYE DISORDERS | 2 (0.2) | 0.5 | (0.1, 1.8) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| Eye pain | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |
| Eyelid rash | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |
| Retinal haemorrhage | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| GASTROINTESTINAL DISORDERS | 12 (1.2) | 3.0 | (1.5, 5.2) | 7 (0.7) | 1.8 | (0.7, 3.6) |
| Abdominal pain | 1 (0.1) | 0.2 | (0.0, 1.4) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| Aphthous ulcer | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |
| Diarrhoea | 2 (0.2) | 0.5 | (0.1, 1.8) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| Lip swelling | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |
| Mouth swelling | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| Mouth ulceration | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| Nausea | 5 (0.5) | 1.2 | (0.4, 2.9) | 2 (0.2) | 0.5 | (0.1, 1.8) |
| Oral mucosal blistering | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |
| Rectal prolapse | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |
| Tooth impacted | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| Toothache | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| Vomiting | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |
| GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS | 16 (1.6) | 4.0 | (2.3, 6.5) | 11 (1.1) | 2.8 | (1.4, 5.0) |
| Chills | 2 (0.2) | 0.5 | (0.1, 1.8) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| Fatigue | 8 (0.8) | 2.0 | (0.9, 3.9) | 4 (0.4) | 1.0 | (0.3, 2.6) |
| Injection site pain | 7 (0.7) | 1.7 | (0.7, 3.6) | 7 (0.7) | 1.8 | (0.7, 3.6) |
| Injection site swelling | 2 (0.2) | 0.5 | (0.1, 1.8) | 0 | 0.0 | (0.0, 0.9) |
| Nodule | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |
| Oedema peripheral | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| Peripheral swelling | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |
| Pyrexia | 6 (0.6) | 1.5 | (0.6, 3.3) | 0 | 0.0 | (0.0, 0.9) |
| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
| | BNT162b2 (30 µg) (Na=997, TEb=4.0) | | | Placebo (Na=996, TEb=4.0) | | |
| nc (%) | IRd | (95% CIe) | nc (%) | IRd | (95% CIe) | |
| IMMUNE SYSTEM DISORDERS | 1 (0.1) | 0.2 | (0.0, 1.4) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| Food allergy | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| Seasonal allergy | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |
| INFECTIONS AND INFESTATIONS | 7 (0.7) | 1.7 | (0.7, 3.6) | 9 (0.9) | 2.3 | (1.0, 4.3) |
| Anal abscess | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |
| Appendicitis | 0 | 0.0 | (0.0, 0.9) | 2 (0.2) | 0.5 | (0.1, 1.8) |
| Candida infection | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| Cellulitis | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.4) |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| Conjunctivitis | 0 | 0.0 | (0.0, 0.9) | 2 (0.2) | 0.5 | (0.1, 1.8) |
| Ear infection | 3 (0.3) | 0.7 | (0.2, 2.2) | 0 | 0.0 | (0.0, 0.9) |
| Focal peritonitis | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| Infectious mononucleosis | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| Otitis externa | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |
| Otitis media | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |
| Pilonidal cyst | 1 (0.1) | 0.2 | (0.0, 1.4) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| Subcutaneous abscess | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| Tinea capitis | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |
| Vulvovaginal mycotic infection | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |
| INJURY, POISONING AND PROCEDURAL COMPLICATIONS | 15 (1.5) | 3.7 | (2.1, 6.2) | 22 (2.2) | 5.6 | (3.5, 8.4) |
| Accident | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |
| Bone contusion | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |
| Clavicle fracture | 1 (0.1) | 0.2 | (0.0, 1.4) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| Concussion | 3 (0.3) | 0.7 | (0.2, 2.2) | 4 (0.4) | 1.0 | (0.3, 2.6) |
| Contusion | 2 (0.2) | 0.5 | (0.1, 1.8) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| Fall | 2 (0.2) | 0.5 | (0.1, 1.8) | 4 (0.4) | 1.0 | (0.3, 2.6) |
| Femur fracture | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |
| Foot fracture | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| Hand fracture | 1 (0.1) | 0.2 | (0.0, 1.4) | 3 (0.3) | 0.8 | (0.2, 2.2) |
| Humerus fracture | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| Ligament sprain | 1 (0.1) | 0.2 | (0.0, 1.4) | 4 (0.4) | 1.0 | (0.3, 2.6) |
| Meniscus injury | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |
| Muscle strain | 1 (0.1) | 0.2 | (0.0, 1.4) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| Patella fracture | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| Procedural pain | 2 (0.2) | 0.5 | (0.1, 1.8) | 3 (0.3) | 0.8 | (0.2, 2.2) |
| Radius fracture | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |
| Skin laceration | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.4) |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
| | BNT162b2 (30 µg) (Na=997, TEb=4.0) | | | Placebo (Na=996, TEb=4.0) | | |
| nc (%) | IRd | (95% CIe) | nc (%) | IRd | (95% CIe) | |
| Tibia fracture | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| Tooth fracture | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| Upper limb fracture | 1 (0.1) | 0.2 | (0.0, 1.4) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| INVESTIGATIONS | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |
| SARS-CoV-2 antibody test positive | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |
| MUSCULOSKELETAL AND CONNECTIVE TISSUE DISORDERS | 8 (0.8) | 2.0 | (0.9, 3.9) | 11 (1.1) | 2.8 | (1.4, 5.0) |
| Arthralgia | 2 (0.2) | 0.5 | (0.1, 1.8) | 2 (0.2) | 0.5 | (0.1, 1.8) |
| Back pain | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| Joint swelling | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| Musculoskeletal chest pain | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |
| Myalgia | 3 (0.3) | 0.7 | (0.2, 2.2) | 2 (0.2) | 0.5 | (0.1, 1.8) |
| Neck pain | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| Osteochondrosis | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |
| Pain in extremity | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |
| Tendonitis | 0 | 0.0 | (0.0, 0.9) | 4 (0.4) | 1.0 | (0.3, 2.6) |
| NEOPLASMS BENIGN, MALIGNANT AND UNSPECIFIED (INCL CYSTS AND POLYPS) | 1 (0.1) | 0.2 | (0.0, 1.4) | 2 (0.2) | 0.5 | (0.1, 1.8) |
| Fibroadenoma of breast | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| Hair follicle tumour benign | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |
| Skin papilloma | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| NERVOUS SYSTEM DISORDERS | 11 (1.1) | 2.7 | (1.4, 4.9) | 10 (1.0) | 2.5 | (1.2, 4.6) |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| Dizziness | 2 (0.2) | 0.5 | (0.1, 1.8) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| Headache | 4 (0.4) | 1.0 | (0.3, 2.6) | 4 (0.4) | 1.0 | (0.3, 2.6) |
| Migraine | 3 (0.3) | 0.7 | (0.2, 2.2) | 0 | 0.0 | (0.0, 0.9) |
| Presyncope | 1 (0.1) | 0.2 | (0.0, 1.4) | 4 (0.4) | 1.0 | (0.3, 2.6) |
| Syncope | 1 (0.1) | 0.2 | (0.0, 1.4) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| PSYCHIATRIC DISORDERS | 15 (1.5) | 3.7 | (2.1, 6.2) | 12 (1.2) | 3.0 | (1.6, 5.3) |
| Anxiety | 3 (0.3) | 0.7 | (0.2, 2.2) | 6 (0.6) | 1.5 | (0.6, 3.3) |
| Attention deficit hyperactivity disorder | 2 (0.2) | 0.5 | (0.1, 1.8) | 4 (0.4) | 1.0 | (0.3, 2.6) |
| Depression | 6 (0.6) | 1.5 | (0.6, 3.3) | 3 (0.3) | 0.8 | (0.2, 2.2) |
| Disorientation | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |
| Generalised anxiety disorder | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |
| Obsessive-compulsive disorder | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| Panic attack | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |
| Sleep terror | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |
| Suicidal ideation | 3 (0.3) | 0.7 | (0.2, 2.2) | 0 | 0.0 | (0.0, 0.9) |
| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
| | BNT162b2 (30 µg) (Na=997, TEb=4.0) | | Placebo (Na=996, TEb=4.0) | | | |
| nc (%) | IRd | (95% CIe) | nc (%) | IRd | (95% CIe) | |
| Tic | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |
| RENAL AND URINARY DISORDERS | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| Dysuria | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| REPRODUCTIVE SYSTEM AND BREAST DISORDERS | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |
| Amenorrhoea | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |
| RESPIRATORY, THORACIC AND MEDIASTINAL DISORDERS | 3 (0.3) | 0.7 | (0.2, 2.2) | 8 (0.8) | 2.0 | (0.9, 4.0) |
| Epistaxis | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| Nasal congestion | 2 (0.2) | 0.5 | (0.1, 1.8) | 3 (0.3) | 0.8 | (0.2, 2.2) |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| Rhinorrhoea | 2 (0.2) | 0.5 | (0.1, 1.8) | 4 (0.4) | 1.0 | (0.3, 2.6) |
| Sneezing | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |
| SKIN AND SUBCUTANEOUS TISSUE DISORDERS | 8 (0.8) | 2.0 | (0.9, 3.9) | 9 (0.9) | 2.3 | (1.0, 4.3) |
| Acne | 2 (0.2) | 0.5 | (0.1, 1.8) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| Dermatitis contact | 1 (0.1) | 0.2 | (0.0, 1.4) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| Eczema | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| Rash | 3 (0.3) | 0.7 | (0.2, 2.2) | 2 (0.2) | 0.5 | (0.1, 1.8) |
| Rash maculopapular | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| Urticaria | 2 (0.2) | 0.5 | (0.1, 1.8) | 4 (0.4) | 1.0 | (0.3, 2.6) |
| SURGICAL AND MEDICAL PROCEDURES | 1 (0.1) | 0.2 | (0.0, 1.4) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| Wisdom teeth removal | 1 (0.1) | 0.2 | (0.0, 1.4) | 1 (0.1) | 0.3 | (0.0, 1.4) |

## 14.27 Incidence Rates of at Least 1 Adverse Event From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term, by Race – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Race: White

**Table 14.27.** Incidence Rates of at Least 1 Adverse Event From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term, by Race – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Race: White

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| | BNT162b2 (30 µg) (N=970, TEb=3.9) | | | Placebo (N=962, TEb=3.8) | | |
| | n° (%) | IRd | (95% CIe) | n° (%) | IRd | (95% CIe) |
| Any event | 83 (8.6) | 21.3 | (17.0, 26.4) | 100 (10.4) | 26.2 | (21.3, 31.9) |
| BLOOD AND LYMPHATIC SYSTEM DISORDERS | 6 (0.6) | 1.5 | (0.6, 3.4) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| Lymphadenopathy | 6 (0.6) | 1.5 | (0.6, 3.4) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| CONGENITAL, FAMILIAL AND GENETIC DISORDERS | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| Spine malformation | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| EAR AND LABYRINTH DISORDERS | 1 (0.1) | 0.3 | (0.0, 1.4) | 2 (0.2) | 0.5 | (0.1, 1.9) |
| Cerumen impaction | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| Ear pain | 1 (0.1) | 0.3 | (0.0, 1.4) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| EYE DISORDERS | 2 (0.2) | 0.5 | (0.1, 1.9) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| Eye pain | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| Eyelid rash | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| Retinal haemorrhage | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| GASTROINTESTINAL DISORDERS | 11 (1.1) | 3.3 | (1.8, 5.7) | 7 (0.7) | 1.8 | (0.7, 3.8) |
| Abdominal pain | 1 (0.1) | 0.3 | (0.0, 1.4) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| Aphthous ulcer | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| Constipation | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| Diarrhoea | 3 (0.3) | 0.8 | (0.2, 2.3) | 0 | 0.0 | (0.0, 1.0) |
| Gastritis | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| Lip swelling | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| Mouth swelling | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| Mouth ulceration | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| Nausea | 5 (0.5) | 1.3 | (0.4, 3.0) | 3 (0.3) | 0.8 | (0.2, 2.3) |
| Oral mucosal blistering | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| Rectal prolapse | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| Tooth impacted | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| Toothache | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| Vomiting | 1 (0.1) | 0.3 | (0.0, 1.4) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS | 15 (1.5) | 3.9 | (2.2, 6.4) | 10 (1.0) | 2.6 | (1.3, 4.8) |
| Chills | 1 (0.1) | 0.3 | (0.0, 1.4) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| Fatigue | 7 (0.7) | 1.8 | (0.7, 3.7) | 4 (0.4) | 1.0 | (0.3, 2.7) |
| Injection site pain | 7 (0.7) | 1.8 | (0.7, 3.7) | 6 (0.6) | 1.6 | (0.6, 3.4) |
| Injection site swelling | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| Nodule | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
| | BNT162b2 (30 µg) (N=970, TEb=3.9) | | Placebo (N=962, TEb=3.8) | | | |
| n c (%) | IR d | (95% CIc) | n c (%) | IR d | (95% CIc) | |
| Oedema peripheral | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| Pyrexia | 5 (0.5) | 1.3 | (0.4, 3.0) | 0 | 0.0 | (0.0, 1.0) |
| Vessel puncture site pain | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| IMMUNE SYSTEM DISORDERS | 1 (0.1) | 0.3 | (0.0, 1.4) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| Food allergy | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| Seasonal allergy | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| INFECTIONS AND INFESTATIONS | 9 (0.9) | 2.3 | (1.1, 4.4) | 7 (0.7) | 1.8 | (0.7, 3.8) |
| Anal abscess | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| Appendicitis | 0 | 0.0 | (0.0, 0.9) | 2 (0.2) | 0.5 | (0.1, 1.9) |
| Candida infection | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.5) |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| Conjunctivitis | 0 | 0.0 | (0.0, 0.9) | 2 (0.2) | 0.5 | (0.1, 1.9) |
| Ear infection | 3 (0.3) | 0.8 | (0.2, 2.3) | 0 | 0.0 | (0.0, 1.0) |
| Focal peritonitis | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| Infectious mononucleosis | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| Otitis externa | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| Otitis media | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| Paronychia | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| Pilonidal cyst | 1 (0.1) | 0.3 | (0.0, 1.4) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| Tinea capitis | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| Vulval abscess | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| Vulvovaginal mycotic infection | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| INJURY, POISONING AND PROCEDURAL COMPLICATIONS | 14 (1.4) | 3.6 | (2.0, 6.0) | 24 (2.5) | 6.3 | (4.0, 9.4) |
| Accident | 1 (0.1) | 0.3 | (0.0, 1.4) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| Ankle fracture | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| Bone contusion | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| Clavicle fracture | 1 (0.1) | 0.3 | (0.0, 1.4) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| Concussion | 3 (0.3) | 0.8 | (0.2, 2.3) | 4 (0.4) | 1.0 | (0.3, 2.7) |
| Contusion | 1 (0.1) | 0.3 | (0.0, 1.4) | 2 (0.2) | 0.5 | (0.1, 1.9) |
| Fall | 1 (0.1) | 0.3 | (0.0, 1.4) | 5 (0.5) | 1.3 | (0.4, 3.1) |
| Femur fracture | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| Foot fracture | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| Hand fracture | 1 (0.1) | 0.3 | (0.0, 1.4) | 3 (0.3) | 0.8 | (0.2, 2.3) |
| Humerus fracture | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| Ligament sprain | 1 (0.1) | 0.3 | (0.0, 1.4) | 4 (0.4) | 1.0 | (0.3, 2.7) |
| Lip injury | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| Meniscus injury | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
| | BNT162b2 (30 µg) (N=970, TEb=3.9) | | | Placebo (N=962, TEb=3.8) | | |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| n° (%) | IRd | (95% CIe) | n° (%) | IRd | (95% CIe) | |
| Muscle strain | 1 (0.1) | 0.3 | (0.0, 1.4) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| Patella fracture | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| Procedural pain | 2 (0.2) | 0.5 | (0.1, 1.9) | 3 (0.3) | 0.8 | (0.2, 2.3) |
| Radius fracture | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| Skin laceration | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| Tibia fracture | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| Tooth fracture | 0 | 0.0 | (0.0, 0.9) | 2 (0.2) | 0.5 | (0.1, 1.9) |
| Upper limb fracture | 1 (0.1) | 0.3 | (0.0, 1.4) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| INVESTIGATIONS | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| SARS-CoV-2 antibody test positive | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| MUSCULOSKELETAL AND CONNECTIVE TISSUE DISORDERS | 8 (0.8) | 2.1 | (0.9, 4.1) | 13 (1.4) | 3.4 | (1.8, 5.8) |
| Arthralgia | 2 (0.2) | 0.5 | (0.1, 1.9) | 4 (0.4) | 1.0 | (0.3, 2.7) |
| Back pain | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| Joint swelling | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| Musculoskeletal chest pain | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| Myalgia | 3 (0.3) | 0.8 | (0.2, 2.3) | 2 (0.2) | 0.5 | (0.1, 1.9) |
| Neck pain | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| Osteochondrosis | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| Pain in extremity | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| Tendonitis | 0 | 0.0 | (0.0, 0.9) | 4 (0.4) | 1.0 | (0.3, 2.7) |
| NEOPLASMS BENIGN, MALIGNANT AND UNSPECIFIED (INCL CYSTS AND POLYPS) | 1 (0.1) | 0.3 | (0.0, 1.4) | 2 (0.2) | 0.5 | (0.1, 1.9) |
| Hair follicle tumour benign | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| Melanocytic naevus | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.5) |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| Skin papilloma | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| NERVOUS SYSTEM DISORDERS | 11 (1.1) | 2.8 | (1.4, 5.1) | 12 (1.2) | 3.1 | (1.6, 5.5) |
| Dizziness | 2 (0.2) | 0.5 | (0.1, 1.9) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| Headache | 5 (0.5) | 1.3 | (0.4, 3.0) | 6 (0.6) | 1.6 | (0.6, 3.4) |
| Migraine | 2 (0.2) | 0.5 | (0.1, 1.9) | 0 | 0.0 | (0.0, 1.0) |
| Presyncope | 1 (0.1) | 0.3 | (0.0, 1.4) | 4 (0.4) | 1.0 | (0.3, 2.7) |
| Syncope | 1 (0.1) | 0.3 | (0.0, 1.4) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| PSYCHIATRIC DISORDERS | 14 (1.4) | 3.6 | (2.0, 6.0) | 13 (1.4) | 3.4 | (1.8, 5.8) |
| Anxiety | 3 (0.3) | 0.8 | (0.2, 2.3) | 6 (0.6) | 1.6 | (0.6, 3.4) |
| Attention deficit hyperactivity disorder | 2 (0.2) | 0.5 | (0.1, 1.9) | 4 (0.4) | 1.0 | (0.3, 2.7) |
| Conversion disorder | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| Depression | 4 (0.4) | 1.0 | (0.3, 2.6) | 3 (0.3) | 0.8 | (0.2, 2.3) |
| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
| | BNT162b2 (30 µg) (N=970, TEb=3.9) | | | Placebo (N=962, TEb=3.8) | | |
| n° (%) | IRd | (95% CIe) | n° (%) | IRd | (95% CIe) | |
| Disorientation | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| Generalised anxiety disorder | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| Obsessive-compulsive disorder | 0 | 0.0 | (0.0, 0.9) | 2 (0.2) | 0.5 | (0.1, 1.9) |
| Panic attack | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| Sleep terror | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| Suicidal ideation | 3 (0.3) | 0.8 | (0.2, 2.3) | 0 | 0.0 | (0.0, 1.0) |
| Tic | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| RENAL AND URINARY DISORDERS | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| Dysuria | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| REPRODUCTIVE SYSTEM AND BREAST DISORDERS | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| Amenorrhoea | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| RESPIRATORY, THORACIC AND MEDIASTINAL DISORDERS | 3 (0.3) | 0.8 | (0.2, 2.3) | 7 (0.7) | 1.8 | (0.7, 3.8) |
| Epistaxis | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| Nasal congestion | 2 (0.2) | 0.5 | (0.1, 1.9) | 3 (0.3) | 0.8 | (0.2, 2.3) |
| Rhinorrhoea | 2 (0.2) | 0.5 | (0.1, 1.9) | 3 (0.3) | 0.8 | (0.2, 2.3) |
| Sneezing | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| SKIN AND SUBCUTANEOUS TISSUE DISORDERS | 9 (0.9) | 2.3 | (1.1, 4.4) | 13 (1.4) | 3.4 | (1.8, 5.8) |
| Acne | 2 (0.2) | 0.5 | (0.1, 1.9) | 2 (0.2) | 0.5 | (0.1, 1.9) |
| Dermatitis contact | 2 (0.2) | 0.5 | (0.1, 1.9) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| Eczema | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| Rash | 3 (0.3) | 0.8 | (0.2, 2.3) | 4 (0.4) | 1.0 | (0.3, 2.7) |
| Rash maculopapular | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| Seborrhoeic dermatitis | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| Urticaria | 2 (0.2) | 0.5 | (0.1, 1.9) | 5 (0.5) | 1.3 | (0.4, 3.1) |
| SURGICAL AND MEDICAL PROCEDURES | 1 (0.1) | 0.3 | (0.0, 1.4) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| Wisdom teeth removal | 1 (0.1) | 0.3 | (0.0, 1.4) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
| BNT162b2 (30 µg) (Na=970, TEb=3.9) | | | Placebo (Na=962, TEb=3.8) | | | |
| nc (%) | IRd | (95% CIc) | nc (%) | IRd | (95% CIc) | |

Note: MedDRA (v24.0) coding dictionary applied.Note: All Others = American Indian or Alaska Native, Asian, Native Hawaiian or other Pacific Islander, multiracial, and not reported race categories.a. N = number of subjects in the specified group. This...

## 14.28 Incidence Rates of at Least 1 Adverse Event From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term, by Race – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Race: Black or African American

**Table 14.28.** Incidence Rates of at Least 1 Adverse Event From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term, by Race – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Race: Black or African American

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| | BNT162b2 (30 µg) (Na=52, TEb=0.2) | | | Placebo (Na=57, TEb=0.3) | | |
| nc (%) | IRd | (95% CIe) | nc (%) | IRd | (95% CIe) | |
| Any event | 3 (5.8) | 12.1 | (2.5, 35.4) | 3 (5.3) | 11.5 | (2.4, 33.6) |
| BLOOD AND LYMPHATIC SYSTEM DISORDERS | 0 | 0.0 | (0.0, 14.9) | 1 (1.8) | 3.8 | (0.1, 21.3) |
| Lymphadenopathy | 0 | 0.0 | (0.0, 14.9) | 1 (1.8) | 3.8 | (0.1, 21.3) |
| EAR AND LABYRINTH DISORDERS | 0 | 0.0 | (0.0, 14.9) | 1 (1.8) | 3.8 | (0.1, 21.3) |
| Conductive deafness | 0 | 0.0 | (0.0, 14.9) | 1 (1.8) | 3.8 | (0.1, 21.3) |
| GASTROINTESTINAL DISORDERS | 0 | 0.0 | (0.0, 14.9) | 1 (1.8) | 3.8 | (0.1, 21.3) |
| Diarrhoea | 0 | 0.0 | (0.0, 14.9) | 1 (1.8) | 3.8 | (0.1, 21.3) |
| GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS | 1 (1.9) | 4.0 | (0.1, 22.5) | 0 | 0.0 | (0.0, 14.1) |
| Peripheral swelling | 1 (1.9) | 4.0 | (0.1, 22.5) | 0 | 0.0 | (0.0, 14.1) |
| INFECTIONS AND INFESTATIONS | 1 (1.9) | 4.0 | (0.1, 22.5) | 0 | 0.0 | (0.0, 14.1) |
| Body tinea | 1 (1.9) | 4.0 | (0.1, 22.5) | 0 | 0.0 | (0.0, 14.1) |
| INJURY, POISONING AND PROCEDURAL COMPLICATIONS | 1 (1.9) | 4.0 | (0.1, 22.5) | 0 | 0.0 | (0.0, 14.1) |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| Contusion | 1 (1.9) | 4.0 | (0.1, 22.5) | 0 | 0.0 | (0.0, 14.1) |
| Fall | 1 (1.9) | 4.0 | (0.1, 22.5) | 0 | 0.0 | (0.0, 14.1) |
| PSYCHIATRIC DISORDERS | 1 (1.9) | 4.0 | (0.1, 22.5) | 0 | 0.0 | (0.0, 14.1) |
| Anxiety | 1 (1.9) | 4.0 | (0.1, 22.5) | 0 | 0.0 | (0.0, 14.1) |
| Depression | 1 (1.9) | 4.0 | (0.1, 22.5) | 0 | 0.0 | (0.0, 14.1) |
| RESPIRATORY, THORACIC AND MEDIASTINAL DISORDERS | 0 | 0.0 | (0.0, 14.9) | 1 (1.8) | 3.8 | (0.1, 21.3) |
| Rhinorrhoea | 0 | 0.0 | (0.0, 14.9) | 1 (1.8) | 3.8 | (0.1, 21.3) |

## 14.29 Incidence Rates of at Least 1 Adverse Event From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term, by Race – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Race: All Others

**Table 14.29.** Incidence Rates of at Least 1 Adverse Event From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term, by Race – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Race: All Others

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | |
|---|---|---|---|---|---|
| | BNT162b2 (30 µg) (N=109, TEb=0.4) | | | Placebo (N=110, TEb=0.4) | |
| | n° (%) | IRd | (95% CIe) | n° (%) | IRd | (95% CIe) |
| Any event | 9 (8.3) | 20.8 | (9.5, 39.4) | 10 (9.1) | 23.3 |
| BLOOD AND LYMPHATIC SYSTEM DISORDERS | 3 (2.8) | 6.9 | (1.4, 20.2) | 0 | 0.0 |
| Lymphadenopathy | 3 (2.8) | 6.9 | (1.4, 20.2) | 0 | 0.0 |
| GASTROINTESTINAL DISORDERS | 1 (0.9) | 2.3 | (0.1, 12.8) | 0 | 0.0 |
| Abdominal pain | 1 (0.9) | 2.3 | (0.1, 12.8) | 0 | 0.0 |
| GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS | 1 (0.9) | 2.3 | (0.1, 12.8) | 2 (1.8) | 4.7 |
| Chills | 1 (0.9) | 2.3 | (0.1, 12.8) | 0 | 0.0 |
| Fatigue | 1 (0.9) | 2.3 | (0.1, 12.8) | 0 | 0.0 |
| Injection site pain | 1 (0.9) | 2.3 | (0.1, 12.8) | 2 (1.8) | 4.7 |
| Injection site swelling | 1 (0.9) | 2.3 | (0.1, 12.8) | 0 | 0.0 |
| Pyrexia | 1 (0.9) | 2.3 | (0.1, 12.8) | 0 | 0.0 |
| INFECTIONS AND INFESTATIONS | 0 | 0.0 | (0.0, 8.5) | 2 (1.8) | 4.7 |
| Cellulitis | 0 | 0.0 | (0.0, 8.5) | 1 (0.9) | 2.3 |
| Subcutaneous abscess | 0 | 0.0 | (0.0, 8.5) | 1 (0.9) | 2.3 |
| INJURY, POISONING AND PROCEDURAL COMPLICATIONS | 0 | 0.0 | (0.0, 8.5) | 1 (0.9) | 2.3 |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | |
|---|---|---|---|---|---|
| Hand fracture | 0 | 0.0 | (0.0, 8.5) | 1 (0.9) | 2.3 |
| MUSCULOSKELETAL AND CONNECTIVE TISSUE DISORDERS | 0 | 0.0 | (0.0, 8.5) | 1 (0.9) | 2.3 |
| Musculoskeletal chest pain | 0 | 0.0 | (0.0, 8.5) | 1 (0.9) | 2.3 |
| NEOPLASMS BENIGN, MALIGNANT AND UNSPECIFIED (INCL CYSTS AND POLYPS) | 0 | 0.0 | (0.0, 8.5) | 1 (0.9) | 2.3 |
| Fibroadenoma of breast | 0 | 0.0 | (0.0, 8.5) | 1 (0.9) | 2.3 |
| NERVOUS SYSTEM DISORDERS | 2 (1.8) | 4.6 | (0.6, 16.7) | 1 (0.9) | 2.3 |
| Headache | 0 | 0.0 | (0.0, 8.5) | 1 (0.9) | 2.3 |
| Migraine | 1 (0.9) | 2.3 | (0.1, 12.8) | 0 | 0.0 |
| Paraesthesia | 1 (0.9) | 2.3 | (0.1, 12.8) | 0 | 0.0 |
| PSYCHIATRIC DISORDERS | 2 (1.8) | 4.6 | (0.6, 16.7) | 0 | 0.0 |
| Depression | 1 (0.9) | 2.3 | (0.1, 12.8) | 0 | 0.0 |
| Suicidal ideation | 1 (0.9) | 2.3 | (0.1, 12.8) | 0 | 0.0 |
| SKIN AND SUBCUTANEOUS TISSUE DISORDERS | 0 | 0.0 | (0.0, 8.5) | 3 (2.7) | 7.0 |
| Acne | 0 | 0.0 | (0.0, 8.5) | 1 (0.9) | 2.3 |
| Pityriasis rosea | 0 | 0.0 | (0.0, 8.5) | 1 (0.9) | 2.3 |
| Rash | 0 | 0.0 | (0.0, 8.5) | 1 (0.9) | 2.3 |
| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | |
| BNT162b2 (30 µg) (Na=109, TEb=0.4) nc (%) | IRd (95% CIe) | Placebo (Na=110, TEb=0.4) nc (%) | IRd (95% CIe) | | |

| System Organ Class Preferred Term | Vaccine Group (as Administered) |
|---|---|
| Note: MedDRA (v24.0) coding dictionary applied.Note: All Others = American Indian or Alaska Native, Asian, Native Hawaiian or other Pacific Islander, multiracial, and not reported race categories.a. N = number of subjects in the specified group. This... | |

## 14.30 Incidence Rates of at Least 1 Adverse Event From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term, by Sex – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Sex: Male

**Table 14.30.** Incidence Rates of at Least 1 Adverse Event From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term, by Sex – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Sex: Male

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| | BNT162b2 (30 µg) (N=567, TEb=2.3) | | | Placebo (N=585, TEb=2.3) | | |
| n^c (%) | IR^d | (95% CI^c) | n^c (%) | IR^d | (95% CI^c) | |
| Any event | 42 (7.4) | 18.2 | (13.1, 24.5) | 57 (9.7) | 24.4 | (18.5, 31.7) |
| BLOOD AND LYMPHATIC SYSTEM DISORDERS | 8 (1.4) | 3.5 | (1.5, 6.8) | 0 | 0.0 | (0.0, 1.6) |
| Lymphadenopathy | 8 (1.4) | 3.5 | (1.5, 6.8) | 0 | 0.0 | (0.0, 1.6) |
| CONGENITAL, FAMILIAL AND GENETIC DISORDERS | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| Spine malformation | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| EAR AND LABYRINTH DISORDERS | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| Ear pain | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| EYE DISORDERS | 2 (0.4) | 0.9 | (0.1, 3.1) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| Eye pain | 1 (0.2) | 0.4 | (0.0, 2.4) | 0 | 0.0 | (0.0, 1.6) |
| Eyelid rash | 1 (0.2) | 0.4 | (0.0, 2.4) | 0 | 0.0 | (0.0, 1.6) |
| Retinal haemorrhage | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| GASTROINTESTINAL DISORDERS | 11 (1.9) | 2.6 | (1.0, 5.6) | 3 (0.5) | 1.3 | (0.3, 3.8) |
| Aphthous ulcer | 1 (0.2) | 0.4 | (0.0, 2.4) | 0 | 0.0 | (0.0, 1.6) |
| Diarrhoea | 2 (0.4) | 0.9 | (0.1, 3.1) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| Lip swelling | 1 (0.2) | 0.4 | (0.0, 2.4) | 0 | 0.0 | (0.0, 1.6) |
| Mouth swelling | 1 (0.2) | 0.4 | (0.0, 2.4) | 0 | 0.0 | (0.0, 1.6) |
| Nausea | 1 (0.2) | 0.4 | (0.0, 2.4) | 0 | 0.0 | (0.0, 1.6) |
| Oral mucosal blistering | 1 (0.2) | 0.4 | (0.0, 2.4) | 0 | 0.0 | (0.0, 1.6) |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| Tooth impacted | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| Toothache | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| Vomiting | 1 (0.2) | 0.4 | (0.0, 2.4) | 0 | 0.0 | (0.0, 1.6) |
| GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS | 10 (1.8) | 4.3 | (2.1, 8.0) | 7 (1.2) | 3.0 | (1.2, 6.2) |
| Fatigue | 5 (0.9) | 2.2 | (0.7, 5.0) | 2 (0.3) | 0.9 | (0.1, 3.1) |
| Injection site pain | 5 (0.9) | 2.2 | (0.7, 5.0) | 4 (0.7) | 1.7 | (0.5, 4.4) |
| Injection site swelling | 1 (0.2) | 0.4 | (0.0, 2.4) | 0 | 0.0 | (0.0, 1.6) |
| Nodule | 1 (0.2) | 0.4 | (0.0, 2.4) | 0 | 0.0 | (0.0, 1.6) |
| Oedema peripheral | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| Pyrexia | 3 (0.5) | 1.3 | (0.3, 3.8) | 0 | 0.0 | (0.0, 1.6) |
| IMMUNE SYSTEM DISORDERS | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| Food allergy | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| INFECTIONS AND INFESTATIONS | 3 (0.5) | 1.3 | (0.3, 3.8) | 7 (1.2) | 3.0 | (1.2, 6.2) |
| Anal abscess | 1 (0.2) | 0.4 | (0.0, 2.4) | 0 | 0.0 | (0.0, 1.6) |
| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
| BNT162b2 (30 µg) (Na=567, TEb=2.3) | | | Placebo (Na=585, TEb=2.3) | | | |
| nc (%) | IRd | (95% CIe) | nc (%) | IRd | (95% CIe) | |
| Appendicitis | 0 | 0.0 | (0.0, 1.6) | 2 (0.3) | 0.9 | (0.1, 3.1) |
| Candida infection | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| Cellulitis | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| Conjunctivitis | 0 | 0.0 | (0.0, 1.6) | 2 (0.3) | 0.9 | (0.1, 3.1) |
| Ear infection | 1 (0.2) | 0.4 | (0.0, 2.4) | 0 | 0.0 | (0.0, 1.6) |
| Focal peritonitis | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| Paronychia | 1 (0.2) | 0.4 | (0.0, 2.4) | 0 | 0.0 | (0.0, 1.6) |
| Pilonidal cyst | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| INJURY, POISONING AND PROCEDURAL COMPLICATIONS | 8 (1.4) | 3.5 | (1.5, 6.8) | 13 (2.2) | 5.6 | (3.0, 9.5) |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| Accident | 1 (0.2) | 0.4 | (0.0, 2.4) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| Bone contusion | 1 (0.2) | 0.4 | (0.0, 2.4) | 0 | 0.0 | (0.0, 1.6) |
| Clavicle fracture | 1 (0.2) | 0.4 | (0.0, 2.4) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| Concussion | 1 (0.2) | 0.4 | (0.0, 2.4) | 0 | 0.0 | (0.0, 1.6) |
| Contusion | 1 (0.2) | 0.4 | (0.0, 2.4) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| Fall | 1 (0.2) | 0.4 | (0.0, 2.4) | 2 (0.3) | 0.9 | (0.1, 3.1) |
| Hand fracture | 0 | 0.0 | (0.0, 1.6) | 2 (0.3) | 0.9 | (0.1, 3.1) |
| Humerus fracture | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| Ligament sprain | 0 | 0.0 | (0.0, 1.6) | 2 (0.3) | 0.9 | (0.1, 3.1) |
| Lip injury | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| Meniscus injury | 1 (0.2) | 0.4 | (0.0, 2.4) | 0 | 0.0 | (0.0, 1.6) |
| Muscle strain | 1 (0.2) | 0.4 | (0.0, 2.4) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| Patella fracture | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| Procedural pain | 0 | 0.0 | (0.0, 1.6) | 2 (0.3) | 0.9 | (0.1, 3.1) |
| Radius fracture | 1 (0.2) | 0.4 | (0.0, 2.4) | 0 | 0.0 | (0.0, 1.6) |
| Skin laceration | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| Tibia fracture | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| Tooth fracture | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| Upper limb fracture | 1 (0.2) | 0.4 | (0.0, 2.4) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| INVESTIGATIONS | 1 (0.2) | 0.4 | (0.0, 2.4) | 0 | 0.0 | (0.0, 1.6) |
| SARS-CoV-2 antibody test positive | 1 (0.2) | 0.4 | (0.0, 2.4) | 0 | 0.0 | (0.0, 1.6) |
| MUSCULOSKELETAL AND CONNECTIVE TISSUE DISORDERS | 4 (0.7) | 1.7 | (0.5, 4.4) | 8 (1.4) | 3.4 | (1.5, 6.8) |
| Arthralgia | 1 (0.2) | 0.4 | (0.0, 2.4) | 2 (0.3) | 0.9 | (0.1, 3.1) |
| Joint swelling | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| Myalgia | 2 (0.4) | 0.9 | (0.1, 3.1) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| Neck pain | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
| | BNT162b2 (30 µg) (Na=567, TEb=2.3) | | | Placebo (Na=585, TEb=2.3) | | |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| nc (%) | IRd | (95% CIe) | nc (%) | IRd | (95% CIe) | |
| Pain in extremity | 1 (0.2) | 0.4 | (0.0, 2.4) | 0 | 0.0 | (0.0, 1.6) |
| Tendonitis | 0 | 0.0 | (0.0, 1.6) | 3 (0.5) | 1.3 | (0.3, 3.8) |
| NEOPLASMS BENIGN, MALIGNANT AND UNSPECIFIED (INCL CYSTS AND POLYPS) | 1 (0.2) | 0.4 | (0.0, 2.4) | 0 | 0.0 | (0.0, 1.6) |
| Hair follicle tumour benign | 1 (0.2) | 0.4 | (0.0, 2.4) | 0 | 0.0 | (0.0, 1.6) |
| NERVOUS SYSTEM DISORDERS | 5 (0.9) | 2.2 | (0.7, 5.0) | 5 (0.9) | 2.1 | (0.7, 5.0) |
| Dizziness | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| Headache | 2 (0.4) | 0.9 | (0.1, 3.1) | 2 (0.3) | 0.9 | (0.1, 3.1) |
| Paraesthesia | 1 (0.2) | 0.4 | (0.0, 2.4) | 0 | 0.0 | (0.0, 1.6) |
| Presyncope | 1 (0.2) | 0.4 | (0.0, 2.4) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| Syncope | 1 (0.2) | 0.4 | (0.0, 2.4) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| PSYCHIATRIC DISORDERS | 5 (0.9) | 2.2 | (0.7, 5.0) | 3 (0.5) | 1.3 | (0.3, 3.8) |
| Anxiety | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| Depression | 2 (0.4) | 0.9 | (0.1, 3.1) | 2 (0.3) | 0.9 | (0.1, 3.1) |
| Disorientation | 1 (0.2) | 0.4 | (0.0, 2.4) | 0 | 0.0 | (0.0, 1.6) |
| Obsessive-compulsive disorder | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| Panic attack | 1 (0.2) | 0.4 | (0.0, 2.4) | 0 | 0.0 | (0.0, 1.6) |
| Suicidal ideation | 1 (0.2) | 0.4 | (0.0, 2.4) | 0 | 0.0 | (0.0, 1.6) |
| RESPIRATORY, THORACIC AND MEDIASTINAL DISORDERS | 1 (0.2) | 0.4 | (0.0, 2.4) | 2 (0.3) | 0.9 | (0.1, 3.1) |
| Nasal congestion | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| Rhinorrhoea | 1 (0.2) | 0.4 | (0.0, 2.4) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| SKIN AND SUBCUTANEOUS TISSUE DISORDERS | 3 (0.5) | 1.3 | (0.3, 3.8) | 11 (1.9) | 4.7 | (2.4, 8.4) |
| Acne | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.4 | (0.0, 2.4) |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| Dermatitis contact | 1 (0.2) | 0.4 | (0.0, 2.4) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| Eczema | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| Pityriasis rosea | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| Rash | 2 (0.4) | 0.9 | (0.1, 3.1) | 3 (0.5) | 1.3 | (0.3, 3.8) |
| Rash maculo-papular | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| Urticaria | 0 | 0.0 | (0.0, 1.6) | 4 (0.7) | 1.7 | (0.5, 4.4) |
| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
| BNT162b2 (30 µg) (Na=567, TEb=2.3) | | | Placebo (Na=585, TEb=2.3) | | | |
| nc (%) | IRd | (95% CIe) | nc (%) | IRd | (95% CIe) | |

Note: MedDRA (v24.0) coding dictionary applied.a. N = number of subjects in the specified group. This value is the denominator for the percentage calculations.b. TE = total exposure time in 100 person-years across all subjects in the specified group....

## 14.31 Incidence Rates of at Least 1 Adverse Event From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term, by Sex – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Sex: Female

**Table 14.31.** Incidence Rates of at Least 1 Adverse Event From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term, by Sex – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Sex: Female

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| | BNT162b2 (30 µg) (Na=564, TEb=2.3) | | | Placebo (Na=544, TEb=2.2) | | |
| nc (%) | IRd | (95% CIe) | nc (%) | IRd | (95% CIe) | |
| Any event | 53 (9.4) | 23.5 | (17.6, 30.7) | 56 (10.3) | 25.7 | (19.4, 33.4) |
| BLOOD AND LYMPHATIC SYSTEM DISORDERS | 1 (0.2) | 0.4 | (0.0, 2.5) | 2 (0.4) | 0.9 | (0.1, 3.3) |
| Lymphadenopathy | 1 (0.2) | 0.4 | (0.0, 2.5) | 2 (0.4) | 0.9 | (0.1, 3.3) |
| EAR AND LABYRINTH DISORDERS | 1 (0.2) | 0.4 | (0.0, 2.5) | 2 (0.4) | 0.9 | (0.1, 3.3) |
| Cerumen impaction | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.5 | (0.0, 2.6) |
| Conductive deafness | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.5 | (0.0, 2.6) |
| Ear pain | 1 (0.2) | 0.4 | (0.0, 2.5) | 0 | 0.0 | (0.0, 1.7) |
| GASTROINTESTINAL DISORDERS | 8 (1.4) | 3.5 | (1.5, 7.0) | 5 (0.9) | 2.3 | (0.7, 5.4) |
| Abdominal pain | 2 (0.4) | 0.9 | (0.1, 3.2) | 1 (0.2) | 0.5 | (0.0, 2.6) |
| Constipation | 1 (0.2) | 0.4 | (0.0, 2.5) | 0 | 0.0 | (0.0, 1.7) |
| Diarrhoea | 1 (0.2) | 0.4 | (0.0, 2.5) | 0 | 0.0 | (0.0, 1.7) |
| Gastritis | 1 (0.2) | 0.4 | (0.0, 2.5) | 0 | 0.0 | (0.0, 1.7) |
| Mouth ulceration | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.5 | (0.0, 2.6) |
| Nausea | 4 (0.7) | 1.8 | (0.5, 4.5) | 3 (0.6) | 1.4 | (0.3, 4.0) |
| Rectal prolapse | 1 (0.2) | 0.4 | (0.0, 2.5) | 0 | 0.0 | (0.0, 1.7) |
| Vomiting | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.5 | (0.0, 2.6) |
| GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS | 7 (1.2) | 3.1 | (1.2, 6.4) | 5 (0.9) | 2.3 | (0.7, 5.4) |
| Chills | 2 (0.4) | 0.9 | (0.1, 3.2) | 1 (0.2) | 0.5 | (0.0, 2.6) |
| Fatigue | 3 (0.5) | 1.3 | (0.3, 3.9) | 2 (0.4) | 0.9 | (0.1, 3.3) |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| Injection site pain | 3 (0.5) | 1.3 | (0.3, 3.9) | 4 (0.7) | 1.8 | (0.5, 4.7) |
| Injection site swelling | 1 (0.2) | 0.4 | (0.0, 2.5) | 0 | 0.0 | (0.0, 1.7) |
| Peripheral swelling | 1 (0.2) | 0.4 | (0.0, 2.5) | 0 | 0.0 | (0.0, 1.7) |
| Pyrexia | 3 (0.5) | 1.3 | (0.3, 3.9) | 0 | 0.0 | (0.0, 1.7) |
| Vessel puncture site pain | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.5 | (0.0, 2.6) |
| IMMUNE SYSTEM DISORDERS | 1 (0.2) | 0.4 | (0.0, 2.5) | 0 | 0.0 | (0.0, 1.7) |
| Seasonal allergy | 1 (0.2) | 0.4 | (0.0, 2.5) | 0 | 0.0 | (0.0, 1.7) |
| INFECTIONS AND INFESTATIONS | 7 (1.2) | 3.1 | (1.2, 6.4) | 2 (0.4) | 0.9 | (0.1, 3.3) |
| Body tinea | 1 (0.2) | 0.4 | (0.0, 2.5) | 0 | 0.0 | (0.0, 1.7) |
| Ear infection | 2 (0.4) | 0.9 | (0.1, 3.2) | 0 | 0.0 | (0.0, 1.7) |
| Infectious mononucleosis | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.5 | (0.0, 2.6) |
| Otitis externa | 1 (0.2) | 0.4 | (0.0, 2.5) | 0 | 0.0 | (0.0, 1.7) |
| Otitis media | 1 (0.2) | 0.4 | (0.0, 2.5) | 0 | 0.0 | (0.0, 1.7) |
| Pilonidal cyst | 1 (0.2) | 0.4 | (0.0, 2.5) | 0 | 0.0 | (0.0, 1.7) |
| Subcutaneous abscess | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.5 | (0.0, 2.6) |
| Tinea capitis | 1 (0.2) | 0.4 | (0.0, 2.5) | 0 | 0.0 | (0.0, 1.7) |
| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
| | BNT162b2 (30 µg) (N°=564, TEb=2.3) | | | Placebo (N°=544, TEb=2.2) | | |
| nc(%) | IRd | (95% CIe) | nc(%) | IRd | (95% CIe) | |
| Vulval abscess | 1 (0.2) | 0.4 | (0.0, 2.5) | 0 | 0.0 | (0.0, 1.7) |
| Vulvovaginal mycotic infection | 1 (0.2) | 0.4 | (0.0, 2.5) | 0 | 0.0 | (0.0, 1.7) |
| INJURY, POISONING AND PROCEDURAL COMPLICATIONS | 7 (1.2) | 3.1 | (1.2, 6.4) | 12 (2.2) | 5.5 | (2.8, 9.6) |
| Ankle fracture | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.5 | (0.0, 2.6) |
| Concussion | 2 (0.4) | 0.9 | (0.1, 3.2) | 4 (0.7) | 1.8 | (0.5, 4.7) |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| Contusion | 1 (0.2) | 0.4 | (0.0, 2.5) | 1 (0.2) | 0.5 | (0.0, 2.6) |
| Fall | 1 (0.2) | 0.4 | (0.0, 2.5) | 3 (0.6) | 1.4 | (0.3, 4.0) |
| Femur fracture | 1 (0.2) | 0.4 | (0.0, 2.5) | 0 | 0.0 | (0.0, 1.7) |
| Foot fracture | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.5 | (0.0, 2.6) |
| Hand fracture | 1 (0.2) | 0.4 | (0.0, 2.5) | 2 (0.4) | 0.9 | (0.1, 3.3) |
| Ligament sprain | 1 (0.2) | 0.4 | (0.0, 2.5) | 2 (0.4) | 0.9 | (0.1, 3.3) |
| Procedural pain | 2 (0.4) | 0.9 | (0.1, 3.2) | 1 (0.2) | 0.5 | (0.0, 2.6) |
| Tooth fracture | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.5 | (0.0, 2.6) |
| MUSCULOSKELETAL AND CONNECTIVE TISSUE DISORDERS | 4 (0.7) | 1.8 | (0.5, 4.5) | 6 (1.1) | 2.8 | (1.0, 6.0) |
| Arthralgia | 1 (0.2) | 0.4 | (0.0, 2.5) | 2 (0.4) | 0.9 | (0.1, 3.3) |
| Back pain | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.5 | (0.0, 2.6) |
| Musculoskeletal chest pain | 1 (0.2) | 0.4 | (0.0, 2.5) | 1 (0.2) | 0.5 | (0.0, 2.6) |
| Myalgia | 1 (0.2) | 0.4 | (0.0, 2.5) | 1 (0.2) | 0.5 | (0.0, 2.6) |
| Osteochondrosis | 1 (0.2) | 0.4 | (0.0, 2.5) | 0 | 0.0 | (0.0, 1.7) |
| Tendonitis | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.5 | (0.0, 2.6) |
| NEOPLASMS BENIGN, MALIGNANT AND UNSPECIFIED (INCL CYSTS AND POLYPS) | 0 | 0.0 | (0.0, 1.6) | 3 (0.6) | 1.4 | (0.3, 4.0) |
| Fibroadenoma of breast | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.5 | (0.0, 2.6) |
| Melanocytic naevus | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.5 | (0.0, 2.6) |
| Skin papilloma | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.5 | (0.0, 2.6) |
| NERVOUS SYSTEM DISORDERS | 8 (1.4) | 3.5 | (1.5, 7.0) | 8 (1.5) | 3.7 | (1.6, 7.2) |
| Dizziness | 2 (0.4) | 0.9 | (0.1, 3.2) | 0 | 0.0 | (0.0, 1.7) |
| Headache | 3 (0.5) | 1.3 | (0.3, 3.9) | 5 (0.9) | 2.3 | (0.7, 5.4) |
| Migraine | 3 (0.5) | 1.3 | (0.3, 3.9) | 0 | 0.0 | (0.0, 1.7) |
| Presyncope | 0 | 0.0 | (0.0, 1.6) | 3 (0.6) | 1.4 | (0.3, 4.0) |
| PSYCHIATRIC DISORDERS | 12 (2.1) | 5.3 | (2.7, 9.3) | 10 (1.8) | 4.6 | (2.2, 8.5) |
| Anxiety | 4 (0.7) | 1.8 | (0.5, 4.5) | 5 (0.9) | 2.3 | (0.7, 5.4) |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| Attention deficit hyperactivity disorder | 2 (0.4) | 0.9 | (0.1, 3.2) | 4 (0.7) | 1.8 | (0.5, 4.7) |
| Conversion disorder | 1 (0.2) | 0.4 | (0.0, 2.5) | 0 | 0.0 | (0.0, 1.7) |
| Depression | 4 (0.7) | 1.8 | (0.5, 4.5) | 1 (0.2) | 0.5 | (0.0, 2.6) |
| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
| | BNT162b2 (30 µg) (N°=564, TEb=2.3) | | | Placebo (N°=544, TEb=2.2) | | |
| nc(%) | IRd | (95% CIc) | nc(%) | IRd | (95% CIc) | |
| Generalised anxiety disorder | 1 (0.2) | 0.4 | (0.0, 2.5) | 0 | 0.0 | (0.0, 1.7) |
| Obsessive-compulsive disorder | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.5 | (0.0, 2.6) |
| Sleep terror | 1 (0.2) | 0.4 | (0.0, 2.5) | 0 | 0.0 | (0.0, 1.7) |
| Suicidal ideation | 3 (0.5) | 1.3 | (0.3, 3.9) | 0 | 0.0 | (0.0, 1.7) |
| Tic | 1 (0.2) | 0.4 | (0.0, 2.5) | 0 | 0.0 | (0.0, 1.7) |
| RENAL AND URINARY DISORDERS | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.5 | (0.0, 2.6) |
| Dysuria | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.5 | (0.0, 2.6) |
| REPRODUCTIVE SYSTEM AND BREAST DISORDERS | 1 (0.2) | 0.4 | (0.0, 2.5) | 0 | 0.0 | (0.0, 1.7) |
| Amenorrhoea | 1 (0.2) | 0.4 | (0.0, 2.5) | 0 | 0.0 | (0.0, 1.7) |
| RESPIRATORY, THORACIC AND MEDIASTINAL DISORDERS | 2 (0.4) | 0.9 | (0.1, 3.2) | 6 (1.1) | 2.8 | (1.0, 6.0) |
| Epistaxis | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.5 | (0.0, 2.6) |
| Nasal congestion | 2 (0.4) | 0.9 | (0.1, 3.2) | 2 (0.4) | 0.9 | (0.1, 3.3) |
| Rhinorrhoea | 1 (0.2) | 0.4 | (0.0, 2.5) | 3 (0.6) | 1.4 | (0.3, 4.0) |
| Sneezing | 1 (0.2) | 0.4 | (0.0, 2.5) | 0 | 0.0 | (0.0, 1.7) |
| SKIN AND SUBCUTANEOUS TISSUE DISORDERS | 6 (1.1) | 2.7 | (1.0, 5.8) | 5 (0.9) | 2.3 | (0.7, 5.4) |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| Acne | 2 (0.4) | 0.9 | (0.1, 3.2) | 2 (0.4) | 0.9 | (0.1, 3.3) |
| Dermatitis contact | 1 (0.2) | 0.4 | (0.0, 2.5) | 0 | 0.0 | (0.0, 1.7) |
| Rash | 1 (0.2) | 0.4 | (0.0, 2.5) | 2 (0.4) | 0.9 | (0.1, 3.3) |
| Seborrhoeic dermatitis | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.5 | (0.0, 2.6) |
| Urticaria | 2 (0.4) | 0.9 | (0.1, 3.2) | 1 (0.2) | 0.5 | (0.0, 2.6) |
| SURGICAL AND MEDICAL PROCEDURES | 1 (0.2) | 0.4 | (0.0, 2.5) | 1 (0.2) | 0.5 | (0.0, 2.6) |
| Wisdom teeth removal | 1 (0.2) | 0.4 | (0.0, 2.5) | 1 (0.2) | 0.5 | (0.0, 2.6) |

## 14.32 Incidence Rates of at Least 1 Related Adverse Event From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population

**Table 14.32.** Incidence Rates of at Least 1 Related Adverse Event From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| | BNT162b2 (30 µg) (Nα=1131, TEb=4.6) | | | Placebo (Nα=1129, TEb=4.5) | | |
| | n° (%) | IRd | (95% CIe) | n° (%) | IRd | (95% CIe) |
| Any event | 36 (3.2) | 7.9 | (5.5, 10.9) | 24 (2.1) | 5.3 | (3.4, 7.9) |
| BLOOD AND LYMPHATIC SYSTEM DISORDERS | 7 (0.6) | 1.5 | (0.6, 3.2) | 1 (0.1) | 0.2 | (0.0, 1.2) |
| Lymphadenopathy | 7 (0.6) | 1.5 | (0.6, 3.2) | 1 (0.1) | 0.2 | (0.0, 1.2) |
| EAR AND LABYRINTH DISORDERS | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.2) |
| Conductive deafness | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.2) |
| EYE DISORDERS | 1 (0.1) | 0.2 | (0.0, 1.2) | 0 | 0.0 | (0.0, 0.8) |
| Eyelid rash | 1 (0.1) | 0.2 | (0.0, 1.2) | 0 | 0.0 | (0.0, 0.8) |
| GASTROINTESTINAL DISORDERS | 11 (1.0) | 2.4 | (1.2, 4.3) | 2 (0.2) | 0.4 | (0.1, 1.6) |
| Abdominal pain | 2 (0.2) | 0.4 | (0.1, 1.6) | 0 | 0.0 | (0.0, 0.8) |
| Diarrhoea | 2 (0.2) | 0.4 | (0.1, 1.6) | 1 (0.1) | 0.2 | (0.0, 1.2) |
| Lip swelling | 1 (0.1) | 0.2 | (0.0, 1.2) | 0 | 0.0 | (0.0, 0.8) |
| Mouth swelling | 1 (0.1) | 0.2 | (0.0, 1.2) | 0 | 0.0 | (0.0, 0.8) |
| Nausea | 5 (0.4) | 1.1 | (0.4, 2.6) | 1 (0.1) | 0.2 | (0.0, 1.2) |
| Oral mucosal blistering | 1 (0.1) | 0.2 | (0.0, 1.2) | 0 | 0.0 | (0.0, 0.8) |
| Vomiting | 1 (0.1) | 0.2 | (0.0, 1.2) | 1 (0.1) | 0.2 | (0.0, 1.2) |
| GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS | 16 (1.4) | 3.5 | (2.0, 5.7) | 10 (0.9) | 2.2 | (1.1, 4.1) |
| Chills | 2 (0.2) | 0.4 | (0.1, 1.6) | 1 (0.1) | 0.2 | (0.0, 1.2) |
| Fatigue | 8 (0.7) | 1.7 | (0.8, 3.4) | 3 (0.3) | 0.7 | (0.1, 1.9) |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| Injection site pain | 8 (0.7) | 1.7 | (0.8, 3.4) | 8 (0.7) | 1.8 | (0.8, 3.5) |
| Injection site swelling | 2 (0.2) | 0.4 | (0.1, 1.6) | 0 | 0.0 | (0.0, 0.8) |
| Peripheral swelling | 1 (0.1) | 0.2 | (0.0, 1.2) | 0 | 0.0 | (0.0, 0.8) |
| Pyrexia | 6 (0.5) | 1.3 | (0.5, 2.9) | 0 | 0.0 | (0.0, 0.8) |
| INVESTIGATIONS | 1 (0.1) | 0.2 | (0.0, 1.2) | 0 | 0.0 | (0.0, 0.8) |
| SARS-CoV-2 antibody test positive | 1 (0.1) | 0.2 | (0.0, 1.2) | 0 | 0.0 | (0.0, 0.8) |
| MUSCULOSKELETAL AND CONNECTIVE TISSUE DISORDERS | 4 (0.4) | 0.9 | (0.2, 2.2) | 1 (0.1) | 0.2 | (0.0, 1.2) |
| Musculoskeletal chest pain | 1 (0.1) | 0.2 | (0.0, 1.2) | 0 | 0.0 | (0.0, 0.8) |
| Myalgia | 3 (0.3) | 0.7 | (0.1, 1.9) | 1 (0.1) | 0.2 | (0.0, 1.2) |
| NERVOUS SYSTEM DISORDERS | 5 (0.4) | 1.1 | (0.4, 2.6) | 5 (0.4) | 1.1 | (0.4, 2.6) |
| Dizziness | 1 (0.1) | 0.2 | (0.0, 1.2) | 1 (0.1) | 0.2 | (0.0, 1.2) |
| Headache | 3 (0.3) | 0.7 | (0.1, 1.9) | 3 (0.3) | 0.7 | (0.1, 1.9) |
| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
| | BNT162b2 (30 µg) (Na=1131, TEb=4.6) | | Placebo (Na=1129, TEb=4.5) | | | |
| nc (%) | IRd | (95% CIc) | nc (%) | IRd | (95% CIc) | |
| Migraine | 1 (0.1) | 0.2 | (0.0, 1.2) | 0 | 0.0 | (0.0, 0.8) |
| Presyncope | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.2) |
| PSYCHIATRIC DISORDERS | 1 (0.1) | 0.2 | (0.0, 1.2) | 0 | 0.0 | (0.0, 0.8) |
| Disorientation | 1 (0.1) | 0.2 | (0.0, 1.2) | 0 | 0.0 | (0.0, 0.8) |
| RESPIRATORY, THORACIC AND MEDIASTINAL DISORDERS | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.2) |
| Rhinorrhoea | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.2) |
| SKIN AND SUBCUTANEOUS TISSUE DISORDERS | 2 (0.2) | 0.4 | (0.1, 1.6) | 7 (0.6) | 1.6 | (0.6, 3.2) |
| Rash | 1 (0.1) | 0.2 | (0.0, 1.2) | 2 (0.2) | 0.4 | (0.1, 1.6) |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| Rash maculo-papular | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.2) |
| Urticaria | 1 (0.1) | 0.2 | (0.0, 1.2) | 4 (0.4) | 0.9 | (0.2, 2.3) |

Note: MedDRA (v24.0) coding dictionary applied.a. N = number of subjects in the specified group. This value is the denominator for the percentage calculations.b. TE = total exposure time in 100 person-years across all subjects in the specified group....

## 14.33 Incidence Rates of at Least 1 Severe Adverse Event From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population

**Table 14.33.** Incidence Rates of at Least 1 Severe Adverse Event From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| | BNT162b2 (30 µg) (Na=1131, TEb=4.6) | | | Placebo (Na=1129, TEb=4.5) | | |
| | nc (%) | IRd | (95% CIe) | nc (%) | IRd | (95% CIe) |
| Any event | 13 (1.1) | 2.8 | (1.5, 4.9) | 5 (0.4) | 1.1 | (0.4, 2.6) |
| GASTROINTESTINAL DISORDERS | 1 (0.1) | 0.2 | (0.0, 1.2) | 0 | 0.0 | (0.0, 0.8) |
| Abdominal pain | 1 (0.1) | 0.2 | (0.0, 1.2) | 0 | 0.0 | (0.0, 0.8) |
| Constipation | 1 (0.1) | 0.2 | (0.0, 1.2) | 0 | 0.0 | (0.0, 0.8) |
| GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS | 2 (0.2) | 0.4 | (0.1, 1.6) | 1 (0.1) | 0.2 | (0.0, 1.2) |
| Fatigue | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.2) |
| Pyrexia | 2 (0.2) | 0.4 | (0.1, 1.6) | 0 | 0.0 | (0.0, 0.8) |
| INFECTIONS AND INFESTATIONS | 1 (0.1) | 0.2 | (0.0, 1.2) | 1 (0.1) | 0.2 | (0.0, 1.2) |
| Anal abscess | 1 (0.1) | 0.2 | (0.0, 1.2) | 0 | 0.0 | (0.0, 0.8) |
| Appendicitis | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.2) |
| INJURY, POISONING AND PROCEDURAL COMPLICATIONS | 2 (0.2) | 0.4 | (0.1, 1.6) | 2 (0.2) | 0.4 | (0.1, 1.6) |
| Femur fracture | 1 (0.1) | 0.2 | (0.0, 1.2) | 0 | 0.0 | (0.0, 0.8) |
| Patella fracture | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.2) |
| Procedural pain | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.2) |
| Upper limb fracture | 1 (0.1) | 0.2 | (0.0, 1.2) | 0 | 0.0 | (0.0, 0.8) |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| MUSCULOSKELETAL AND CONNECTIVE TISSUE DISORDERS | 1 (0.1) | 0.2 | (0.0, 1.2) | 0 | 0.0 | (0.0, 0.8) |
| Arthralgia | 1 (0.1) | 0.2 | (0.0, 1.2) | 0 | 0.0 | (0.0, 0.8) |
| NERVOUS SYSTEM DISORDERS | 2 (0.2) | 0.4 | (0.1, 1.6) | 1 (0.1) | 0.2 | (0.0, 1.2) |
| Headache | 1 (0.1) | 0.2 | (0.0, 1.2) | 1 (0.1) | 0.2 | (0.0, 1.2) |
| Migraine | 1 (0.1) | 0.2 | (0.0, 1.2) | 0 | 0.0 | (0.0, 0.8) |
| PSYCHIATRIC DISORDERS | 4 (0.4) | 0.9 | (0.2, 2.2) | 0 | 0.0 | (0.0, 0.8) |
| Anxiety | 1 (0.1) | 0.2 | (0.0, 1.2) | 0 | 0.0 | (0.0, 0.8) |
| Depression | 2 (0.2) | 0.4 | (0.1, 1.6) | 0 | 0.0 | (0.0, 0.8) |
| Suicidal ideation | 2 (0.2) | 0.4 | (0.1, 1.6) | 0 | 0.0 | (0.0, 0.8) |
| SKIN AND SUBCUTANEOUS TISSUE DISORDERS | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.2) |
| Urticaria | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.2) |
| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
| | BNT162b2 (30 µg) (N°=1131, TEb=4.6) | | Placebo (N°=1129, TEb=4.5) | | | |
| | n° (%) | IRd | (95% CIe) | n° (%) | IRd | (95% CIe) |

Note: MedDRA (v24.0) coding dictionary applied. a. N = number of subjects in the specified group. This value is the denominator for the percentage calculations. b. TE = total exposure time in 100 person-years across all subjects in the specified grou...

## 14.34 Incidence Rates of at Least 1 Life-Threatening Adverse Event From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population

**Table 14.34.** Incidence Rates of at Least 1 Life-Threatening Adverse Event From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| | BNT162b2 (30 µg) (Na=1131, TEb=4.6) | | | Placebo (Na=1129, TEb=4.5) | | |
| nc (%) | IRd | (95% CIe) | nc (%) | IRd | (95% CIe) | |
| Any event | 2 (0.2) | 0.4 | (0.1, 1.6) | 1 (0.1) | 0.2 | (0.0, 1.2) |
| GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS | 1 (0.1) | 0.2 | (0.0, 1.2) | 0 | 0.0 | (0.0, 0.8) |
| Pyrexia | 1 (0.1) | 0.2 | (0.0, 1.2) | 0 | 0.0 | (0.0, 0.8) |
| INFECTIONS AND INFESTATIONS | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.2) |
| Appendicitis | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.2) |
| Focal peritonitis | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.2) |
| PSYCHIATRIC DISORDERS | 1 (0.1) | 0.2 | (0.0, 1.2) | 0 | 0.0 | (0.0, 0.8) |
| Suicidal ideation | 1 (0.1) | 0.2 | (0.0, 1.2) | 0 | 0.0 | (0.0, 0.8) |

## 14.35 Incidence Rates of at Least 1 Adverse Event From Unblinding Date to Data Cutoff Date (02SEP2021), by System Organ Class and Preferred Term – Open-Label Follow-up Period – Subjects Who Originally Received BNT162b2 – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population

**Table 14.35.** Incidence Rates of at Least 1 Adverse Event From Unblinding Date to Data Cutoff Date (02SEP2021), by System Organ Class and Preferred Term – Open-Label Follow-up Period – Subjects Who Originally Received BNT162b2 – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | |
|---|---|---|---|
| BNT162b2 (30 µg) (N°=1107, TE°=3.3) | | | |
| n° (%) | IRd | (95% CIe) | |
| Any event | 18 (1.6) | 5.4 | (3.2, 8.5) |
| CARDIAC DISORDERS | 1 (0.1) | 0.3 | (0.0, 1.7) |
| Tachycardia | 1 (0.1) | 0.3 | (0.0, 1.7) |
| CONGENITAL, FAMILIAL AND GENETIC DISORDERS | 1 (0.1) | 0.3 | (0.0, 1.7) |
| Syringomyelia | 1 (0.1) | 0.3 | (0.0, 1.7) |
| GASTROINTESTINAL DISORDERS | 3 (0.3) | 0.9 | (0.2, 2.6) |
| Abdominal pain upper | 2 (0.2) | 0.6 | (0.1, 2.2) |
| Aphthous ulcer | 1 (0.1) | 0.3 | (0.0, 1.7) |
| Nausea | 1 (0.1) | 0.3 | (0.0, 1.7) |
| GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS | 4 (0.4) | 1.2 | (0.3, 3.1) |
| Fatigue | 2 (0.2) | 0.6 | (0.1, 2.2) |
| Injection site pain | 3 (0.3) | 0.9 | (0.2, 2.6) |
| Pain | 1 (0.1) | 0.3 | (0.0, 1.7) |
| Pyrexia | 2 (0.2) | 0.6 | (0.1, 2.2) |
| INFECTIONS AND INFESTATIONS | 3 (0.3) | 0.9 | (0.2, 2.6) |
| Appendicitis | 2 (0.2) | 0.6 | (0.1, 2.2) |
| Otitis media | 1 (0.1) | 0.3 | (0.0, 1.7) |
| INJURY, POISONING AND PROCEDURAL COMPLICATIONS | 2 (0.2) | 0.6 | (0.1, 2.2) |
| Hand fracture | 1 (0.1) | 0.3 | (0.0, 1.7) |
| Upper limb fracture | 1 (0.1) | 0.3 | (0.0, 1.7) |
| NERVOUS SYSTEM DISORDERS | 6 (0.5) | 1.8 | (0.7, 3.9) |
| Dizziness | 2 (0.2) | 0.6 | (0.1, 2.2) |
| Headache | 2 (0.2) | 0.6 | (0.1, 2.2) |
| Presyncope | 2 (0.2) | 0.6 | (0.1, 2.2) |
| Syncope | 1 (0.1) | 0.3 | (0.0, 1.7) |
| PSYCHIATRIC DISORDERS | 1 (0.1) | 0.3 | (0.0, 1.7) |
| Anxiety | 1 (0.1) | 0.3 | (0.0, 1.7) |
| System Organ Class Preferred Term | Vaccine Group (as Administered) | | |
| BNT162b2 (30 µg) (N°=1107, TEb=3.3) | | | |
| | n° (%) | IRd | (95% CIe) |

| System Organ Class Preferred Term | Vaccine Group (as Administered) |
|---|---|

Note: MedDRA (v24.0) coding dictionary applied. a. N = number of subjects in the specified group. This value is the denominator for the percentage calculations. b. TE = total exposure time in 100 person-years across all subjects in the specified grou...

## 14.36 Incidence Rates of at Least 1 Related Adverse Event From Unblinding Date to Data Cutoff Date (02SEP2021), by System Organ Class and Preferred Term – Open-Label Follow-up Period – Subjects Who Originally Received BNT162b2 – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population

**Table 14.36.** Incidence Rates of at Least 1 Related Adverse Event From Unblinding Date to Data Cutoff Date (02SEP2021), by System Organ Class and Preferred Term – Open-Label Follow-up Period – Subjects Who Originally Received BNT162b2 – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | |
|---|---|---|---|
| na (%) | IRd | (95% CIe) | |
| Any event | 4 (0.4) | 1.2 | (0.3, 3.1) |
| GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS | 4 (0.4) | 1.2 | (0.3, 3.1) |
| Fatigue | 2 (0.2) | 0.6 | (0.1, 2.2) |
| Injection site pain | 3 (0.3) | 0.9 | (0.2, 2.6) |
| Pain | 1 (0.1) | 0.3 | (0.0, 1.7) |
| Pyrexia | 2 (0.2) | 0.6 | (0.1, 2.2) |
| NERVOUS SYSTEM DISORDERS | 2 (0.2) | 0.6 | (0.1, 2.2) |
| Dizziness | 1 (0.1) | 0.3 | (0.0, 1.7) |
| Headache | 2 (0.2) | 0.6 | (0.1, 2.2) |

Note: MedDRA (v24.0) coding dictionary applied.a. N = number of subjects in the specified group. This value is the denominator for the percentage calculations.b. TE = total exposure time in 100 person-years across all subjects in the specified group....

## 14.37 Incidence Rates of at Least 1 Severe Adverse Event From Unblinding Date to Data Cutoff Date (02SEP2021), by System Organ Class and Preferred Term – Open-Label Follow-up Period – Subjects Who Originally Received BNT162b2 – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population

**Table 14.37.** Incidence Rates of at Least 1 Severe Adverse Event From Unblinding Date to Data Cutoff Date (02SEP2021), by System Organ Class and Preferred Term – Open-Label Follow-up Period – Subjects Who Originally Received BNT162b2 – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | |
|---|---|---|---|
| BNT162b2 (30 µg) (Na=1107, TEb=3.3) nc (%) | IRd | (95% CIe) | |
| Any event | 3 (0.3) | 0.9 | (0.2, 2.6) |
| CONGENITAL, FAMILIAL AND GENETIC DISORDERS | 1 (0.1) | 0.3 | (0.0, 1.7) |
| Syringomyelia | 1 (0.1) | 0.3 | (0.0, 1.7) |
| GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS | 2 (0.2) | 0.6 | (0.1, 2.2) |
| Pyrexia | 2 (0.2) | 0.6 | (0.1, 2.2) |

Note: MedDRA (v24.0) coding dictionary applied.a. N = number of subjects in the specified group. This value is the denominator for the percentage calculations.b. TE = total exposure time in 100 person-years across all subjects in the specified group....

## 14.38 Number % of Subjects Reporting at Least 1 Related Adverse Event From Dose 1 to 6 Months After Dose 2, by System Organ Class and Preferred Term - Subjects With at Least 6 Months of Follow-up Time After Dose 2 - Phase 2/3 Subjects 12 Through 15 Years of Age (Subjects Who Originally Received BNT162b2) - Safety Population

**Table 14.38.** Number % of Subjects Reporting at Least 1 Related Adverse Event From Dose 1 to 6 Months After Dose 2, by System Organ Class and Preferred Term - Subjects With at Least 6 Months of Follow-up Time After Dose 2 - Phase 2/3 Subjects 12 Through 15 Years of Age (Subjects Who Originally Received BNT162b2) - Safety Population

| System Organ Class Preferred Term | Vaccine Group (as Administered) | |
|---|---|---|
| BNT162b2 (30 µg) (N°=1113) n° (%) | (95% CI#) | |
| Any event | 34 (3.1) | (2.1, 4.2) |
| BLOOD AND LYMPHATIC SYSTEM DISORDERS | 7 (0.6) | (0.3, 1.3) |
| Lymphadenopathy | 7 (0.6) | (0.3, 1.3) |
| GASTROINTESTINAL DISORDERS | 11 (1.0) | (0.5, 1.8) |
| Nausea | 5 (0.4) | (0.1, 1.0) |
| Abdominal pain | 2 (0.2) | (0.0, 0.6) |
| Diarrhoea | 2 (0.2) | (0.0, 0.6) |
| Lip swelling | 1 (0.1) | (0.0, 0.5) |
| Mouth swelling | 1 (0.1) | (0.0, 0.5) |
| Oral mucosal blistering | 1 (0.1) | (0.0, 0.5) |
| Vomiting | 1 (0.1) | (0.0, 0.5) |
| GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS | 15 (1.3) | (0.8, 2.2) |
| Fatigue | 8 (0.7) | (0.3, 1.4) |
| Injection site pain | 8 (0.7) | (0.3, 1.4) |
| Pyrexia | 5 (0.4) | (0.1, 1.0) |
| Chills | 2 (0.2) | (0.0, 0.6) |
| Injection site swelling | 2 (0.2) | (0.0, 0.6) |
| Peripheral swelling | 1 (0.1) | (0.0, 0.5) |
| INVESTIGATIONS | 1 (0.1) | (0.0, 0.5) |
| SARS-CoV-2 antibody test positive | 1 (0.1) | (0.0, 0.5) |
| MUSCULOSKELETAL AND CONNECTIVE TISSUE DISORDERS | 4 (0.4) | (0.1, 0.9) |
| Myalgia | 3 (0.3) | (0.1, 0.8) |
| Musculoskeletal chest pain | 1 (0.1) | (0.0, 0.5) |
| NERVOUS SYSTEM DISORDERS | 5 (0.4) | (0.1, 1.0) |
| Headache | 3 (0.3) | (0.1, 0.8) |
| Dizziness | 1 (0.1) | (0.0, 0.5) |
| Migraine | 1 (0.1) | (0.0, 0.5) |
| PSYCHIATRIC DISORDERS | 1 (0.1) | (0.0, 0.5) |
| Disorientation | 1 (0.1) | (0.0, 0.5) |
| SKIN AND SUBCUTANEOUS TISSUE DISORDERS | 1 (0.1) | (0.0, 0.5) |
| Urticaria | 1 (0.1) | (0.0, 0.5) |

| System Organ Class Preferred Term | Vaccine Group (as Administered) |
|---|---|
| System Organ Class Preferred Term<br>BNT162b2 (30 µg) (N°=1113)<br>nb(%) | Vaccine Group (as Administered)<br>(95% CI#) |
| Note: MedDRA (v24.0) coding dictionary applied. a. N = number of subjects in the specified group. This value is the denominator for the percentage calculations. b. n = Number of subjects reporting at least 1 occurrence of the specified event. For ”an... | |

## 14.39 Number (%) of Subjects Reporting at Least 1 Adverse Event From Dose 3 to 7 Days After Dose 3, by System Organ Class and Preferred Term – Open-Label Follow-up Period – Subjects Who Originally Received Placebo and Then Received BNT162b2 After Unblinding – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population

**Table 14.39.** Number (%) of Subjects Reporting at Least 1 Adverse Event From Dose 3 to 7 Days After Dose 3, by System Organ Class and Preferred Term – Open-Label Follow-up Period – Subjects Who Originally Received Placebo and Then Received BNT162b2 After Unblinding – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population

| System Organ Class Preferred Term | Vaccine Group (as Administered) | |
|---|---|---|
| BNT162b2 (30 µg) (N=1010) | | |
| nb(%) | (95% CIc) | |
| Any event | 198 (19.6) | (17.2, 22.2) |
| BLOOD AND LYMPHATIC SYSTEM DISORDERS | 1 (0.1) | (0.0, 0.6) |
| Lymphadenitis | 1 (0.1) | (0.0, 0.6) |
| EAR AND LABYRINTH DISORDERS | 1 (0.1) | (0.0, 0.6) |
| Motion sickness | 1 (0.1) | (0.0, 0.6) |
| GASTROINTESTINAL DISORDERS | 9 (0.9) | (0.4, 1.7) |
| Nausea | 7 (0.7) | (0.3, 1.4) |
| Vomiting | 2 (0.2) | (0.0, 0.7) |
| Abdominal pain upper | 1 (0.1) | (0.0, 0.6) |
| GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS | 178 (17.6) | (15.3, 20.1) |
| Injection site pain | 131 (13.0) | (11.0, 15.2) |
| Fatigue | 68 (6.7) | (5.3, 8.5) |
| Chills | 29 (2.9) | (1.9, 4.1) |
| Pyrexia | 28 (2.8) | (1.8, 4.0) |
| Pain | 17 (1.7) | (1.0, 2.7) |
| Injection site erythema | 3 (0.3) | (0.1, 0.9) |
| Injection site swelling | 3 (0.3) | (0.1, 0.9) |
| Injection site bruising | 2 (0.2) | (0.0, 0.7) |
| Malaise | 2 (0.2) | (0.0, 0.7) |
| Injection site hypoesthesia | 1 (0.1) | (0.0, 0.6) |
| Non-cardiac chest pain | 1 (0.1) | (0.0, 0.6) |
| Thirst | 1 (0.1) | (0.0, 0.6) |
| INFECTIONS AND INFESTATIONS | 1 (0.1) | (0.0, 0.6) |
| Paronychia | 1 (0.1) | (0.0, 0.6) |
| Skin candida | 1 (0.1) | (0.0, 0.6) |
| INJURY, POISONING AND PROCEDURAL COMPLICATIONS | 2 (0.2) | (0.0, 0.7) |
| Joint injury | 1 (0.1) | (0.0, 0.6) |
| Muscle strain | 1 (0.1) | (0.0, 0.6) |
| INVESTIGATIONS | 2 (0.2) | (0.0, 0.7) |
| Body temperature increased | 2 (0.2) | (0.0, 0.7) |
| METABOLISM AND NUTRITION DISORDERS | 1 (0.1) | (0.0, 0.6) |
| Decreased appetite | 1 (0.1) | (0.0, 0.6) |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | |
|---|---|---|
| MUSCULOSKELETAL AND CONNECTIVE TISSUE DISORDERS | 24 (2.4) | (1.5, 3.5) |
| System Organ Class Preferred Term | Vaccine Group (as Administered) | |
| BNT162b2 (30 µg) (N=1010) | | |
| nb (%) | (95% CIc) | |
| Myalgia | 18 (1.8) | (1.1, 2.8) |
| Pain in extremity | 5 (0.5) | (0.2, 1.2) |
| Musculoskeletal stiffness | 1 (0.1) | (0.0, 0.6) |
| NERVOUS SYSTEM DISORDERS | 42 (4.2) | (3.0, 5.6) |
| Headache | 40 (4.0) | (2.8, 5.4) |
| Dizziness | 2 (0.2) | (0.0, 0.7) |
| Syncope | 1 (0.1) | (0.0, 0.6) |
| RESPIRATORY, THORACIC AND MEDIASTINAL DISORDERS | 1 (0.1) | (0.0, 0.6) |
| Rhinorrhoea | 1 (0.1) | (0.0, 0.6) |
| SKIN AND SUBCUTANEOUS TISSUE DISORDERS | 2 (0.2) | (0.0, 0.7) |
| Hyperhidrosis | 1 (0.1) | (0.0, 0.6) |
| Photosensitivity reaction | 1 (0.1) | (0.0, 0.6) |

Note: Dose 3 = first dose of BNT162b2 (30 µg).Note: MedDRA (v24.0) coding dictionary applied.a. N = number of subjects in the specified group. This value is the denominator for the percentage calculations.b. n = Number of subjects reporting at least ...

## 14.40 Number (%) of Subjects Reporting at Least 1 Adverse Event From Dose 4 to 7 Days After Dose 4, by System Organ Class and Preferred Term – Open-Label Follow-up Period – Subjects Who Originally Received Placebo and Then Received BNT162b2 After Unblinding – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population

**Table 14.40.** Number (%) of Subjects Reporting at Least 1 Adverse Event From Dose 4 to 7 Days After Dose 4, by System Organ Class and Preferred Term – Open-Label Follow-up Period – Subjects Who Originally Received Placebo and Then Received BNT162b2 After Unblinding – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population

| System Organ Class Preferred Term | Vaccine Group (as Administered) | |
|---|---|---|
| BNT162b2 (30 µg) (N°=992) | | |
| n° (%) | (95% CI°) | |
| Any event | 138 (13.9) | (11.8, 16.2) |
| BLOOD AND LYMPHATIC SYSTEM DISORDERS | 1 (0.1) | (0.0, 0.6) |
| Lymphadenopathy | 1 (0.1) | (0.0, 0.6) |
| CARDIAC DISORDERS | 1 (0.1) | (0.0, 0.6) |
| Myocarditis | 1 (0.1) | (0.0, 0.6) |
| GASTROINTESTINAL DISORDERS | 12 (1.2) | (0.6, 2.1) |
| Nausea | 7 (0.7) | (0.3, 1.4) |
| Vomiting | 5 (0.5) | (0.2, 1.2) |
| Abdominal pain upper | 1 (0.1) | (0.0, 0.6) |
| Diarrhoea | 1 (0.1) | (0.0, 0.6) |
| GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS | 120 (12.1) | (10.1, 14.3) |
| Injection site pain | 60 (6.0) | (4.6, 7.7) |
| Fatigue | 56 (5.6) | (4.3, 7.3) |
| Pyrexia | 41 (4.1) | (3.0, 5.6) |
| Chills | 21 (2.1) | (1.3, 3.2) |
| Pain | 21 (2.1) | (1.3, 3.2) |
| Malaise | 5 (0.5) | (0.2, 1.2) |
| Injection site bruising | 2 (0.2) | (0.0, 0.7) |
| Injection site erythema | 2 (0.2) | (0.0, 0.7) |
| Axillary pain | 1 (0.1) | (0.0, 0.6) |
| Chest discomfort | 1 (0.1) | (0.0, 0.6) |
| Chest pain | 1 (0.1) | (0.0, 0.6) |
| Injection site reaction | 1 (0.1) | (0.0, 0.6) |
| Injection site swelling | 1 (0.1) | (0.0, 0.6) |
| INFECTIONS AND INFESTATIONS | 3 (0.3) | (0.1, 0.9) |
| Appendicitis | 1 (0.1) | (0.0, 0.6) |
| Pharyngitis streptococcal | 1 (0.1) | (0.0, 0.6) |
| Sinusitis | 1 (0.1) | (0.0, 0.6) |
| INVESTIGATIONS | 2 (0.2) | (0.0, 0.7) |
| Body temperature increased | 2 (0.2) | (0.0, 0.7) |
| METABOLISM AND NUTRITION DISORDERS | 1 (0.1) | (0.0, 0.6) |
| Vitamin D deficiency | 1 (0.1) | (0.0, 0.6) |
| MUSCULOSKELETAL AND CONNECTIVE TISSUE DISORDERS | 25 (2.5) | (1.6, 3.7) |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | |
|---|---|---|
| System Organ Class Preferred Term | Vaccine Group (as Administered) | |
| BNT162b2 (30 µg) (N°=992) | | |
| n° (%) | (95% CI#) | |
| Myalgia | 21 (2.1) | (1.3, 3.2) |
| Pain in extremity | 3 (0.3) | (0.1, 0.9) |
| Arthralgia | 2 (0.2) | (0.0, 0.7) |
| Musculoskeletal chest pain | 1 (0.1) | (0.0, 0.6) |
| Neck pain | 1 (0.1) | (0.0, 0.6) |
| NEOPLASMS BENIGN, MALIGNANT AND UNSPECIFIED (INCL CYSTS AND POLYPS) | 1 (0.1) | (0.0, 0.6) |
| Skin papilloma | 1 (0.1) | (0.0, 0.6) |
| NERVOUS SYSTEM DISORDERS | 40 (4.0) | (2.9, 5.5) |
| Headache | 38 (3.8) | (2.7, 5.2) |
| Dizziness | 2 (0.2) | (0.0, 0.7) |
| Epilepsy | 1 (0.1) | (0.0, 0.6) |
| PSYCHIATRIC DISORDERS | 1 (0.1) | (0.0, 0.6) |
| Major depression | 1 (0.1) | (0.0, 0.6) |
| RENAL AND URINARY DISORDERS | 1 (0.1) | (0.0, 0.6) |
| Dysuria | 1 (0.1) | (0.0, 0.6) |
| RESPIRATORY, THORACIC AND MEDIASTINAL DISORDERS | 1 (0.1) | (0.0, 0.6) |
| Cough | 1 (0.1) | (0.0, 0.6) |
| Nasal congestion | 1 (0.1) | (0.0, 0.6) |

Note: Dose 4 = second dose of BNT162b2 (30 µg).Note: Subjects who did not receive Dose 4 or who received a different vaccine at Dose 3 and Dose 4 were excluded from this table.Note: MedDRA (v24.0) coding dictionary applied.a. N = number of subjects i...

## 14.41 Incidence Rates of at Least 1 Related Adverse Event From Dose 3 to Data Cutoff Date (02SEP2021), by System Organ Class and Preferred Term – Open-Label Follow-up Period – Subjects Who Originally Received Placebo and Then Received BNT162b2 After Unblinding – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population

**Table 14.41.** Incidence Rates of at Least 1 Related Adverse Event From Dose 3 to Data Cutoff Date (02SEP2021), by System Organ Class and Preferred Term – Open-Label Follow-up Period – Subjects Who Originally Received Placebo and Then Received BNT162b2 After Unblinding – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | |
|---|---|---|---|
| BNT162b2 (30 µg) (Na=1010, TEb=2.9) | | | |
| nc (%) | IRd | (95% CIe) | |
| Any event | 242 (24.0) | 82.5 | (72.4, 93.5) |
| BLOOD AND LYMPHATIC SYSTEM DISORDERS | 2 (0.2) | 0.7 | (0.1, 2.5) |
| Lymphadenitis | 1 (0.1) | 0.3 | (0.0, 1.9) |
| Lymphadenopathy | 1 (0.1) | 0.3 | (0.0, 1.9) |
| GASTROINTESTINAL DISORDERS | 18 (1.8) | 6.1 | (3.6, 9.7) |
| Abdominal pain upper | 2 (0.2) | 0.7 | (0.1, 2.5) |
| Diarrhoea | 1 (0.1) | 0.3 | (0.0, 1.9) |
| Nausea | 12 (1.2) | 4.1 | (2.1, 7.1) |
| Vomiting | 6 (0.6) | 2.0 | (0.8, 4.4) |
| GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS | 223 (22.1) | 76.0 | (66.3, 86.6) |
| Axillary pain | 1 (0.1) | 0.3 | (0.0, 1.9) |
| Chills | 45 (4.5) | 15.3 | (11.2, 20.5) |
| Fatigue | 104 (10.3) | 35.4 | (29.0, 42.9) |
| Injection site bruising | 3 (0.3) | 1.0 | (0.2, 3.0) |
| Injection site erythema | 5 (0.5) | 1.7 | (0.6, 4.0) |
| Injection site hypoesthesia | 1 (0.1) | 0.3 | (0.0, 1.9) |
| Injection site pain | 157 (15.5) | 53.5 | (45.5, 62.5) |
| Injection site reaction | 1 (0.1) | 0.3 | (0.0, 1.9) |
| Injection site swelling | 4 (0.4) | 1.4 | (0.4, 3.5) |
| Malaise | 7 (0.7) | 2.4 | (1.0, 4.9) |
| Non-cardiac chest pain | 1 (0.1) | 0.3 | (0.0, 1.9) |
| Pain | 35 (3.5) | 11.9 | (8.3, 16.6) |
| Pyrexia | 63 (6.2) | 21.5 | (16.5, 27.5) |
| Thirst | 1 (0.1) | 0.3 | (0.0, 1.9) |
| INFECTIONS AND INFESTATIONS | 1 (0.1) | 0.3 | (0.0, 1.9) |
| Appendicitis | 1 (0.1) | 0.3 | (0.0, 1.9) |
| INVESTIGATIONS | 3 (0.3) | 1.0 | (0.2, 3.0) |
| Body temperature increased | 3 (0.3) | 1.0 | (0.2, 3.0) |
| METABOLISM AND NUTRITION DISORDERS | 1 (0.1) | 0.3 | (0.0, 1.9) |
| Decreased appetite | 1 (0.1) | 0.3 | (0.0, 1.9) |
| MUSCULOSKELETAL AND CONNECTIVE TISSUE DISORDERS | 46 (4.6) | 15.7 | (11.5, 20.9) |
| Arthralgia | 2 (0.2) | 0.7 | (0.1, 2.5) |

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | |
|---|---|---|---|
| Musculoskeletal chest pain | 1 (0.1) | 0.3 | (0.0, 1.9) |
| System Organ Class Preferred Term | Vaccine Group (as Administered) | | |
| BNT162b2 (30 µg) (Na=1010, TEb=2.9) | | | |
| n° (%) | IRd | (95% CIe) | |
| Musculoskeletal stiffness | 1 (0.1) | 0.3 | (0.0, 1.9) |
| Myalgia | 37 (3.7) | 12.6 | (8.9, 17.4) |
| Pain in extremity | 8 (0.8) | 2.7 | (1.2, 5.4) |
| NERVOUS SYSTEM DISORDERS | 73 (7.2) | 24.9 | (19.5, 31.3) |
| Dizziness | 4 (0.4) | 1.4 | (0.4, 3.5) |
| Headache | 70 (6.9) | 23.8 | (18.6, 30.1) |
| Syncope | 1 (0.1) | 0.3 | (0.0, 1.9) |
| RESPIRATORY, THORACIC AND MEDIASTINAL DISORDERS | 1 (0.1) | 0.3 | (0.0, 1.9) |
| Rhinorrhoea | 1 (0.1) | 0.3 | (0.0, 1.9) |
| SKIN AND SUBCUTANEOUS TISSUE DISORDERS | 1 (0.1) | 0.3 | (0.0, 1.9) |
| Hyperhidrosis | 1 (0.1) | 0.3 | (0.0, 1.9) |

Note: Dose 3 = First dose of BNT162b2 (30 µg). Note: MedDRA (v24.0) coding dictionary applied. a. N = number of subjects in the specified group. This value is the denominator for the percentage calculations. b. TE = total exposure time in 100 person-...

## 14.42 Number (%) of Subjects Reporting at Least 1 Immediate Adverse Event After Vaccination (Dose 3/4), by System Organ Class and Preferred Term – Open-Label Follow-up Period – Subjects Who Originally Received Placebo and Then Received BNT162b2 After Unblinding – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population

**Table 14.42.** Number (%) of Subjects Reporting at Least 1 Immediate Adverse Event After Vaccination (Dose 3/4), by System Organ Class and Preferred Term – Open-Label Follow-up Period – Subjects Who Originally Received Placebo and Then Received BNT162b2 After Unblinding – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population

| System Organ Class Preferred Term | Vaccine Group (as Administered) | |
|---|---|---|
| BNT162b2 (30 µg) (Na=1010) | | |
| nb (%) | (95% CIc) | |
| Any event | 7 (0.7) | (0.3, 1.4) |
| GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS | 7 (0.7) | (0.3, 1.4) |
| Injection site pain | 6 (0.6) | (0.2, 1.3) |
| Injection site erythema | 1 (0.1) | (0.0, 0.6) |

Note: Dose 3 = first dose of BNT162b2 (30 µg), Dose 4 = second dose of BNT162b2 (30 µg). Note: MedDRA (v24.0) coding dictionary applied. Note: Immediate AE refers to an AE reported in the 30-minute observation period after vaccination. a. N = number ...

## 14.43 Incidence Rates of at Least 1 Severe Adverse Event From Dose 3 to Data Cutoff Date (02SEP2021), by System Organ Class and Preferred Term – Open-Label Follow-up Period – Subjects Who Originally Received Placebo and Then Received BNT162b2 After Unblinding – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population

**Table 14.43.** Incidence Rates of at Least 1 Severe Adverse Event From Dose 3 to Data Cutoff Date (02SEP2021), by System Organ Class and Preferred Term – Open-Label Follow-up Period – Subjects Who Originally Received Placebo and Then Received BNT162b2 After Unblinding – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | |
|---|---|---|---|
| BNT162b2 (30 µg) (Na=1010, TEb=2.9) nc (%) | IRd | (95% CIe) | |
| Any event | 12 (1.2) | 4.1 | (2.1, 7.1) |
| CARDIAC DISORDERS | 1 (0.1) | 0.3 | (0.0, 1.9) |
| Myocarditis | 1 (0.1) | 0.3 | (0.0, 1.9) |
| GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS | 6 (0.6) | 2.0 | (0.8, 4.4) |
| Fatigue | 2 (0.2) | 0.7 | (0.1, 2.5) |
| Malaise | 1 (0.1) | 0.3 | (0.0, 1.9) |
| Pyrexia | 3 (0.3) | 1.0 | (0.2, 3.0) |
| INJURY, POISONING AND PROCEDURAL COMPLICATIONS | 3 (0.3) | 1.0 | (0.2, 3.0) |
| Concussion | 1 (0.1) | 0.3 | (0.0, 1.9) |
| Muscle strain | 1 (0.1) | 0.3 | (0.0, 1.9) |
| Sunburn | 1 (0.1) | 0.3 | (0.0, 1.9) |
| MUSCULOSKELETAL AND CONNECTIVE TISSUE DISORDERS | 1 (0.1) | 0.3 | (0.0, 1.9) |
| Musculoskeletal chest pain | 1 (0.1) | 0.3 | (0.0, 1.9) |
| Myalgia | 1 (0.1) | 0.3 | (0.0, 1.9) |
| NERVOUS SYSTEM DISORDERS | 1 (0.1) | 0.3 | (0.0, 1.9) |
| Somnolence | 1 (0.1) | 0.3 | (0.0, 1.9) |
| PSYCHIATRIC DISORDERS | 1 (0.1) | 0.3 | (0.0, 1.9) |
| Major depression | 1 (0.1) | 0.3 | (0.0, 1.9) |

Note: Dose 3 = First dose of BNT162b2 (30 µg). Note: MedDRA (v24.0) coding dictionary applied. a. N = number of subjects in the specified group. This value is the denominator for the percentage calculations. b. TE = total exposure time in 100 person-...

## 14.44 Incidence Rates of at Least 1 Serious Adverse Event From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term, by Baseline SARS-CoV-2 Status – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Baseline SARS-CoV-2 Status: Negative

**Table 14.44.** Incidence Rates of at Least 1 Serious Adverse Event From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term, by Baseline SARS-CoV-2 Status – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Baseline SARS-CoV-2 Status: Negative

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| | BNT162b2 (30 µg) (Na=1083, TEb=4.4) | | | Placebo (Na=1078, TEb=4.3) | | |
| nc (%) | IRd | (95% CIe) | nc (%) | IRd | (95% CIe) | |
| Any event | 10 (0.9) | 2.3 | (1.1, 4.2) | 2 (0.2) | 0.5 | (0.1, 1.7) |
| GASTROINTESTINAL DISORDERS | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| Abdominal pain | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| Constipation | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| INFECTIONS AND INFESTATIONS | 1 (0.1) | 0.2 | (0.0, 1.3) | 2 (0.2) | 0.5 | (0.1, 1.7) |
| Anal abscess | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| Appendicitis | 0 | 0.0 | (0.0, 0.8) | 2 (0.2) | 0.5 | (0.1, 1.7) |
| Focal peritonitis | 0 | 0.0 | (0.0, 0.8) | 1 (0.1) | 0.2 | (0.0, 1.3) |
| INJURY, POISONING AND PROCEDURAL COMPLICATIONS | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| Femur fracture | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| PSYCHIATRIC DISORDERS | 8 (0.7) | 1.8 | (0.8, 3.6) | 0 | 0.0 | (0.0, 0.9) |
| Anxiety | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| Conversion disorder | 1 (0.1) | 0.2 | (0.0, 1.3) | 0 | 0.0 | (0.0, 0.9) |
| Depression | 3 (0.3) | 0.7 | (0.1, 2.0) | 0 | 0.0 | (0.0, 0.9) |
| Suicidal ideation | 4 (0.4) | 0.9 | (0.2, 2.3) | 0 | 0.0 | (0.0, 0.9) |

## 14.45 Incidence Rates of at Least 1 Serious Adverse Event From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term, by Ethnicity – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Ethnicity: Hispanic/Latino

**Table 14.45.** Incidence Rates of at Least 1 Serious Adverse Event From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term, by Ethnicity – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Ethnicity: Hispanic/Latino

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| | BNT162b2 (30 µg) (Na=132, TEb=0.6) | | | Placebo (Na=130, TEb=0.5) | | |
| nc (%) | IRd | (95% CIe) | nc (%) | IRd | (95% CIe) | |
| Any event | 2 (1.5) | 3.6 | (0.4, 12.8) | 0 | 0.0 | (0.0, 6.9) |
| GASTROINTESTINAL DISORDERS | 1 (0.8) | 1.8 | (0.0, 9.9) | 0 | 0.0 | (0.0, 6.9) |
| Abdominal pain | 1 (0.8) | 1.8 | (0.0, 9.9) | 0 | 0.0 | (0.0, 6.9) |
| Constipation | 1 (0.8) | 1.8 | (0.0, 9.9) | 0 | 0.0 | (0.0, 6.9) |
| PSYCHIATRIC DISORDERS | 2 (1.5) | 3.6 | (0.4, 12.8) | 0 | 0.0 | (0.0, 6.9) |
| Conversion disorder | 1 (0.8) | 1.8 | (0.0, 9.9) | 0 | 0.0 | (0.0, 6.9) |
| Suicidal ideation | 1 (0.8) | 1.8 | (0.0, 9.9) | 0 | 0.0 | (0.0, 6.9) |

## 14.46 Incidence Rates of at Least 1 Serious Adverse Event From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term, by Ethnicity – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Ethnicity: Non-Hispanic/Non-Latino

**Table 14.46.** Incidence Rates of at Least 1 Serious Adverse Event From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term, by Ethnicity – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Ethnicity: Non-Hispanic/Non-Latino

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| | BNT162b2 (30 µg) (Na=997, TEb=4.0) | | | Placebo (Na=996, TEb=4.0) | | |
| nc (%) | IRd | (95% CIe) | nc (%) | IRd | (95% CIe) | |
| Any event | 8 (0.8) | 2.0 | (0.9, 3.9) | 2 (0.2) | 0.5 | (0.1, 1.8) |
| INFECTIONS AND INFESTATIONS | 1 (0.1) | 0.2 | (0.0, 1.4) | 2 (0.2) | 0.5 | (0.1, 1.8) |
| Anal abscess | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |
| Appendicitis | 0 | 0.0 | (0.0, 0.9) | 2 (0.2) | 0.5 | (0.1, 1.8) |
| Focal peritonitis | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.4) |
| INJURY, POISONING AND PROCEDURAL COMPLICATIONS | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |
| Femur fracture | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |
| PSYCHIATRIC DISORDERS | 6 (0.6) | 1.5 | (0.6, 3.3) | 0 | 0.0 | (0.0, 0.9) |
| Anxiety | 1 (0.1) | 0.2 | (0.0, 1.4) | 0 | 0.0 | (0.0, 0.9) |
| Depression | 3 (0.3) | 0.7 | (0.2, 2.2) | 0 | 0.0 | (0.0, 0.9) |
| Suicidal ideation | 3 (0.3) | 0.7 | (0.2, 2.2) | 0 | 0.0 | (0.0, 0.9) |

# 14.47 Incidence Rates of at Least 1 Serious Adverse Event From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term, by Race – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Race: White

**Table 14.47.** Incidence Rates of at Least 1 Serious Adverse Event From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term, by Race – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Race: White

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| | BNT162b2 (30 µg) (Nα=970, TEb=3.9) | | | Placebo (Nα=962, TEb=3.8) | | |
| nc(%) | IRd | (95% CIe) | nc(%) | IRd | (95% CIe) | |
| Any event | 7 (0.7) | 1.8 | (0.7, 3.7) | 2 (0.2) | 0.5 | (0.1, 1.9) |
| GASTROINTESTINAL DISORDERS | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| Abdominal pain | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| Constipation | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| INFECTIONS AND INFESTATIONS | 1 (0.1) | 0.3 | (0.0, 1.4) | 2 (0.2) | 0.5 | (0.1, 1.9) |
| Anal abscess | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| Appendicitis | 0 | 0.0 | (0.0, 0.9) | 2 (0.2) | 0.5 | (0.1, 1.9) |
| Focal peritonitis | 0 | 0.0 | (0.0, 0.9) | 1 (0.1) | 0.3 | (0.0, 1.5) |
| INJURY, POISONING AND PROCEDURAL COMPLICATIONS | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| Femur fracture | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| PSYCHIATRIC DISORDERS | 5 (0.5) | 1.3 | (0.4, 3.0) | 0 | 0.0 | (0.0, 1.0) |
| Conversion disorder | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| Depression | 1 (0.1) | 0.3 | (0.0, 1.4) | 0 | 0.0 | (0.0, 1.0) |
| Suicidal ideation | 3 (0.3) | 0.8 | (0.2, 2.3) | 0 | 0.0 | (0.0, 1.0) |

## 14.48 Incidence Rates of at Least 1 Serious Adverse Event From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term, by Race – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Race: Black or African American

**Table 14.48.** Incidence Rates of at Least 1 Serious Adverse Event From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term, by Race – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Race: Black or African American

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| | BNT162b2 (30 µg) (Na=52, TEb=0.2) | | | Placebo (Na=57, TEb=0.3) | | |
| nc (%) | IRd | (95% CIe) | nc (%) | IRd | (95% CIe) | |
| Any event | 1 (1.9) | 4.0 | (0.1, 22.5) | 0 | 0.0 | (0.0, 14.1) |
| PSYCHIATRIC DISORDERS | 1 (1.9) | 4.0 | (0.1, 22.5) | 0 | 0.0 | (0.0, 14.1) |
| Anxiety | 1 (1.9) | 4.0 | (0.1, 22.5) | 0 | 0.0 | (0.0, 14.1) |
| Depression | 1 (1.9) | 4.0 | (0.1, 22.5) | 0 | 0.0 | (0.0, 14.1) |

## 14.49 Incidence Rates of at Least 1 Serious Adverse Event From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term, by Race – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Race: All Others

**Table 14.49.** Incidence Rates of at Least 1 Serious Adverse Event From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term, by Race – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Race: All Others

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| | BNT162b2 (30 µg) (N°=109, TEb=0.4) | | | Placebo (N°=110, TEb=0.4) | | |
| | n° (%) | IRd | (95% CIe) | n° (%) | IRd | (95% CIe) |
| Any event | 2 (1.8) | 4.6 | (0.6, 16.7) | 0 | 0.0 | (0.0, 8.6) |
| PSYCHIATRIC DISORDERS | 2 (1.8) | 4.6 | (0.6, 16.7) | 0 | 0.0 | (0.0, 8.6) |
| Depression | 1 (0.9) | 2.3 | (0.1, 12.8) | 0 | 0.0 | (0.0, 8.6) |
| Suicidal ideation | 1 (0.9) | 2.3 | (0.1, 12.8) | 0 | 0.0 | (0.0, 8.6) |

## 14.50 Incidence Rates of at Least 1 Serious Adverse Event From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term, by Sex – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Sex: Male

**Table 14.50.** Incidence Rates of at Least 1 Serious Adverse Event From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term, by Sex – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Sex: Male

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| | BNT162b2 (30 µg) (Na=567, TEb=2.3) | | | Placebo (Na=585, TEb=2.3) | | |
| nc (%) | IRd | (95% CIc) | nc (%) | IRd | (95% CIc) | |
| Any event | 3 (0.5) | 1.3 | (0.3, 3.8) | 2 (0.3) | 0.9 | (0.1, 3.1) |
| INFECTIONS AND INFESTATIONS | 1 (0.2) | 0.4 | (0.0, 2.4) | 2 (0.3) | 0.9 | (0.1, 3.1) |
| Anal abscess | 1 (0.2) | 0.4 | (0.0, 2.4) | 0 | 0.0 | (0.0, 1.6) |
| Appendicitis | 0 | 0.0 | (0.0, 1.6) | 2 (0.3) | 0.9 | (0.1, 3.1) |
| Focal peritonitis | 0 | 0.0 | (0.0, 1.6) | 1 (0.2) | 0.4 | (0.0, 2.4) |
| PSYCHIATRIC DISORDERS | 2 (0.4) | 0.9 | (0.1, 3.1) | 0 | 0.0 | (0.0, 1.6) |
| Depression | 1 (0.2) | 0.4 | (0.0, 2.4) | 0 | 0.0 | (0.0, 1.6) |
| Suicidal ideation | 1 (0.2) | 0.4 | (0.0, 2.4) | 0 | 0.0 | (0.0, 1.6) |

## 14.51 Incidence Rates of at Least 1 Serious Adverse Event From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term, by Sex – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Sex: Female

**Table 14.51.** Incidence Rates of at Least 1 Serious Adverse Event From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term, by Sex – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population Sex: Female

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| | BNT162b2 (30 µg) (Na=564, TEb=2.3) | | | Placebo (Na=544, TEb=2.2) | | |
| nc (%) | IRd | (95% CIe) | nc (%) | IRd | (95% CIe) | |
| Any event | 7 (1.2) | 3.1 | (1.2, 6.4) | 0 | 0.0 | (0.0, 1.7) |
| GASTROINTESTINAL DISORDERS | 1 (0.2) | 0.4 | (0.0, 2.5) | 0 | 0.0 | (0.0, 1.7) |
| Abdominal pain | 1 (0.2) | 0.4 | (0.0, 2.5) | 0 | 0.0 | (0.0, 1.7) |
| Constipation | 1 (0.2) | 0.4 | (0.0, 2.5) | 0 | 0.0 | (0.0, 1.7) |
| INJURY, POISONING AND PROCEDURAL COMPLICATIONS | 1 (0.2) | 0.4 | (0.0, 2.5) | 0 | 0.0 | (0.0, 1.7) |
| Femur fracture | 1 (0.2) | 0.4 | (0.0, 2.5) | 0 | 0.0 | (0.0, 1.7) |
| PSYCHIATRIC DISORDERS | 6 (1.1) | 2.7 | (1.0, 5.8) | 0 | 0.0 | (0.0, 1.7) |
| Anxiety | 1 (0.2) | 0.4 | (0.0, 2.5) | 0 | 0.0 | (0.0, 1.7) |
| Conversion disorder | 1 (0.2) | 0.4 | (0.0, 2.5) | 0 | 0.0 | (0.0, 1.7) |
| Depression | 2 (0.4) | 0.9 | (0.1, 3.2) | 0 | 0.0 | (0.0, 1.7) |
| Suicidal ideation | 3 (0.5) | 1.3 | (0.3, 3.9) | 0 | 0.0 | (0.0, 1.7) |

Note: MedDRA (v24.0) coding dictionary applied.a. N = number of subjects in the specified group. This value is the denominator for the percentage calculations.b. TE = total exposure time in 100 person-years across all subjects in the specified group....

## 14.52 Incidence Rates of at Least 1 Serious Adverse Event From Unblinding Date to Data Cutoff Date (02SEP2021), by System Organ Class and Preferred Term – Open-Label Follow-up Period – Subjects Who Originally Received BNT162b2 – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population

**Table 14.52.** Incidence Rates of at Least 1 Serious Adverse Event From Unblinding Date to Data Cutoff Date (02SEP2021), by System Organ Class and Preferred Term – Open-Label Follow-up Period – Subjects Who Originally Received BNT162b2 – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | |
|---|---|---|---|
| BNT162b2 (30 µg) (Na=1107, TEb=3.3) nc (%) | IRd | (95% CIe) | |
| Any event | 4 (0.4) | 1.2 | (0.3, 3.1) |
| CONGENITAL, FAMILIAL AND GENETIC DISORDERS | 1 (0.1) | 0.3 | (0.0, 1.7) |
| Syringomyelia | 1 (0.1) | 0.3 | (0.0, 1.7) |
| INFECTIONS AND INFESTATIONS | 2 (0.2) | 0.6 | (0.1, 2.2) |
| Appendicitis | 2 (0.2) | 0.6 | (0.1, 2.2) |
| INJURY, POISONING AND PROCEDURAL COMPLICATIONS | 1 (0.1) | 0.3 | (0.0, 1.7) |
| Upper limb fracture | 1 (0.1) | 0.3 | (0.0, 1.7) |

Note: MedDRA (v24.0) coding dictionary applied.a. N = number of subjects in the specified group. This value is the denominator for the percentage calculations.b. TE = total exposure time in 100 person-years across all subjects in the specified group....

## 14.53 Incidence Rates of Subjects Withdrawn Because of Adverse Events From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population

**Table 14.53.** Incidence Rates of Subjects Withdrawn Because of Adverse Events From Dose 1 to Unblinding Date, by System Organ Class and Preferred Term – Blinded Placebo-Controlled Follow-up Period – Phase 2/3 Subjects 12 Through 15 Years of Age – Safety Population

| System Organ Class Preferred Term | Vaccine Group (as Administered) | | | | | |
|---|---|---|---|---|---|---|
| | BNT162b2 (30 µg) (Na=1131, TEb=4.6) | | | Placebo (Na=1129, TEb=4.5) | | |
| nc (%) | IRd | (95% CIe) | nc (%) | IRd | (95% CIe) | |
| Any event | 1 (0.1) | 0.2 | (0.0, 1.2) | 0 | 0.0 | (0.0, 0.8) |
| GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS | 1 (0.1) | 0.2 | (0.0, 1.2) | 0 | 0.0 | (0.0, 0.8) |
| Pyrexia | 1 (0.1) | 0.2 | (0.0, 1.2) | 0 | 0.0 | (0.0, 0.8) |

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This section provides a list of articles from the literature pertinent to the evaluation of the study, in accordance with the internationally accepted standards of the 1979 Vancouver Declaration on 'Uniform Requirements for Manuscripts Submitted to Biomedical Journals'.

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# 16. Appendices

This section provides a comprehensive list of all appendices available for the study report. Those appendices submitted with this report are clearly indicated. All appendices available upon request have been finalized at the time of filing of this submission.

## 16.1 List of Appendices

### 16.1.1 Appendices Submitted with the Report

- Appendix 16.2: Protocol and Protocol Amendments
- Appendix 16.3: Sample Case Report Form
- Appendix 16.4: List of IECs or IRBs and Representative Written Information for Patient and Sample Consent Forms
- Appendix 16.5: List and Description of Investigators and Other Important Participants in the Study
- Appendix 16.6: Signatures of Principal or Coordinating Investigator(s) or Sponsor's Responsible Medical Officer
- Appendix 16.7: Listing of Patients Receiving Investigational Product from Specific Batches
- Appendix 16.8: Randomisation Scheme and Codes
- Appendix 16.10: Documentation of Statistical Methods
- Appendix 16.11: Documentation of Inter-laboratory Standardisation Methods and Quality Assurance Procedures
- Appendix 16.14: Listing of Discontinued Patients
- Appendix 16.15: Listing of Protocol Deviations
- Appendix 16.16: Listing of Patients Excluded from the Efficacy Analysis
- Appendix 16.17: Demographic Data
- Appendix 16.20: Adverse Event Listings (Each Patient)
- Appendix 16.22: CRFs for Deaths, Other Serious Adverse Events, and Withdrawals for Adverse Events
- Appendix 16.23: Other CRFs Submitted
- Appendix 16.24: Individual Patient Data Listings (US Archival Listings)

### 16.1.2 Appendices Available on Request

- Appendix 16.9: Audit Certificates
- Appendix 16.12: Publications Based on the Study
- Appendix 16.13: Important Publications Referenced in the Report
- Appendix 16.18: Compliance and/or Drug Concentration Data
- Appendix 16.19: Individual Efficacy Response Data
- Appendix 16.21: Listing of Individual Laboratory Measurements by Patient

## 16.2 Protocol and Protocol Amendments

The final version of the study protocol and all subsequent amendments are included in this appendix.

## 16.3 Sample Case Report Form

A sample of unique pages from the Case Report Form (CRF) used in this study is included in this appendix.

## 16.4 List of IECs or IRBs and Representative Written Information for Patient and Sample Consent Forms

A comprehensive list of all Institutional Ethics Committees (IECs) or Institutional Review Boards (IRBs) that reviewed and approved the study protocol, along with the names of their respective Chairs (where required by regulatory authorities), is provided. Representative written information provided to patients and their sample consent forms are also included.

## 16.5 List and Description of Investigators and Other Important Participants in the Study

This appendix contains a list of all investigators and other important participants involved in the study. Brief curricula vitae (CVs) or equivalent summaries, not exceeding one page, detailing their training and experience relevant to the conduct of the clinical study, are also included.

## 16.6 Signatures of Principal or Coordinating Investigator(s) or Sponsor's Responsible Medical Officer

The original signatures of the principal or coordinating investigator(s) or the sponsor's responsible medical officer, as per regulatory authority requirements, are provided in this appendix.

## 16.7 Listing of Patients Receiving Investigational Product from Specific Batches

This appendix lists the patients who received investigational product from specific batches, where more than one batch was used during the study.

Table 22. Investigational Product Lot Numbers – Interim – Adolescent 6-Month Update

| Investigational Product | Manufacturer | Vendor Lot Number (Manufacturer) | Lot Number (Pfizer) |
|---|---|---|---|
| BNT162b2 (30 µg) | BioNTech | BCV40720-A | PA2074172/P220395-0053L |
| | | BCV40720-A | PA2074998/P220395-0060L |
| | | BCV40720-B | PA2074173/P220395-0051L |
| | | BCV40720-C | PA2074071/P220395-0052L |
| | | ED3938 | PA2074300/P220395-0021L |

| Investigational Product | Manufacturer | Vendor Lot Number (Manufacturer) | Lot Number (Pfizer) |
|---|---|---|---|
| | | ED3938 | PA2074300/P220395-0022L |
| | | ED3938 | PA2074300/P220395-0023L |
| | | EE3813 | NC2075485/P220395-0068L |
| | | EE3813 | NC2075485/P220395-0074L |
| | | EE3813 | NC2075485/P220395-0077L |
| | | EE3813 | PA2074838/P220395-0020L |
| | | EE3813 | PA2074838/P220395-0024L |
| | | ER9449Z | PA2096794/P220395-0079L |
| | | ER9449Z | PA2096794/P220395-0082L |
| | | EE8493Y | PA2087473/P220395-0073L |
| | | EJ0553Z | PA2085061/P220395-0070L |
| Placebo (normal saline 0.9% sodium chloride solution) | Pfizer | DK2074;20-002221 | PA2069407/P220395-0032L |
| | | DK2074;20-002221 | PA2069407/P220395-0033L |
| | | | PA2069407/P220395- |

## 16.8 Randomisation Scheme and Codes

The randomisation scheme and codes, detailing patient identification and their assigned treatment, are provided in this appendix. This includes documentation of the methodology used for randomisation.

## 16.9 Audit Certificates

Audit certificates, if available, are provided in this appendix. These documents confirm the audit of trial processes and data.

## 16.10 Documentation of Statistical Methods

Detailed documentation of all statistical methods and analyses employed in this study is included in this appendix. This encompasses the statistical principles, methods, and proce-

dures adhered to.

## 16.11 Documentation of Inter-laboratory Standardisation Methods and Quality Assurance Procedures

Documentation of any inter-laboratory standardisation methods and quality assurance procedures used in the study, particularly for analytical or diagnostic procedures, is provided in this appendix.

## 16.12 Publications Based on the Study

Any publications arising directly from the data or findings of this study are listed or included in this appendix.

## 16.13 Important Publications Referenced in the Report

A list of important publications that were referenced in the main body of this clinical study report is provided in this appendix.

## 16.14 Listing of Discontinued Patients

This appendix provides a comprehensive listing of all patients who discontinued from the study, along with the reasons for their discontinuation.

In a cohort of 140 participants, 13 participants did not complete the study. The most frequently cited reason for non-completion in this group was the inability to contact participants or their caregivers for follow-up.

For the larger Phase 2/3 analysis, the disposition of 37,796 randomized participants was consistent across the BNT162b2 and placebo groups. A small percentage of participants (0.6% overall) discontinued the study vaccine after the first dose and prior to the second dose.

The most common reasons reported for discontinuation from the study overall included: - **No longer meeting eligibility criteria:** This accounted for 0.3% of participants in the BNT162b2 group and 0.4% in the placebo group. The primary reasons for this category were a previous clinical or microbiological diagnosis of COVID-19. - **Withdrawal by participant:** Participants or their legal representatives requested to withdraw. - **Adverse Event (AE):** Withdrawals due to AEs were reported for 0.1% of participants in both treatment groups.

Specifically, regarding AEs leading to study withdrawal: - Among 37,706 participants with a median of 2 months of follow-up after Dose 2, a total of 34 participants in the BNT162b2 group (0.2%) and 25 participants in the placebo group (0.1%) were withdrawn due to AEs. - The AEs most frequently leading to study withdrawal were categorized under System Organ Classes (SOCs) including gastrointestinal disorders, general disorders and administration site conditions, injury, poisoning and procedural complications, musculoskeletal and connective tissue disorders, and nervous system disorders. - Across all 43,448 enrolled participants included in the safety database up to the data cutoff date, similar percentages of participants were withdrawn due to AEs (0.2% in the BNT162b2 group and 0.1% in the

placebo group). - No participants aged 16 to 17 years experienced an AE that led to their withdrawal from the study.

## 16.15 Listing of Protocol Deviations

This appendix provides a listing of all protocol deviations that occurred during the conduct of the study, along with their classification and any impact on the study results or participant safety.

## 16.16 Listing of Patients Excluded from the Efficacy Analysis

This appendix provides a listing of all patients who were excluded from the efficacy analysis, along with the reasons for their exclusion, as detailed in the Efficacy Populations table.

Table 23. Efficacy Populations - Subjects 12 Through 15 Years of Age - Blinded Placebo-Controlled Follow-up

| Vaccine Group (as Randomized) | BNT162b2 (30 µg) n (%) | Placebo n (%) | Total n (%) |
|---|---|---|---|
| **Randomized** | 1134 (100.0) | 1130 (100.0) | 2264 (100.0) |
| **Dose 1 all-available efficacy population** | 1131 (99.7) | 1129 (99.9) | 2260 (99.8) |
| Subjects without evidence of infection before Dose 1 | 1083 (95.5) | 1078 (95.4) | 2161 (95.5) |
| **Subjects excluded from Dose 1 all-available efficacy population** | **3 (0.3)** | **1 (0.1)** | **4 (0.2)** |
| **Reason for exclusion** | | | |
| Did not receive at least 1 vaccination | 3 (0.3) | 1 (0.1) | 4 (0.2) |
| **Dose 2 all-available efficacy population** | 1123 (99.0) | 1117 (98.8) | 2240 (98.9) |
| Subjects without evidence of infection prior to 7 days after Dose 2 | 1061 (93.6) | 1037 (91.8) | 2098 (92.7) |
| **Subjects excluded from Dose 2 all-available efficacy population** | **11 (1.0)** | **13 (1.2)** | **24 (1.1)** |
| **Reason for exclusion** | | | |
| Did not receive 2 vaccinations | 10 (0.9) | 13 (1.2) | 23 (1.0) |
| Unblinded prior to 7 days after Dose 2 | 1 (0.1) | 0 | 1 (0.0) |
| **Evaluable efficacy (7 days) population** | 1119 (98.7) | 1109 (98.1) | 2228 (98.4) |
| Subjects without evidence of infection prior to 7 days after Dose 2 | 1057 (93.2) | 1030 (91.2) | 2087 (92.2) |

| Vaccine Group (as Randomized) | BNT162b2 (30 µg) n (%) | Placebo n (%) | Total n (%) |
|---|---|---|---|
| **Subjects excluded from evaluable efficacy (7 days) population** | **15 (1.3)** | **21 (1.9)** | **36 (1.6)** |
| **Reason for exclusion** | | | |
| Randomized but did not meet all eligibility criteria | 1 (0.1) | 1 (0.1) | 2 (0.1) |
| Did not receive all vaccinations as randomized or did not receive Dose 2 within the predefined window (19-42 days after Dose 1) | 14 (1.2) | 19 (1.7) | 33 (1.5) |
| Unblinded prior to 7 days after Dose 2 | 1 (0.1) | 0 | 1 (0.0) |
| Had other important protocol deviations on or prior to 7 days after Dose 2 | 0 | 3 (0.3) | 3 (0.1) |

## 16.17 Demographic Data

This appendix presents the demographic characteristics of the study population.

Table 24. Demographic Characteristics - Phase 2/3 Subjects 12 Through 15 Years of Age - Safety Population

| Vaccine Group (as Administered) | BNT162b2 (30 µg) (Na=1131) n (%) | Placebo (Na=1129) n (%) | Total (Na=2260) n (%) |
|---|---|---|---|
| **Sex** | | | |
| Male | 567 (50.1) | 585 (51.8) | 1152 (51.0) |
| Female | 564 (49.9) | 544 (48.2) | 1108 (49.0) |
| **Race** | | | |
| White | 970 (85.8) | 962 (85.2) | 1932 (85.5) |
| Black or African American | 52 (4.6) | 57 (5.0) | 109 (4.8) |
| All others | 109 (9.6) | 110 (9.7) | 219 (9.7) |
| American Indian or Alaska Native | 4 (0.4) | 3 (0.3) | 7 (0.3) |
| Asian | 72 (6.4) | 71 (6.3) | 143 (6.3) |
| Native Hawaiian or other Pacific Islander | 3 (0.3) | 0 | 3 (0.1) |
| Multiracial | 24 (2.1) | 29 (2.6) | 53 (2.3) |
| Not reported | 6 (0.5) | 7 (0.6) | 13 (0.6) |
| **Racial designation** | | | |
| Japanese | 5 (0.4) | 2 (0.2) | 7 (0.3) |
| **Ethnicity** | | | |
| Hispanic/Latino | 132 (11.7) | 130 (11.5) | 262 (11.6) |

| Vaccine Group (as Administered) | BNT162b2 (30 µg) (Na=1131) n (%) | Placebo (Na=1129) n (%) | Total (Na=2260) n (%) |
|---|---|---|---|
| Non-Hispanic/non-Latino | 997 (88.2) | 996 (88.2) | 1993 (88.2) |
| Not reported | 2 (0.2) | 3 (0.3) | 5 (0.2) |
| **Country** | | | |
| USA | 1131 (100.0) | 1129 (100.0) | 2260 (100.0) |
| **Baseline SARS-CoV-2 status** | | | |
| Positive | 46 (4.1) | 50 (4.4) | 96 (4.2) |
| Negative | 1083 (95.8) | 1078 (95.5) | 2161 (95.6) |
| Missing | 2 (0.2) | 1 (0.1) | 3 (0.1) |
| **Comorbidities** | | | |
| Yes | 249 (22.0) | 242 (21.4) | 491 (21.7) |
| No | 882 (78.0) | 887 (78.6) | 1769 (78.3) |
| **Obese** | | | |
| Yes | 143 (12.6) | 128 (11.3) | 271 (12.0) |
| No | 988 (87.4) | 1001 (88.7) | 1989 (88.0) |
| **Age at vaccination (years)** | | | |
| Mean (SD) | 13.6 (1.11) | 13.6 (1.11) | 13.6 (1.11) |
| Median | 14.0 | 14.0 | |
| Min, max | (12, 15) | (12, 15) | (12, 15) |

## 16.18 Compliance and/or Drug Concentration Data

Pharmacokinetic (drug concentration) parameters were not evaluated in this study, as determined to be unnecessary based on relevant WHO guidelines for vaccine candidates.

However, compliance with study intervention was monitored and recorded. For participants dosed at the site, the study intervention was administered directly by the investigator or a designee under medical supervision. The date and time of each administered dose were recorded in the source documents and subsequently in the Case Report Forms (CRFs). Additionally, the dose of study intervention and participant identification were confirmed at the time of dosing by a study site staff member separate from the person administering the intervention.

## 16.19 Individual Efficacy Response Data

Individual patient efficacy response data, which includes details of each participant's outcome with respect to the defined efficacy endpoints, are available upon request to regulatory authorities. This data allows for detailed examination of individual responses to the investigational product or placebo.

The aggregate efficacy results, including Vaccine Efficacy (VE) calculations and case counts, are provided in Section 6 (Efficacy) of the Clinical Study Report. These analyses cover

various populations and time points, demonstrating the overall efficacy profile of the investigational product.

## 16.20 Adverse Event Listings (Each Patient)

This appendix contains a comprehensive listing of all adverse events reported for each patient in the study. While aggregate summaries and analyses of adverse events are presented in Section 6 (Safety) of the Clinical Study Report, detailed individual patient listings of all reported adverse events, including severity, causality assessment, and outcome, are extensive. Therefore, complete individual patient adverse event listings are available upon request to regulatory authorities.

## 16.21 Listing of Individual Laboratory Measurements by Patient

Individual patient listings of all clinical safety laboratory measurements, including hematology and chemistry parameters, as well as immunogenicity assessments (e.g., SARS-CoV-2 neutralization assay, S1-binding IgG level assay, RBD-binding IgG level assay, and N-binding antibody assay), are maintained. These detailed listings are available upon request to regulatory authorities. The protocol outlined specific time points and procedures for collecting and processing these samples, and investigators were responsible for reviewing laboratory reports and documenting any clinically significant abnormal findings in the Case Report Forms (CRFs).

## 16.22 CRFs for Deaths, Other Serious Adverse Events and Withdrawals for AE

This appendix includes copies of the Case Report Forms (CRFs) for all patients who experienced death, other serious adverse events, or withdrew from the study due to an adverse event.

## 16.23 Other CRFs Submitted

This appendix contains copies of any other relevant Case Report Forms (CRFs) submitted as part of this clinical study report, as required by regulatory authorities.

## 16.24 Individual Patient Data Listings (US Archival Listings)

This appendix comprises the comprehensive individual patient data listings prepared for archival purposes, in accordance with US regulatory requirements.